1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
0.1	0010932459	ENGINEERING RELEASED	06/04/2014

SCHEM,MLB,VENUS2, PROTO1A

ENG 06/04/2014

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00388	1	SCHEM,MLB,VENUS2,PROTO1A	SCH	CRITICAL	
820-00426	1	PCBF,MLB,VENUS2,PROTO1A	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=ABBREV
LAST_MODIFIED=Mon Jan 12 16:34:40 2015

DRAWING TITLE
<PART_DESCRIPTION>

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00042	COMMON PARTS,MLB,VENUS2	X425_COMMON
985-00050	DEV BOM,MLB,VENUS2	X425_DEVEL:ENG
639-00682	PCBA,MLB,VENUS2,CTO,16GHYN,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:CTO,RAM:HYNIX_1600,FB_4G_HYNIX
639-00703	PCBA,MLB,VENUS2,CTO,16GMIC,VR-4GMIC,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:CTO,RAM:MICRON_1600,FB_4G_MICRON
639-00739	PCBA,MLB,VENUS2,CTO,16GHYN,VR-4GMIC,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:CTO,RAM:HYNIX_1600,FB_4G_MICRON
639-00740	PCBA,MLB,VENUS2,CTO,16GMIC,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:CTO,RAM:MICRON_1600,FB_4G_HYNIX
639-00798	PCBA,MLB,VENUS2,BEST,16GHYN,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:BEST,RAM:HYNIX_1600,FB_4G_HYNIX
639-00799	PCBA,MLB,VENUS2,BEST,16GMIC,VR-4GMIC,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:BEST,RAM:MICRON_1600,FB_4G_MICRON
639-00800	PCBA,MLB,VENUS2,BEST,16GHYN,VR-4GMIC,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:BEST,RAM:HYNIX_1600,FB_4G_MICRON
639-00801	PCBA,MLB,VENUS2,BEST,16GMIC,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,CPU_CRW:BEST,RAM:MICRON_1600,FB_4G_HYNIX
639-00803	PCBA,MLB,VENUS2,NOCPU,16GMIC,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,GFX_BOM,RAM:MICRON_1600,FB_4G_HYNIX
639-00974	PCBA,MLB,NOGPU,CTO,16GMIC,VR-4GHYN,X425G	BASE_BOM,DEVEL_BOM,CPU_CRW:CTO,RAM:MICRON_1600,FB_4G_HYNIX

X425 BOM Groups

BOM GROUP	BOM OPTIONS
X425_COMMON	ALTERNATE,COMMON,X425_COMMON1,X425_COMMON2,X425_PROGPARTS,ACAPS:A2
X425_COMMON1	CPUMEM:S0,TBTHV:P15V,SKIP_5V3V3:AUDIBLE,CPUPEG:X8X4X4,S2_PWR:S0,SMC_SUSACK:YES
X425_COMMON2	EDP:YES,XDP,SSD_PWR_EN:GPIO,CAM_WAKE:NO,SAMCONN,APCLKRQ:ISOL,CRW_SPRT,WLAN_SW:SIL
X425_PVT	BKLT:PROD,SENSOR_NONPROD:N
X425_PROGPARTS	SMC_PROG:BASE,BOOTROM_PROG:EVT,TBTROM:PROG,DPMUXMCU:PROG
X425_DEVEL:ENG	ALTERNATE,XDP_DEBUG,S0PGOOD_ISL,SENSOR_NONPROD:Y,SENSOR_NONPROD_R,BKLT:ENG,DBGLED,DPMUX_DEBUG,GPU_ROM:YES,SENSOR_GPU_NONPROD:Y
X425_DEVEL:DVT	ALTERNATE,XDP_DEBUG,BKLT:PROD,SENSOR_NONPROD:N,DBGLED
X425_DEVEL:PVT	XDP_DEBUG
GFX_BOM	VENUS:XTA
XDP_DEBUG	XDP_CONN,XDP_PCH

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00058	1	CRW,SR1ZX,PRQ,C0,2.5,47W,4+3E,1.2,6M,BGA	U0500	CRITICAL	CPU_CRW:BEST
337S00059	1	CRW,SR1ZY,PRQ,C0,2.8,47W,4+3E,1.2,6M,BGA	U0500	CRITICAL	CPU_CRW:CTO
337S4542	1	IC,QEWV,LPT-M,HM87,C2,SR199,PRQ,FCBGA	U1100	CRITICAL	
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC,FCBGA288	U2800	CRITICAL	
338S1264	1	IC,BCM15700A2,S2 PCIE CMRA,8X8,208FCBGA	U3900	CRITICAL	
333S0700	1	IC,SDRAM,4GBIT,DDR3L-1600,GEMMA,96B FBGA	U4000	CRITICAL	
333S00032	32	IC,SDRAM,DDR3L-1600,4GBIT,78B FBGA		CRITICAL	HYNIX_1600
333S0660	32	IC,SDRAM,4GBIT,DDR3L-1600,V80A,78P,FBGA		CRITICAL	MICRON_1600
337S00116	1	IC,GPU,VENUS XTAA1,QS,29X29MM,FCBGA962	U8400	CRITICAL	VENUS:XTA
333S00027	4	IC,GDDR5,4GBIT,6GBPS,1.5V,25NM,BGA170	U8800,U8850,U8900,U8950	CRITICAL	FB_4G_HYNIX
333S0766	4	IC,GDDR5,4GBIT,6GBPS,128MX32,25NM,170BGA	U8800,U8850,U8900,U8950	CRITICAL	FB_4G_MICRON

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM:HYNIX_1600	HYNIX_1600,RAMCFG3:H,RAMCFG2:H,RAMCFG1:L,RAMCFG0:L
RAM:MICRON_1600	MICRON_1600,RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:L

COMMON/DEVEL BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00042	1	COMMON PARTS,MLB,VENUS,X425G	BASE	CRITICAL	BASE_BOM
985-00050	1	DEV,MLB,VENUS,X425G	DEVEL	CRITICAL	DEVEL_BOM

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BOM Configuration

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Programmables - All builds

335S0915	1	IC,SERIAL SPI FLASH ROM,4MBIT,50MHZ,USON	U2890	CRITICAL	TBTROM:BLANK
341S00166	1	T29,FALCON RIDGE(V27.1)PROTO0,X425G	U2890	CRITICAL	TBTROM:PROG
335S0724	1	1MBIT SERIAL FLASH 2X3X0.6MM UFDFPN8 PKG	U9101	CRITICAL	GPUROM:BLANK
341S3565	1	IC,EDP MUX-95C,(RENESAS) V3.2.8,DVB,D2	U9600	CRITICAL	DPMUXMCU:PROG
337S4313	1	IC,MCU,H8S/2113,9X9MM,TLP-145V	U9600	CRITICAL	DPMUXMCU:BLANK

SMC

338S1214	1	IC,SMC-B1,40MHZ/50DMIPS,SCPL FW,157BGA	U5000	CRITICAL	SMC_PROG:BLANK
341S00157	1	IC,SMC-B1,EXT (V2.25A9) PROTO 0,X425G	U5000	CRITICAL	SMC_PROG:BASE

EFI ROM

335S00007	1	IC,SERIAL FLASH,64MB,3V,WSON,6X5MM	U6100	CRITICAL	BOOTROM_BLANK:WIN
335S00006	1	IC,SERIAL FLASH,64MB,3V,WSON,6X5MM	U6100	CRITICAL	BOOTROM_BLANK:MAC
341S00239	1	IC,EFI ROM (V0145) EVT,X425	U6100	CRITICAL	BOOTROM_PROG:EVT

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	Diodes alt to Fairchild
128S0311	128S0329		ALL	NEC alt to Sanyo
138S0739	138S0706		ALL	Samsung alt to Murata
197S0481	197S0480		ALL	Epson Alt to NDK
197S0478	197S0479		ALL	NDK Alt to Epson
371S0713	371S0558		ALL	DDS alt to ST
152S0461	152S1645		ALL	Cyntec alt to Vishay
376S1080	376S0820		ALL	Diodes alt to On Semi
155S0667	155S00008		ALL	Panasonic alt to TDK
376S00074	376S0855		ALL	Toshiba alt to Diodes
376S1129	376S0855		ALL	NXP alt to Diodes
376S1089	376S1128		ALL	NXP alt to Diodes
128S0371	128S0376		ALL	Kemet alt to Sanyo
138S0803	138S0639		ALL	Samsung alt to Murata
138S0843	138S0674		ALL	Samsung alt to Murata
138S0846	138S0811		ALL	Samsung alt to Murata
127S0164	127S0162		ALL	Rohm alt to Vishay
138S0732	138S0715		ALL	Rohm alt to Vishay
128S0364	128S0264		ALL	Kemet alt to Sanyo
333S0704	333S0700		ALL	ELPIDA to HYNIX U4000
311S0649	311S0541		ALL	ON alt to Toshiba
376S00014	376S0761		ALL	Toshiba alt to Vishay
740S00003	740S0135		ALL	AEM alt to Tyco
740S00004	740S0134		ALL	AEM alt to Littlefuse
107S00029	107S00030		ALL	TFT alt to Cyntec
128S0398	128S0220		ALL	Kemet alt to Sanyo
128S0386	128S0284		ALL	Kemet alt to Sanyo
311S00008	311S0271		ALL	Diodes alt to NXP
128S0393	128S0334		ALL	Kemet alt to Sanyo
311S00007	311S0426		ALL	Diodes alt to NXP
371S00017	371S0749		ALL	Diodes alt to Onsemi
107S00033	107S00034		ALL	TFT alt to Cyntec
107S0240	107S0255		ALL	TFT alt to Cyntec
107S0248	107S0250		ALL	TFT alt to Cyntec
107S00031	107S00032		ALL	TFT alt to Cyntec
107S0249	107S0251		ALL	TFT alt to Cyntec
107S00037	107S00038		ALL	TFT alt to Cyntec
107S00015	107S00011		ALL	TFT alt to Cyntec
128S00008	128S0380		ALL	NEC alt to Sanyo
311S00060	311S0273		ALL	Diodes alt to NXP
353S00133	353S2741		ALL	ON Semi alt to TI
353S00394	353S2162		ALL	ON Semi alt to TI
376S00086	376S0761		ALL	Diodes alt to Vishay
112S00001	112S0254		ALL	Yageo alt to Cyntec
353S00095	353S3328		ALL	Pericom alt to TI
128S0397	128S0325		ALL	Kemet alt to Sanyo
311S00004	311S0370		ALL	ON Semi alt to NXP

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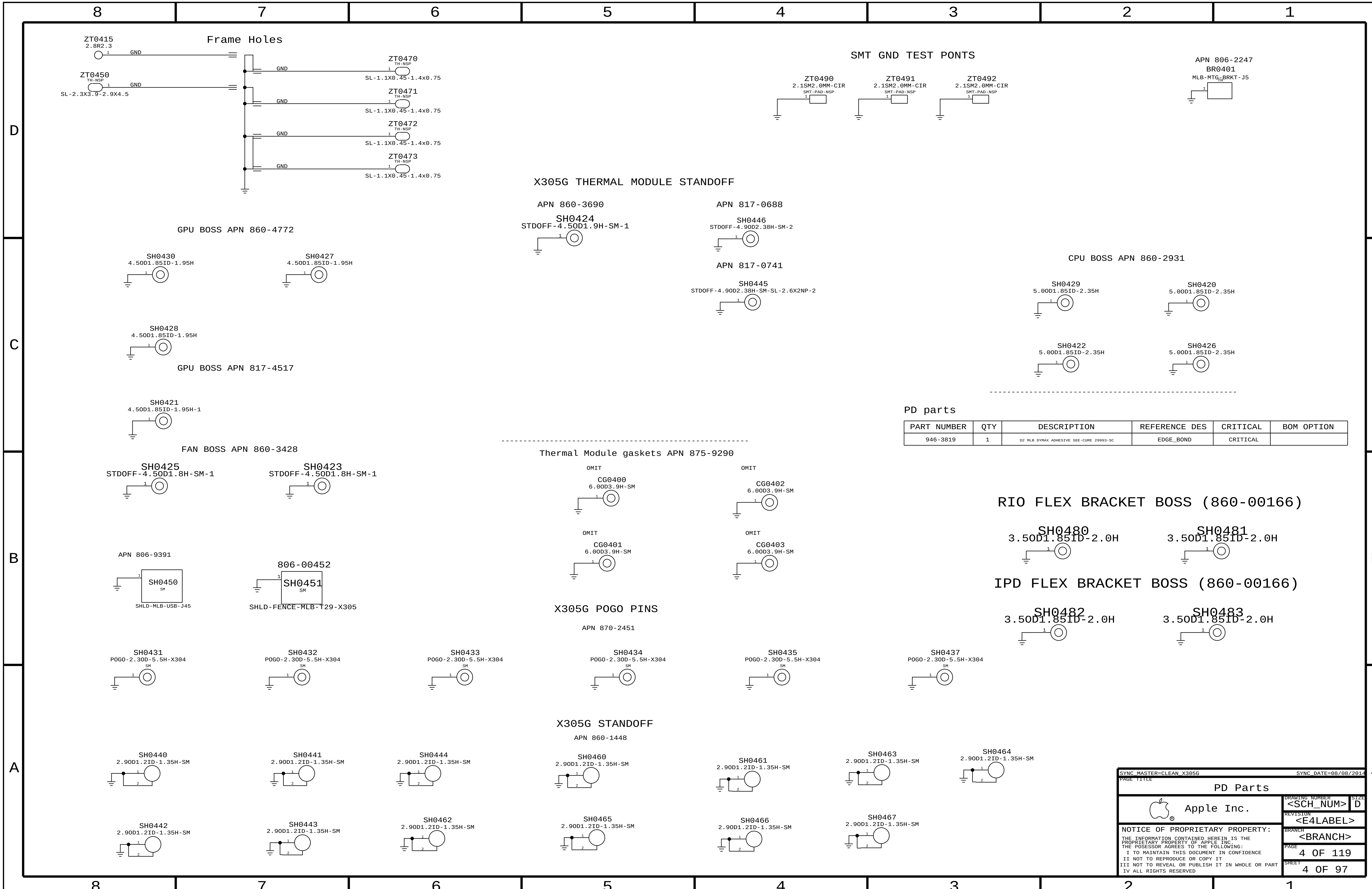

PD parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
946-3819	1	D2 MLB DYMAX ADHESIVE SEE-CURE 29993-SC	EDGE_BOND	CRITICAL	

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PAGE TITLE
PD Parts

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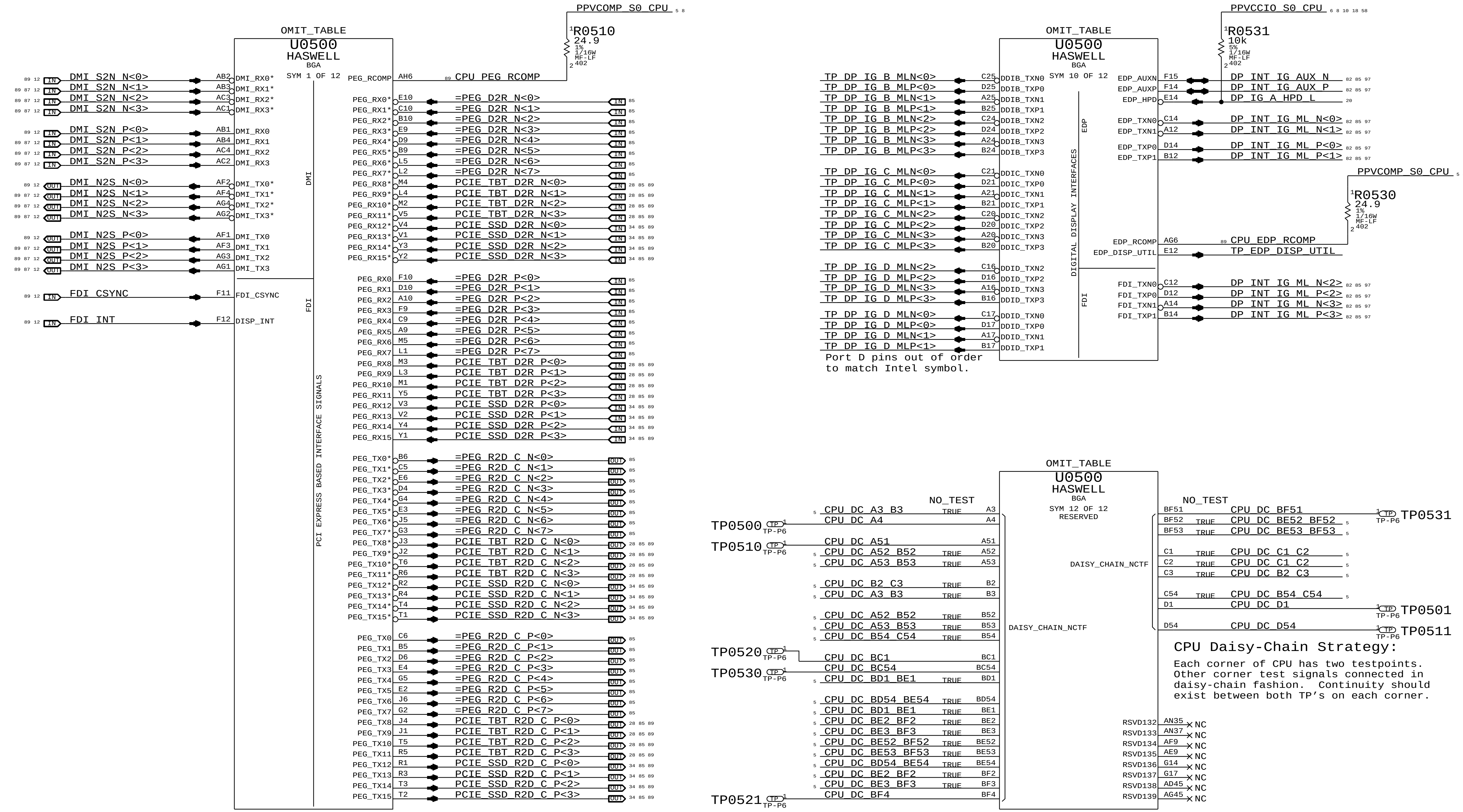

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PAGE TITLE

CPU DMI/PEG/FDI/RSVD

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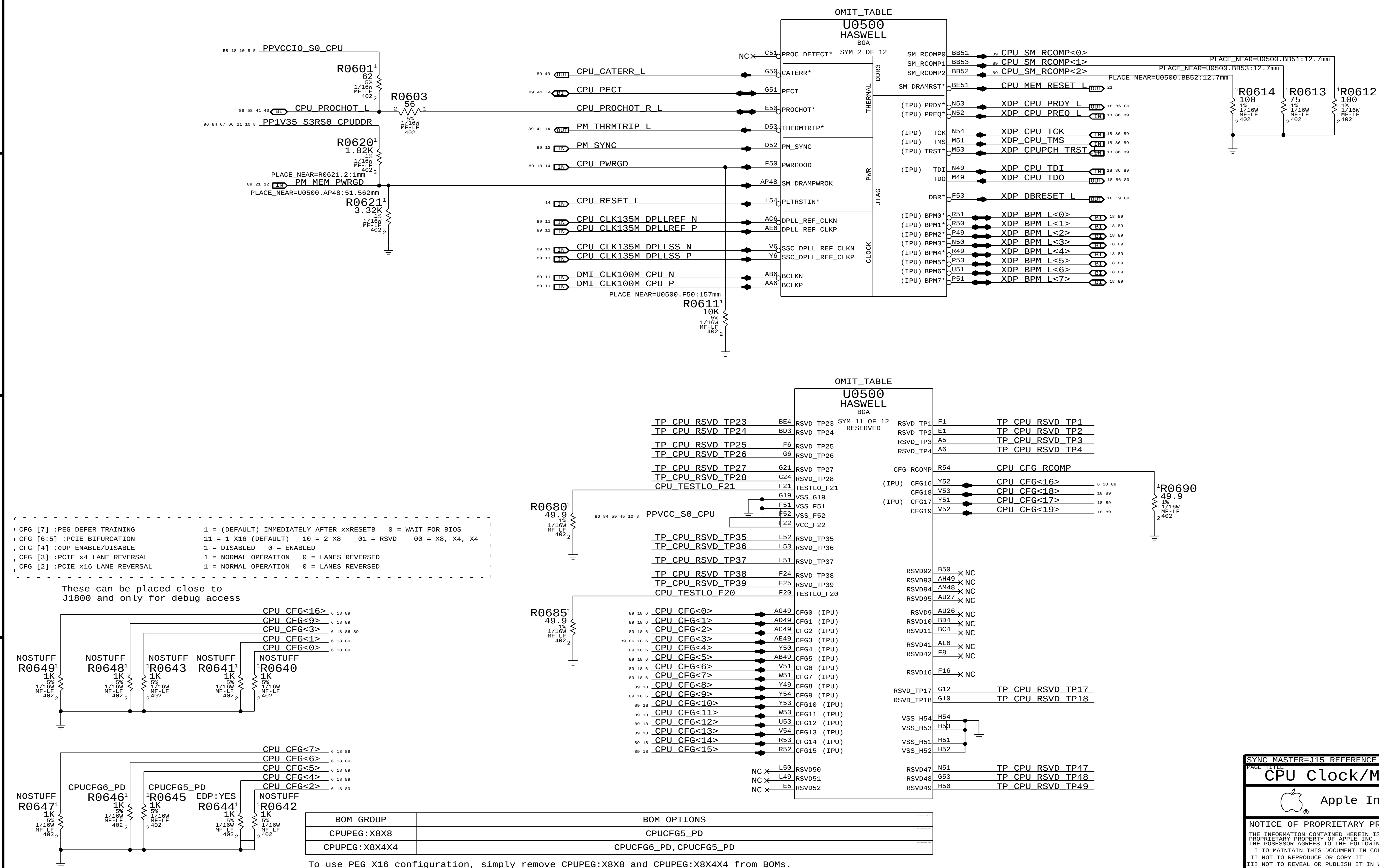

CFG [7] :PEG DEFER TRAINING	1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION	11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE	1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL	1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL	1 = NORMAL OPERATION 0 = LANES REVERSED

These can be placed close to J1800 and only for debug access

BOM GROUP	BOM OPTIONS
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD,CPUCFG5_PD

To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.

SYNC_MASTER=J15_REFERENCE SYNC_DATE=12/18/2012

PAGE TITLE
CPU Clock/Misc/JTAG/CFG

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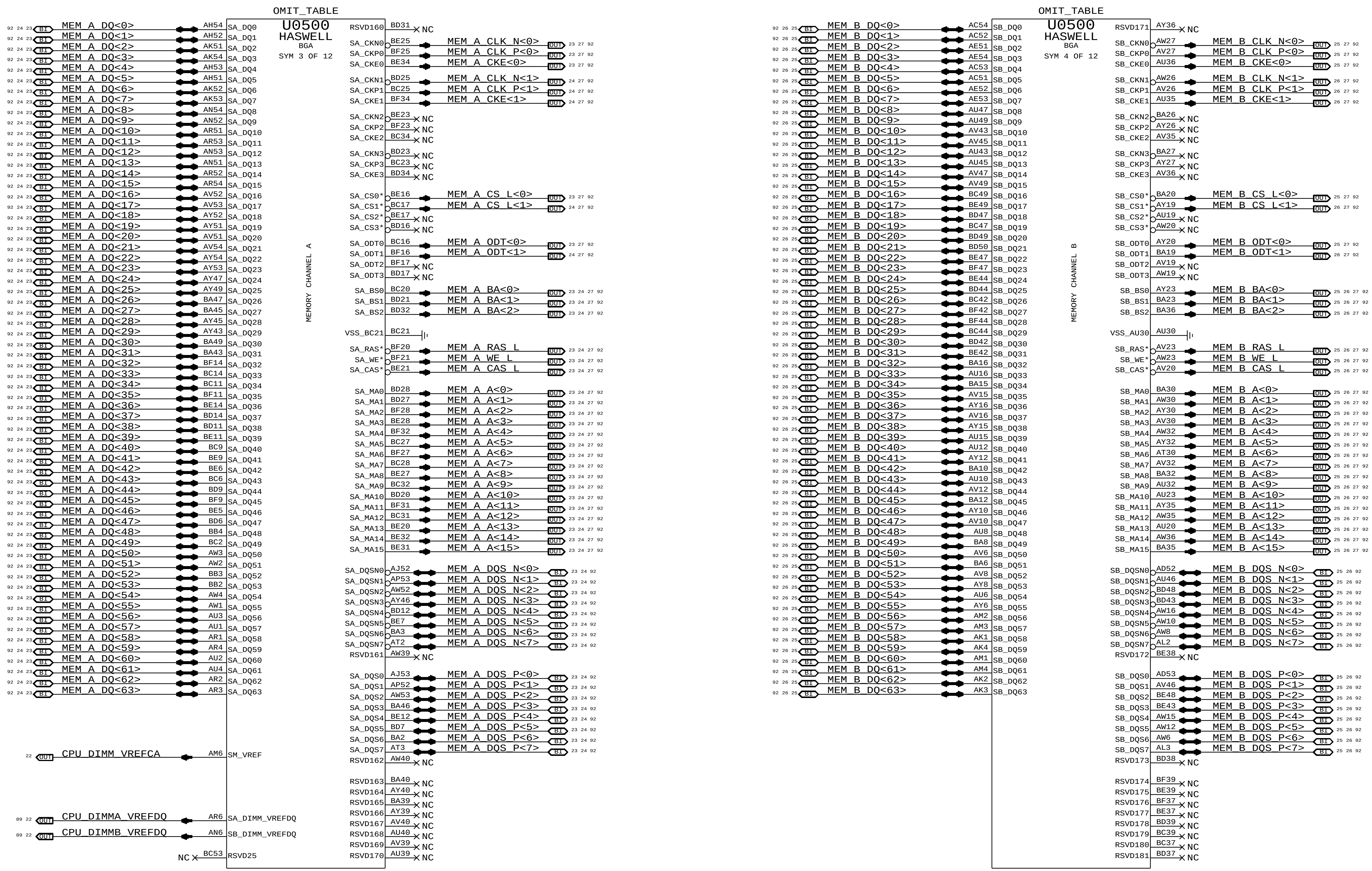

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CPU DDR3 Interfaces

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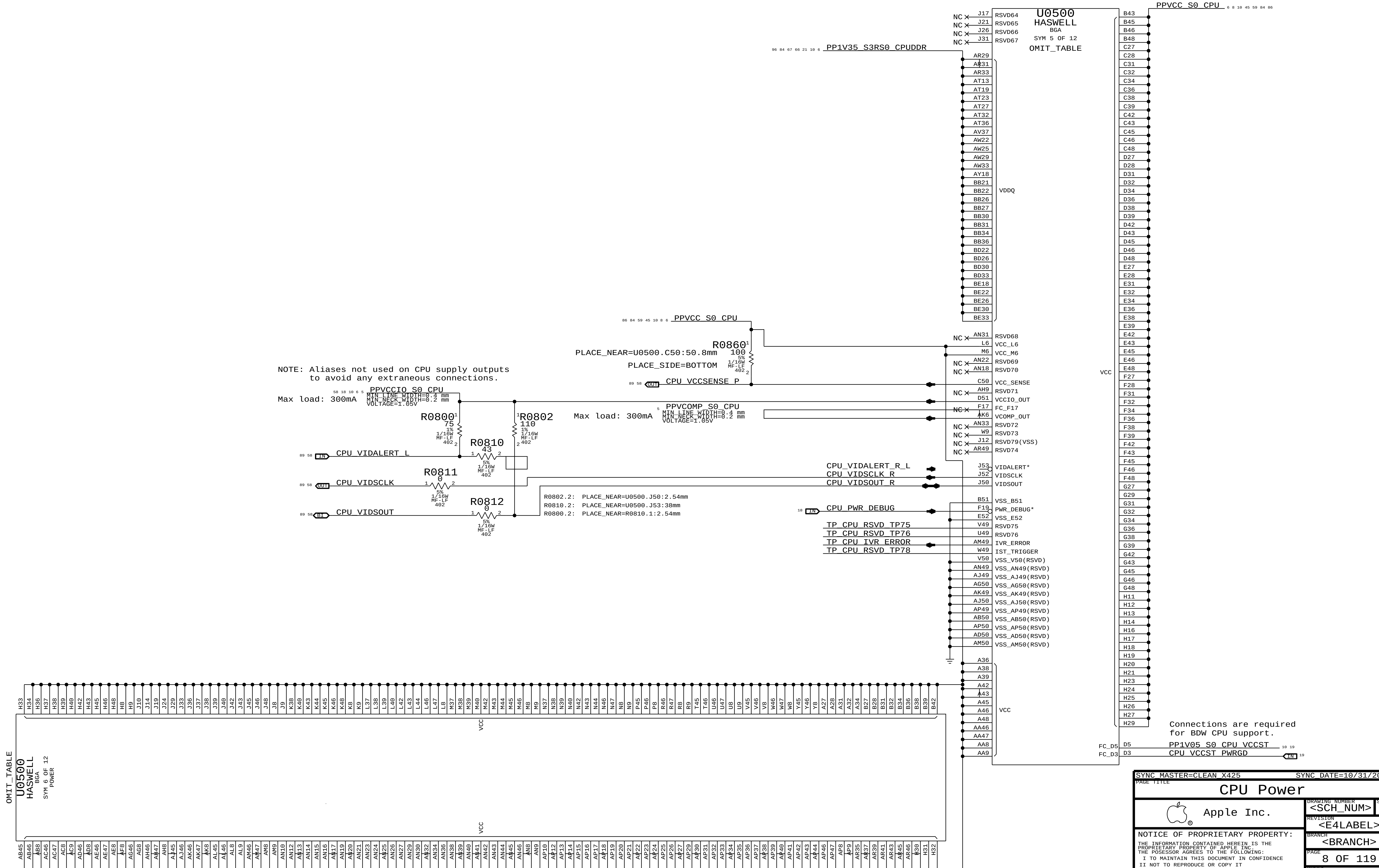

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PAGE TITLE

CPU Power

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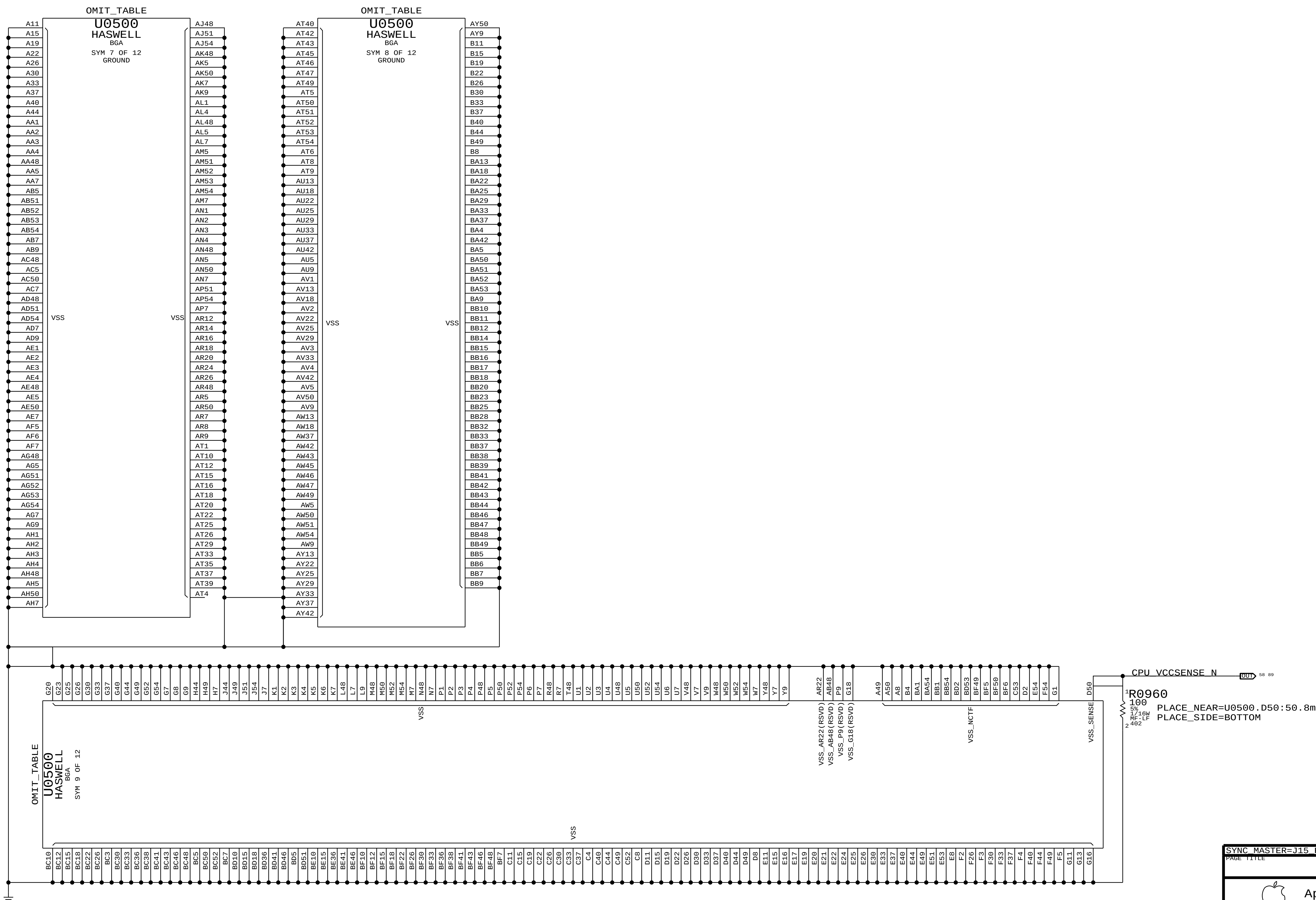

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CPU Ground

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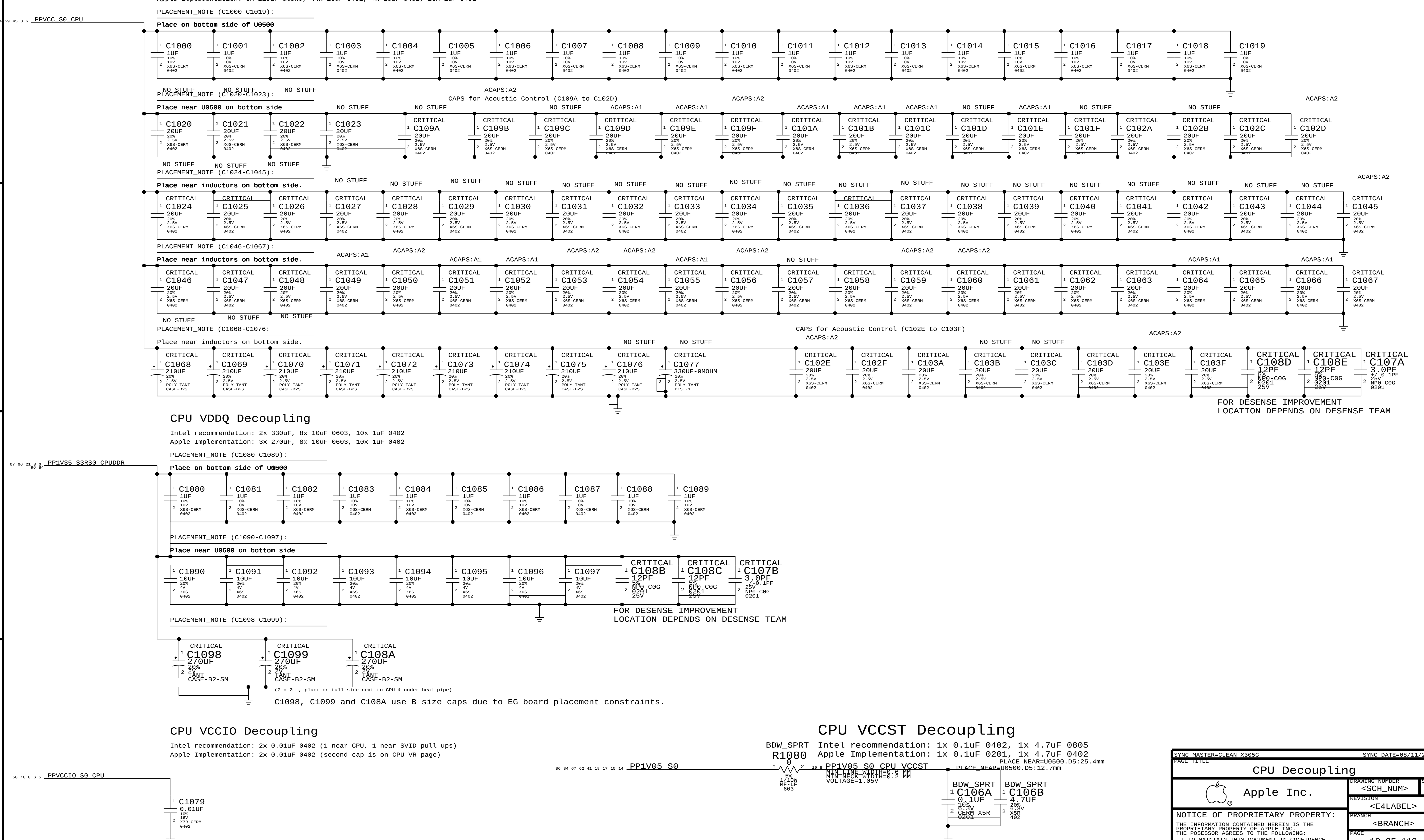
CPU VCORE Decoupling

Intel recommendation: 4x 470uF 4mOhm (3 CPU-side, 1 opposite), 20x 22uF 0805 (10 CPU-side, 10 opposite near edge, 4x 10uF 0603 (2 CPU-side, 2 opposite), 20x 1uF 0402 (under CPU)

Apple Implementation: 9x 210uF 6mOhm, 44x 10uF 0402, 4x 10uF 0402, 20x 1uF 0402



CPU VDDQ Decoupling

Intel recommendation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402

Apple Implementation: 3x 270uF, 8x 10uF 0603, 10x 1uF 0402

C1098, C1099 and C108A use B size caps due to EG board placement constraints.

CPU VCCIO Decoupling

Intel recommendation: 2x 0.01uF 0402 (1 near CPU, 1 near SVID pull-ups)

Apple Implementation: 2x 0.01uF 0402 (second cap is on CPU VR page)

CPU VCCST Decoupling

Intel recommendation: 1x 0.1uF 0402, 1x 4.7uF 0805

Apple Implementation: 1x 0.1uF 0201, 1x 4.7uF 0402

NOTE: Intel decoupling recommendations from Shark Bay Mobile Platform Power Delivery Design Guide (doc #487822, Rev 0.8 dated January 2012), Section 5.

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CPU Decoupling

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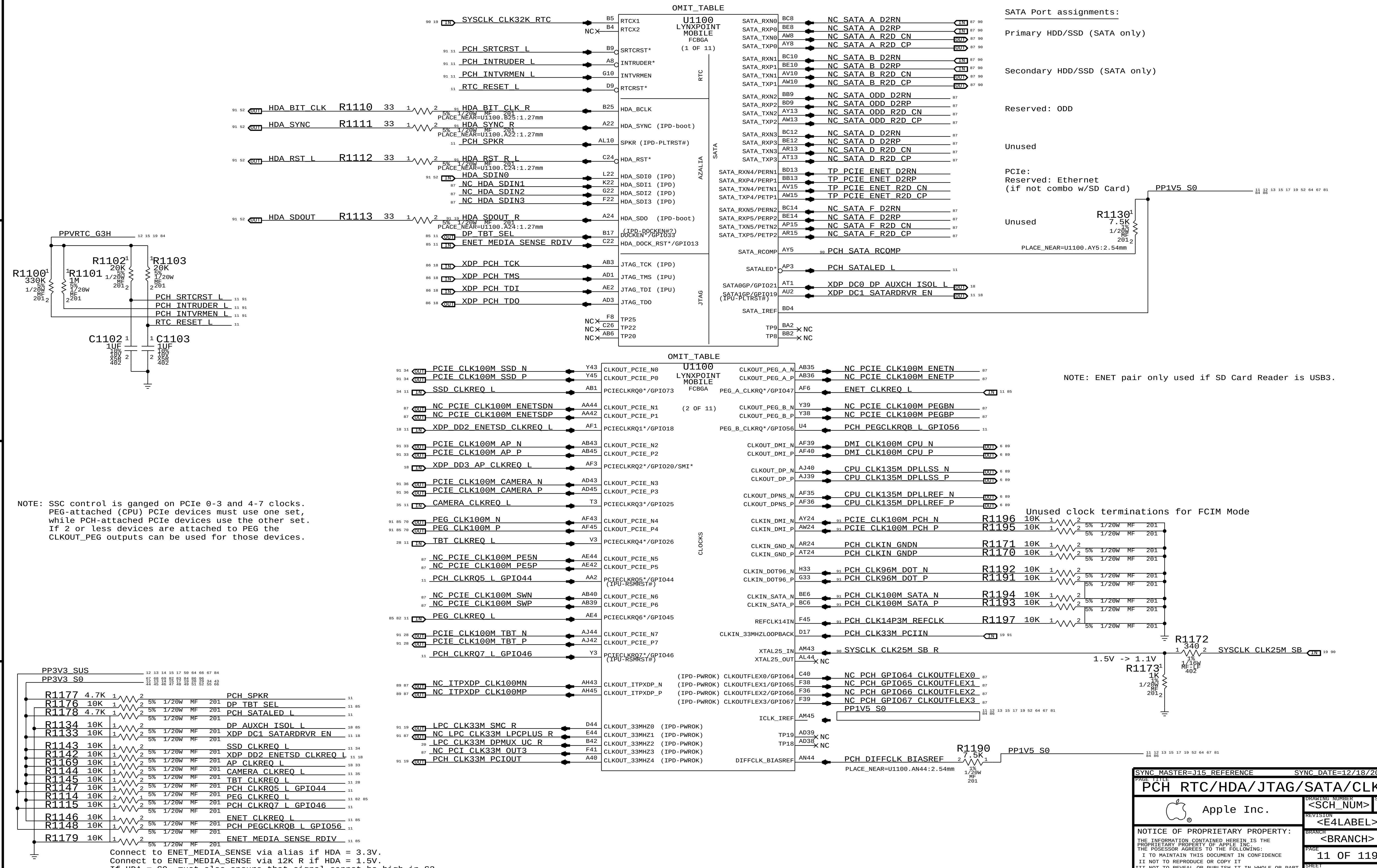

NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

Connect to ENET_MEDIA_SENSE via alias if HDA = 3.3V.
Connect to ENET_MEDIA_SENSE via 12K R if HDA = 1.5V.
If HDA = S0, must also ensure that signal cannot be high in S3.

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PAGE TITLE
PCH RTC/HDA/JTAG/SATA/CLK

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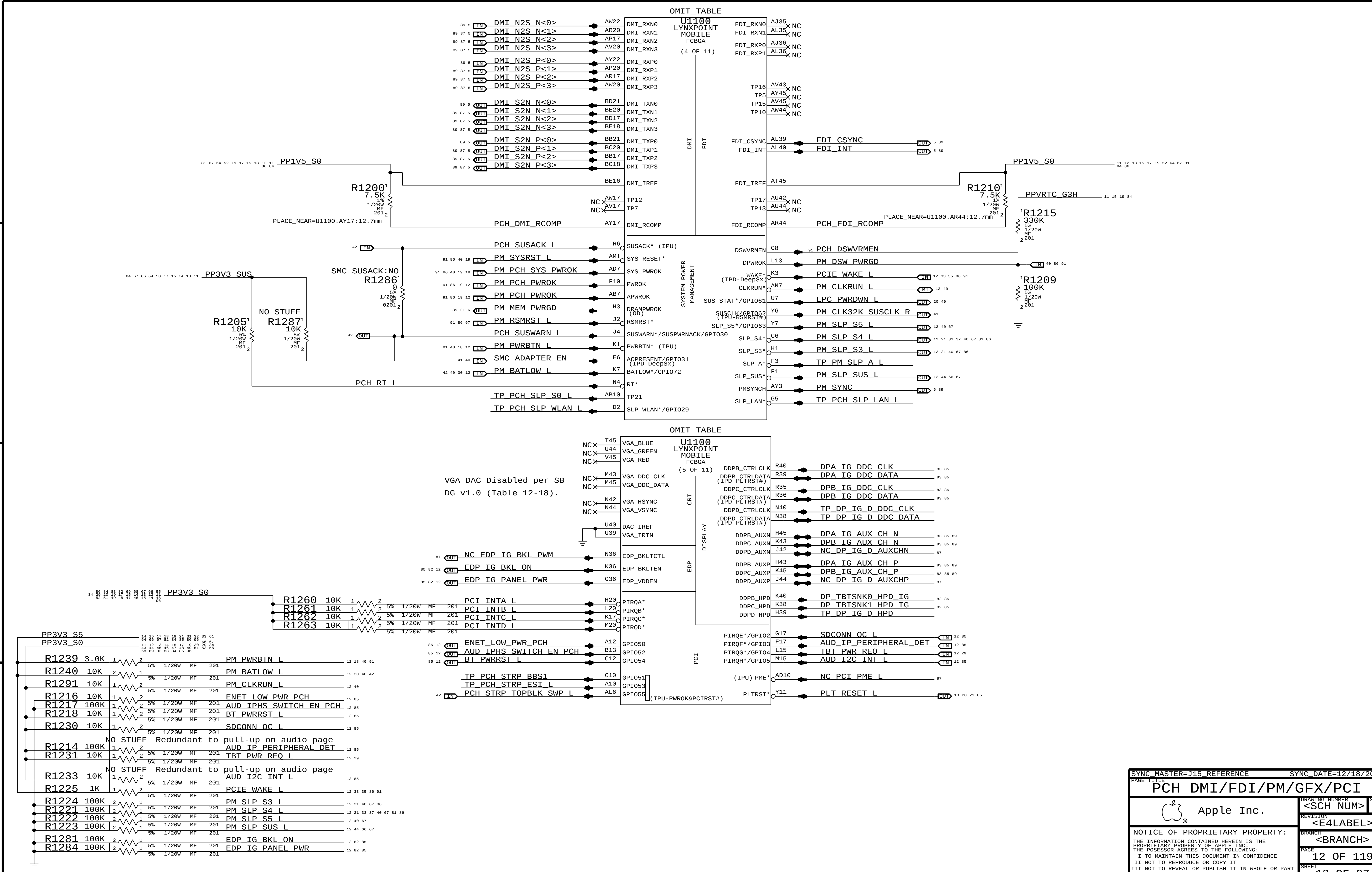

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PAGE TITLE

PCH DMI/FDI/PM/GFX/PCI

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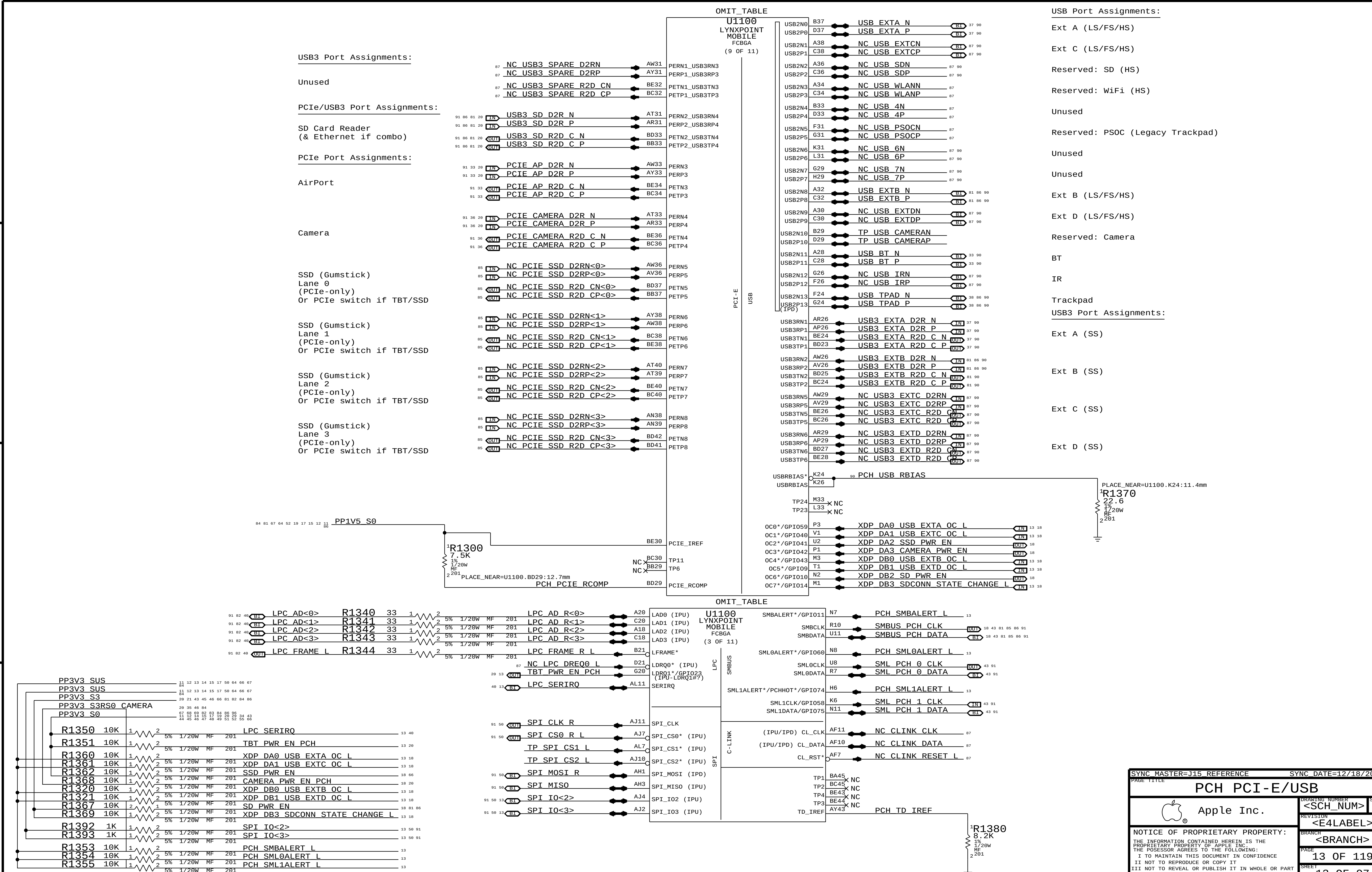

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PCH PCI-E/USB

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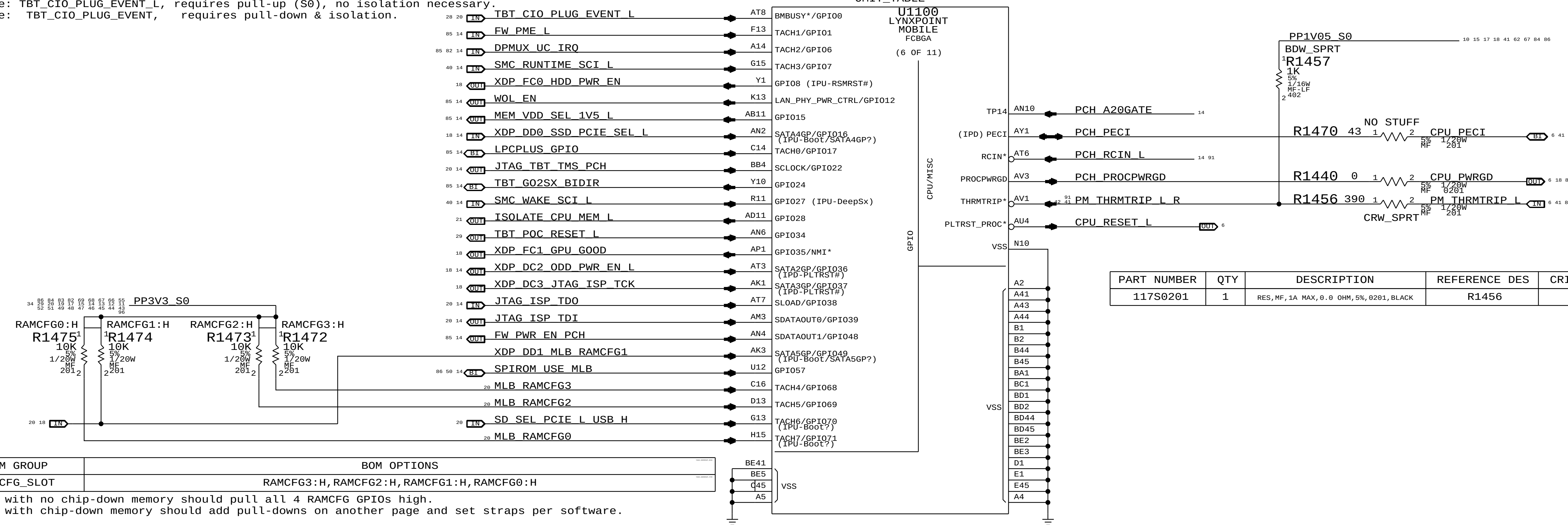
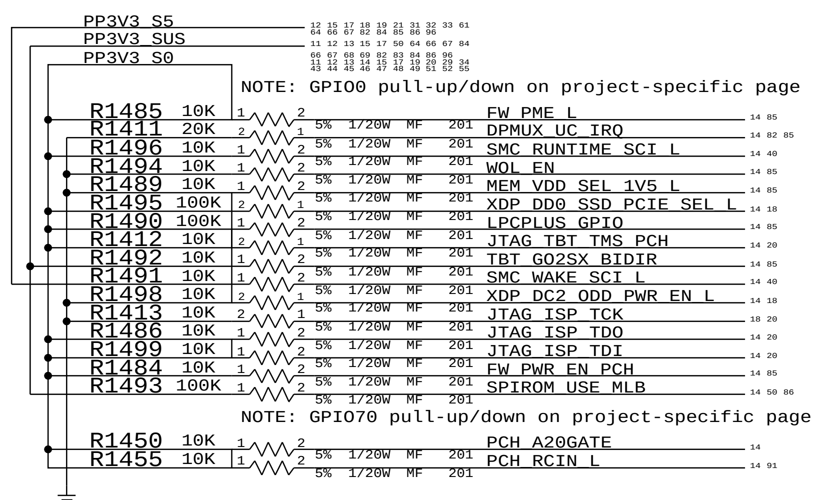
Pull-up/down on chipset support page (depends on TBT controller)
Falcon Ridge: TBT_CIO_PLUG_EVENT_L, requires pull-up (S0), no isolation necessary.
Cactus Ridge: TBT_CIO_PLUG_EVENT, requires pull-down & isolation.



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0201	1	RES,MF,1A MAX,0.0 OHM,5%,0201,BLACK	R1456		BDW_SPRT

BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H

Systems with no chip-down memory should pull all 4 RAMCFG GPIOs high.
Systems with chip-down memory should add pull-downs on another page and set straps per software.



SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/31/2014

PAGE TITLE
PCH GPIO/MISC/NCTF

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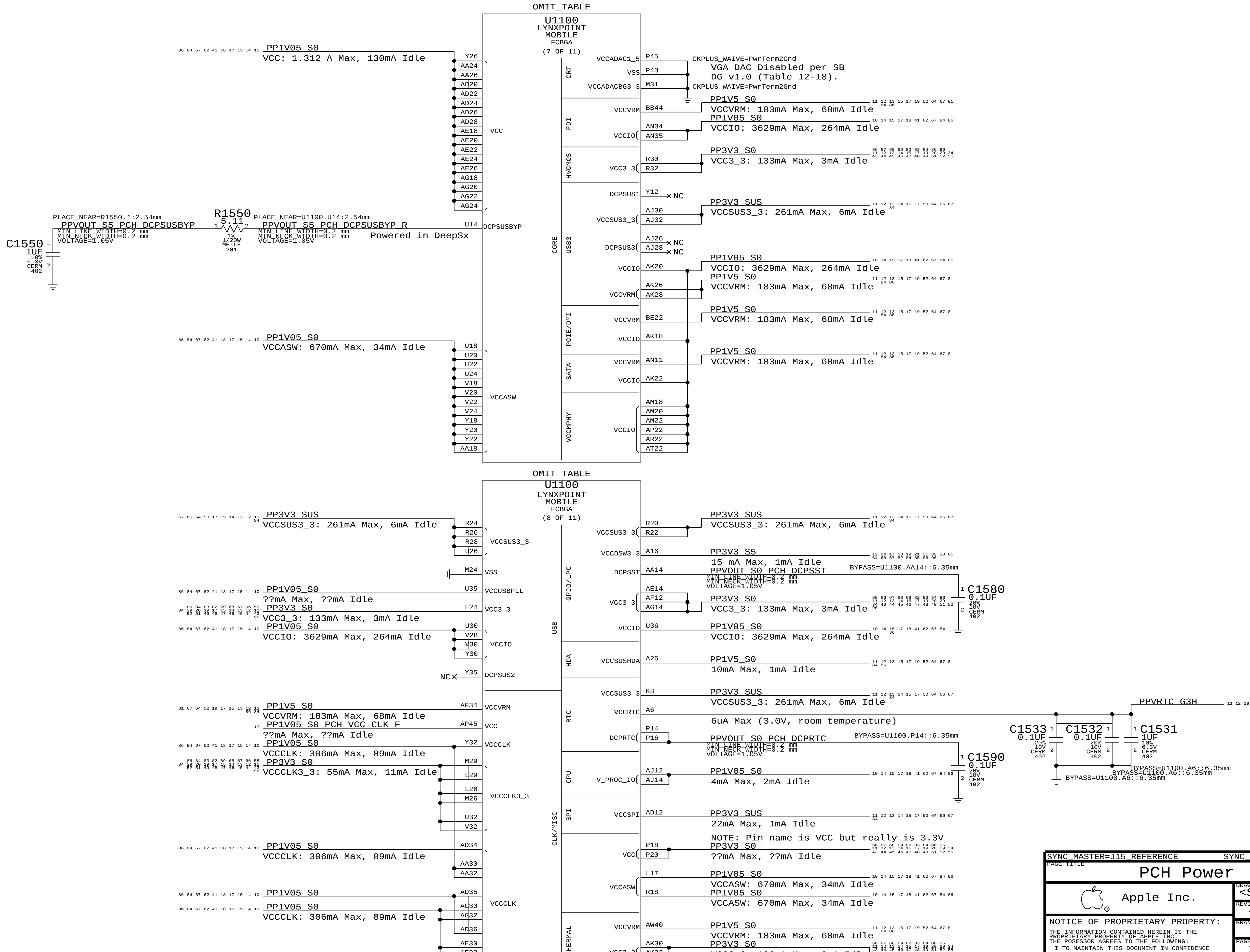

Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC_MASTER=J15_REFERENCE	SYNC_DATE=12/18/2012
PAGE TITLE	PCH Power
Apple Inc.	DRAWING NUMBER <SCH_NUM> SIZE D
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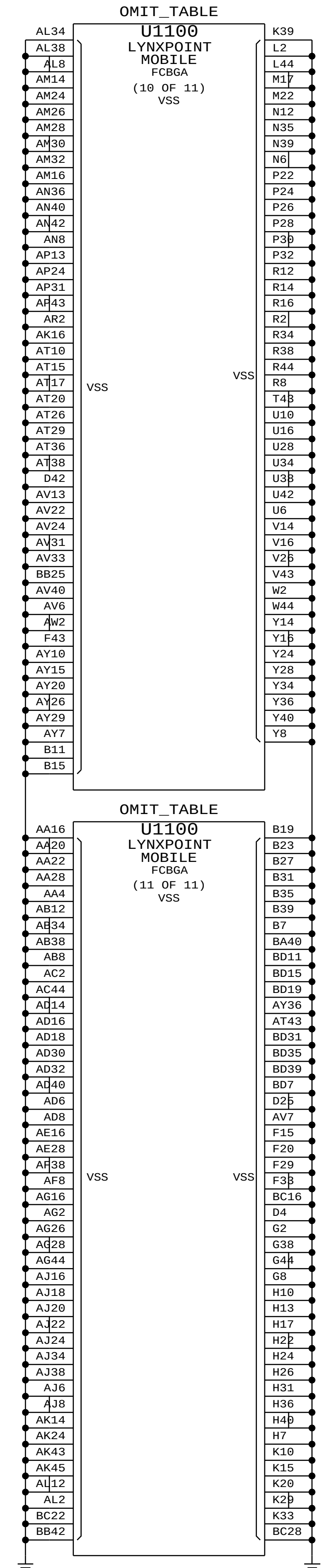

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PCH Grounds

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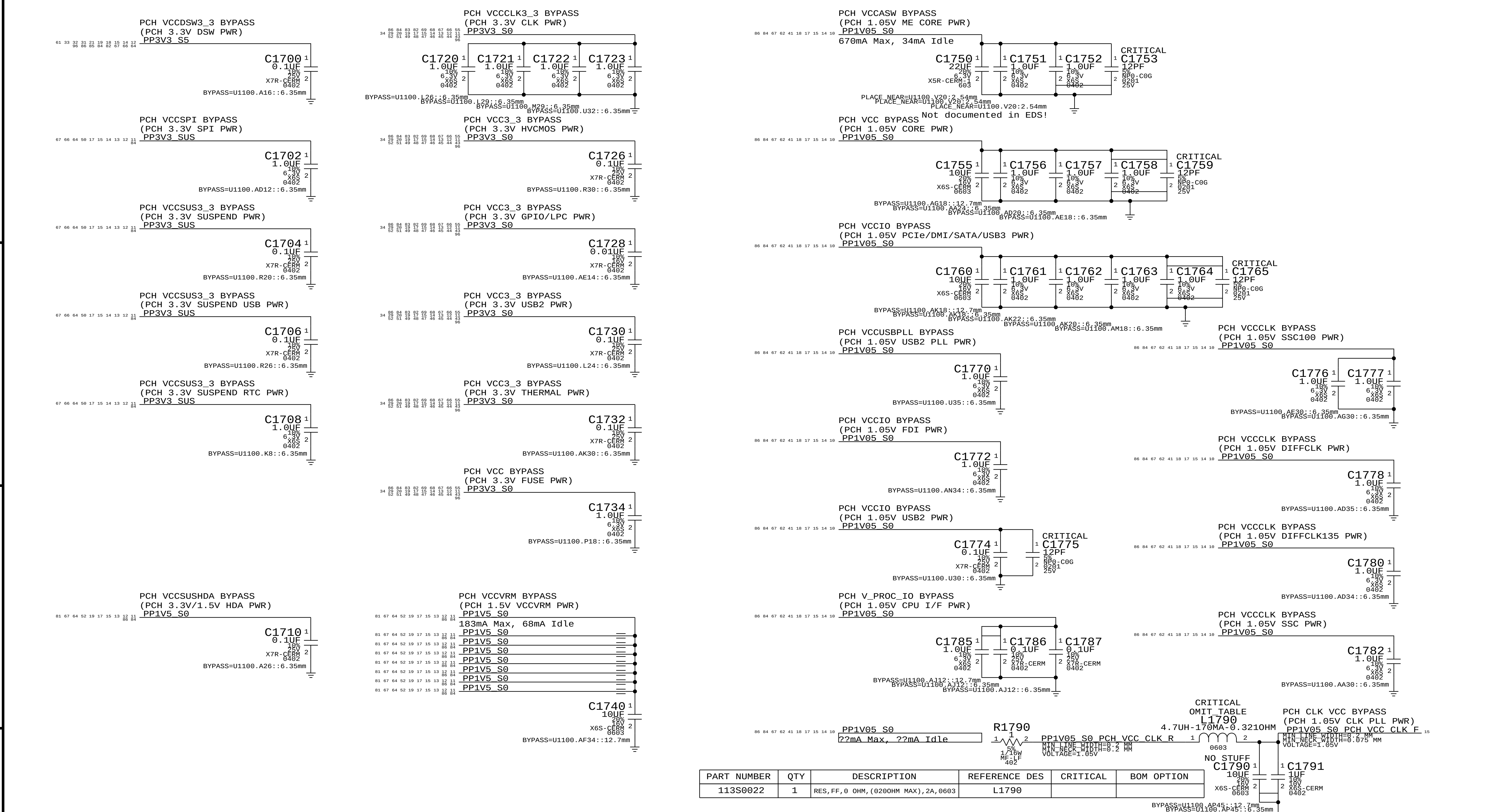

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	1	RES,FF,0 OHM,(020OHM MAX),2A,0603	L1790		

Current data from LPT EDS (doc #486708, Rev 1.0).

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PCH DECOUPLING

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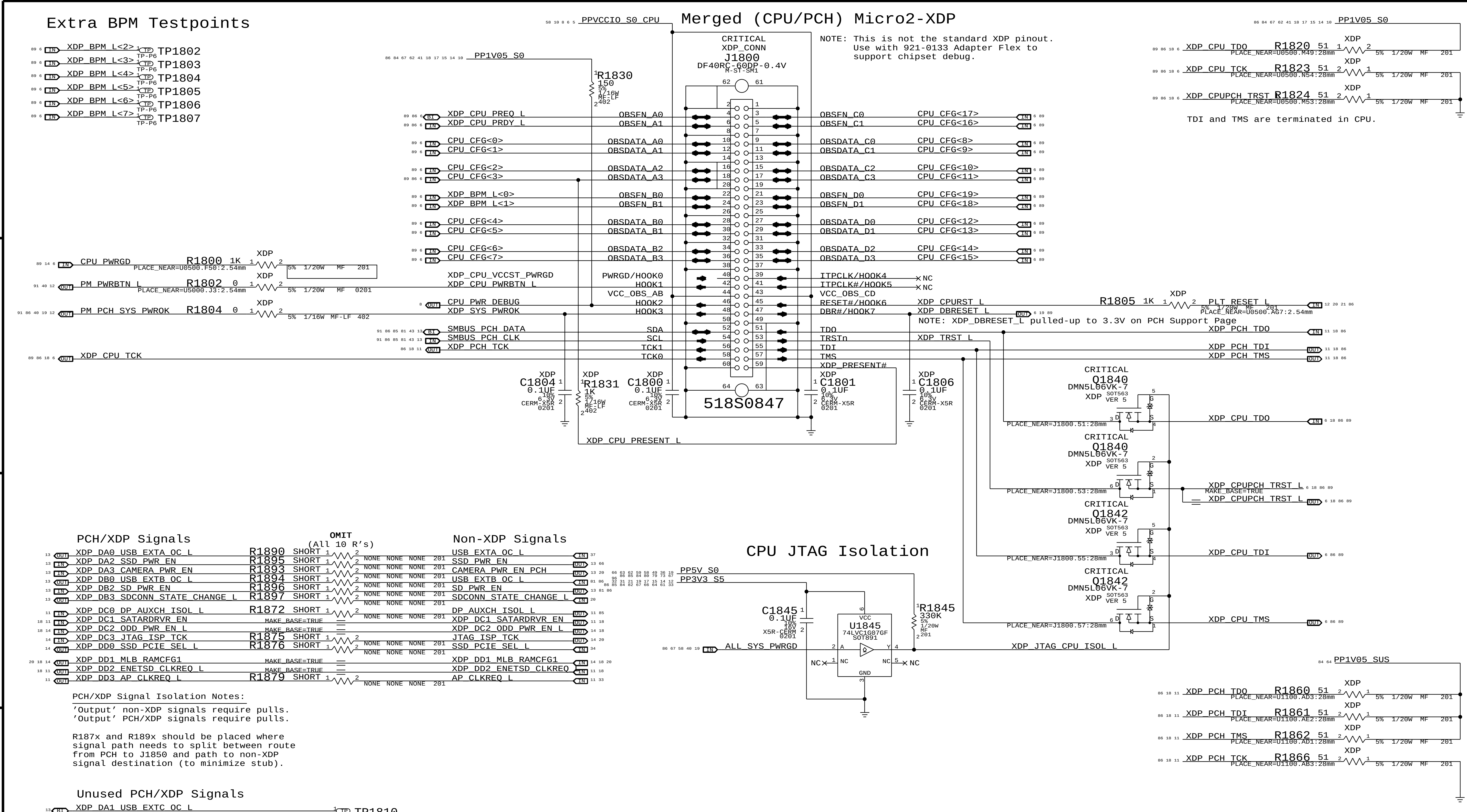

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CPU & PCH XDP

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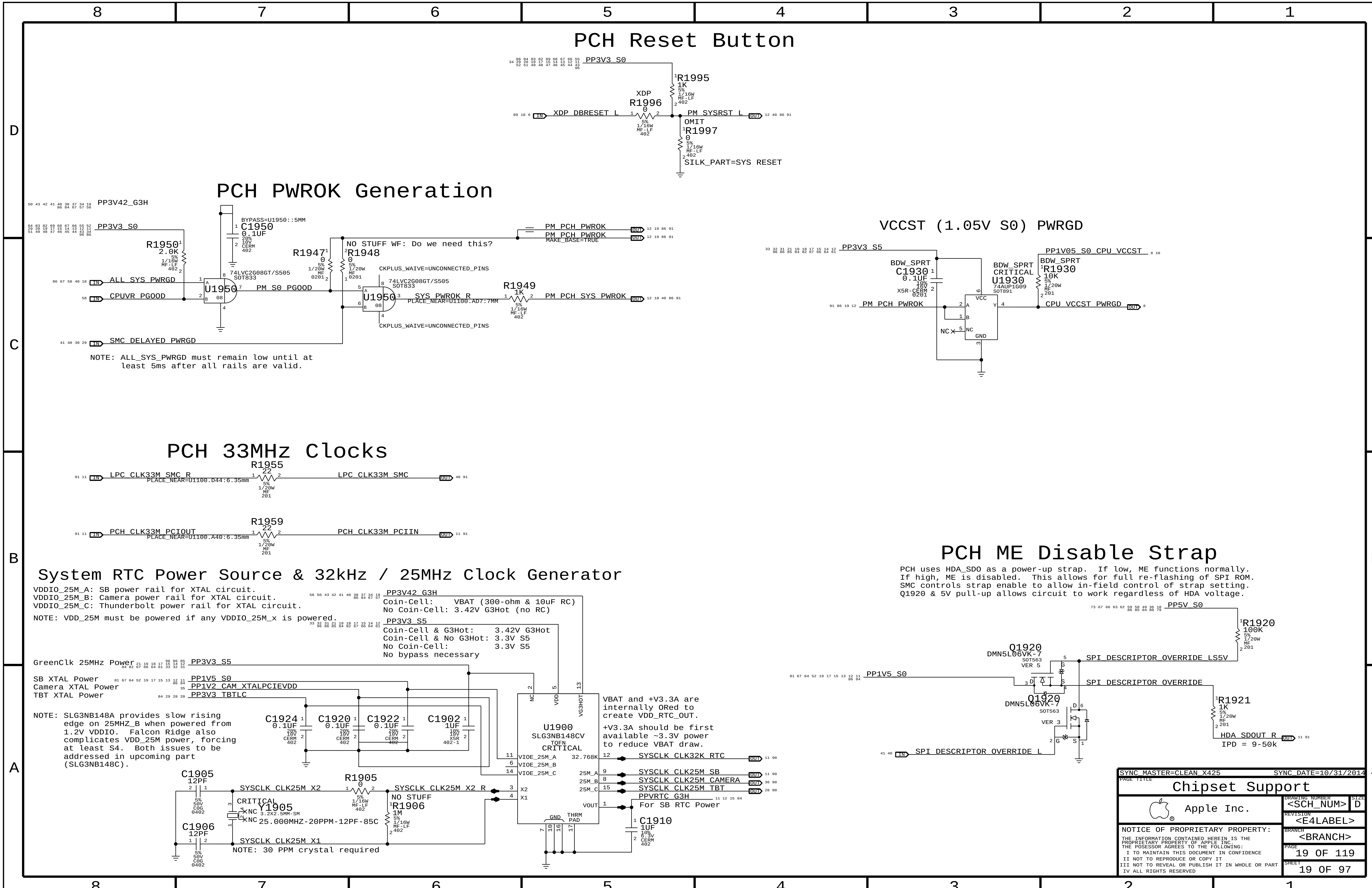

PCH Reset Button
PP3V3_S0
R1995
1K
5%
1/16W
MF-LF
402
XDP
R1996
0
XDP_DBRESET_L
PM_SYSRST_L
OMIT
R1997
0
SILK_PART=SYS RESET
PCH PWROK Generation
PP3V42_G3H
PP3V3_S0
BYPASS=U1950::5MM
C1950
0.1UF
20%
10V
CERM
402
R1950
2.0K
ALL_SYS_PWRGD
CPUVR_PGOOD
74LVC2G08GT/S505
SOT833
U1950
PM_S0_PGOOD
R1947
0
NO STUFF WF: Do we need this?
R1948
0
CKPLUS_WAIVE=UNCONNECTED_PINS
PM_PCH_PWROK
MAKE_BASE=TRUE
R1949
1K
SYS_PWROK_R
PLACE_NEAR=U1100.AD7:7MM
PM_PCH_SYS_PWROK
SMC_DELAYED_PWRGD
NOTE: ALL_SYS_PWRGD must remain low until at least 5ms after all rails are valid.
VCCST (1.05V S0) PWRGD
PP3V3_S5
BDW_SPRT
C1930
0.1UF
X5R-CERM
0201
PP1V05_S0_CPU_VCCST
BDW_SPRT
CRITICAL
U1930
74AUP1G09
SOT891
BDW_SPRT
R1930
10K
PM_PCH_PWROK
CPU_VCCST_PWRGD
VCC
GND
NC
PCH 33MHz Clocks
R1955
22
LPC_CLK33M_SMC_R
PLACE_NEAR=U1100.D44:6.35mm
LPC_CLK33M_SMC
R1959
22
PCH_CLK33M_PCIOUT
PLACE_NEAR=U1100.A40:6.35mm
PCH_CLK33M_PCIIN
System RTC Power Source & 32kHz / 25MHz Clock Generator
VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Camera power rail for XTAL circuit.
VDDIO_25M_C: Thunderbolt power rail for XTAL circuit.
NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.
PP3V42_G3H
Coin-Cell: VBAT (300-ohm & 10uF RC)
No Coin-Cell: 3.42V G3Hot (no RC)
PP3V3_S5
Coin-Cell & G3Hot: 3.42V G3Hot
Coin-Cell & No G3Hot: 3.3V S5
No Coin-Cell: 3.3V S5
No bypass necessary
GreenClk 25MHz Power PP3V3_S5
SB XTAL Power PP1V5_S0
Camera XTAL Power PP1V2_CAM_XTALPCIEVDD
TBT XTAL Power PP3V3_TBTLC
NOTE: SLG3NB148A provides slow rising edge on 25MHZ_B when powered from 1.2V VDDIO. Falcon Ridge also complicates VDD_25M power, forcing at least S4. Both issues to be addressed in upcoming part (SLG3NB148C).
C1924 0.1UF
C1920 0.1UF
C1922 0.1UF
C1902 1UF
U1900
SLG3NB148CV
TQFN
CRITICAL
VIOE_25M_A
VIOE_25M_B
VIOE_25M_C
32.768K
25M_A
25M_B
25M_C
VOUT
VBAT and +V3.3A are internally ORed to create VDD_RTC_OUT.
+V3.3A should be first available ~3.3V power to reduce VBAT draw.
SYSCLK_CLK32K_RTC
SYSCLK_CLK25M_SB
SYSCLK_CLK25M_CAMERA
SYSCLK_CLK25M_TBT
PPVRTC_G3H
For SB RTC Power
C1910
1UF
C1905
12PF
SYSCLK_CLK25M_X2
R1905
0
SYSCLK_CLK25M_X2_R
NO STUFF
R1906
1M
CRITICAL
Y1905
3.2X2.5MM-SM
25.000MHZ-20PPM-12PF-85C
C1906
12PF
SYSCLK_CLK25M_X1
NOTE: 30 PPM crystal required
PCH ME Disable Strap
PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.
Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.
PP5V_S0
R1920
100K
Q1920
DMN5L06VK-7
SOT563
SPI_DESCRIPTOR_OVERRIDE_LS5V
PP1V5_S0
SPI_DESCRIPTOR_OVERRIDE
R1921
1K
HDA_SDOUT_R
IPD = 9-50k
SPI_DESCRIPTOR_OVERRIDE_L
SYNC_MASTER=CLEAN_X425
SYNC_DATE=10/31/2014
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Chipset Support
Apple Inc.
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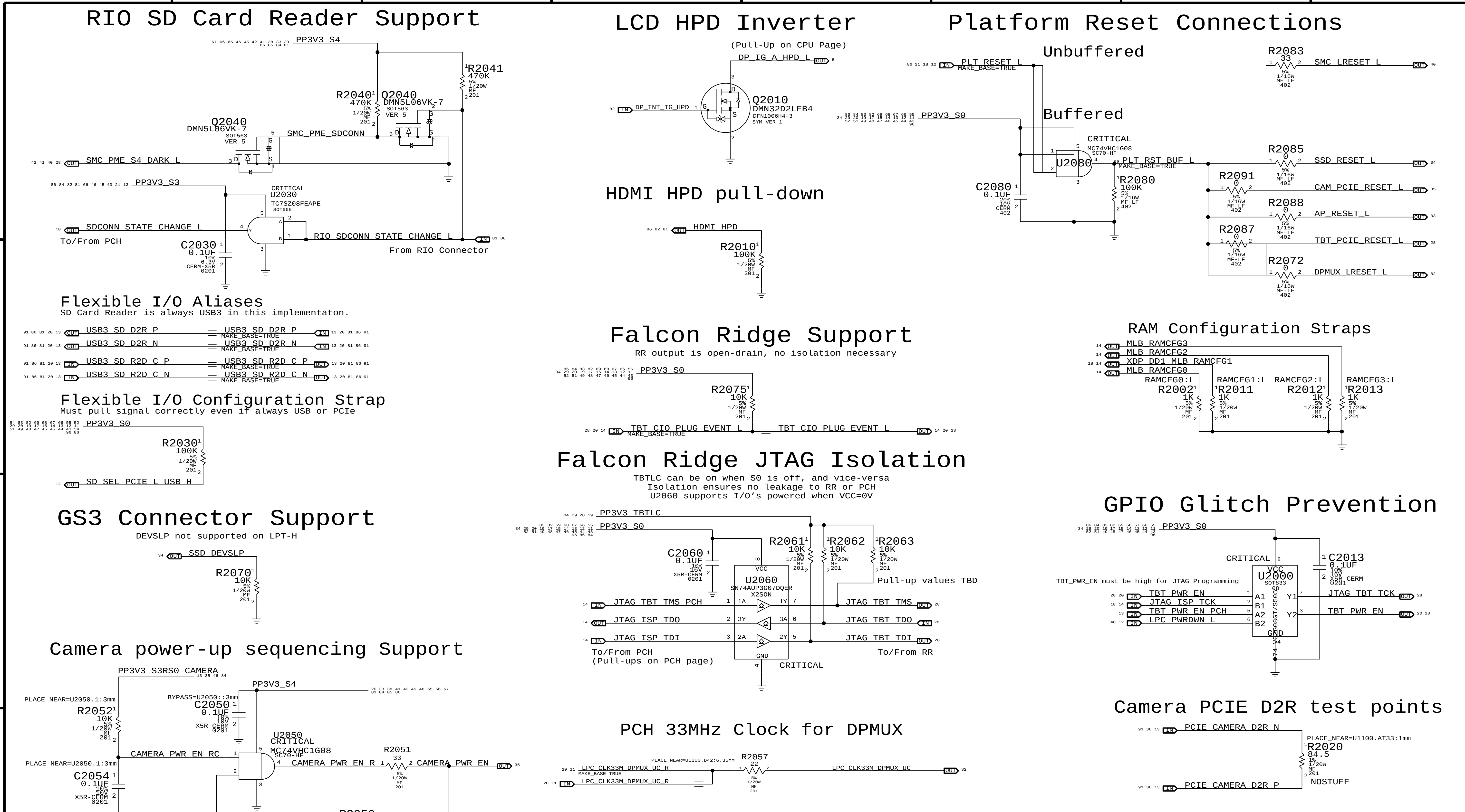

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Project Chipset Support

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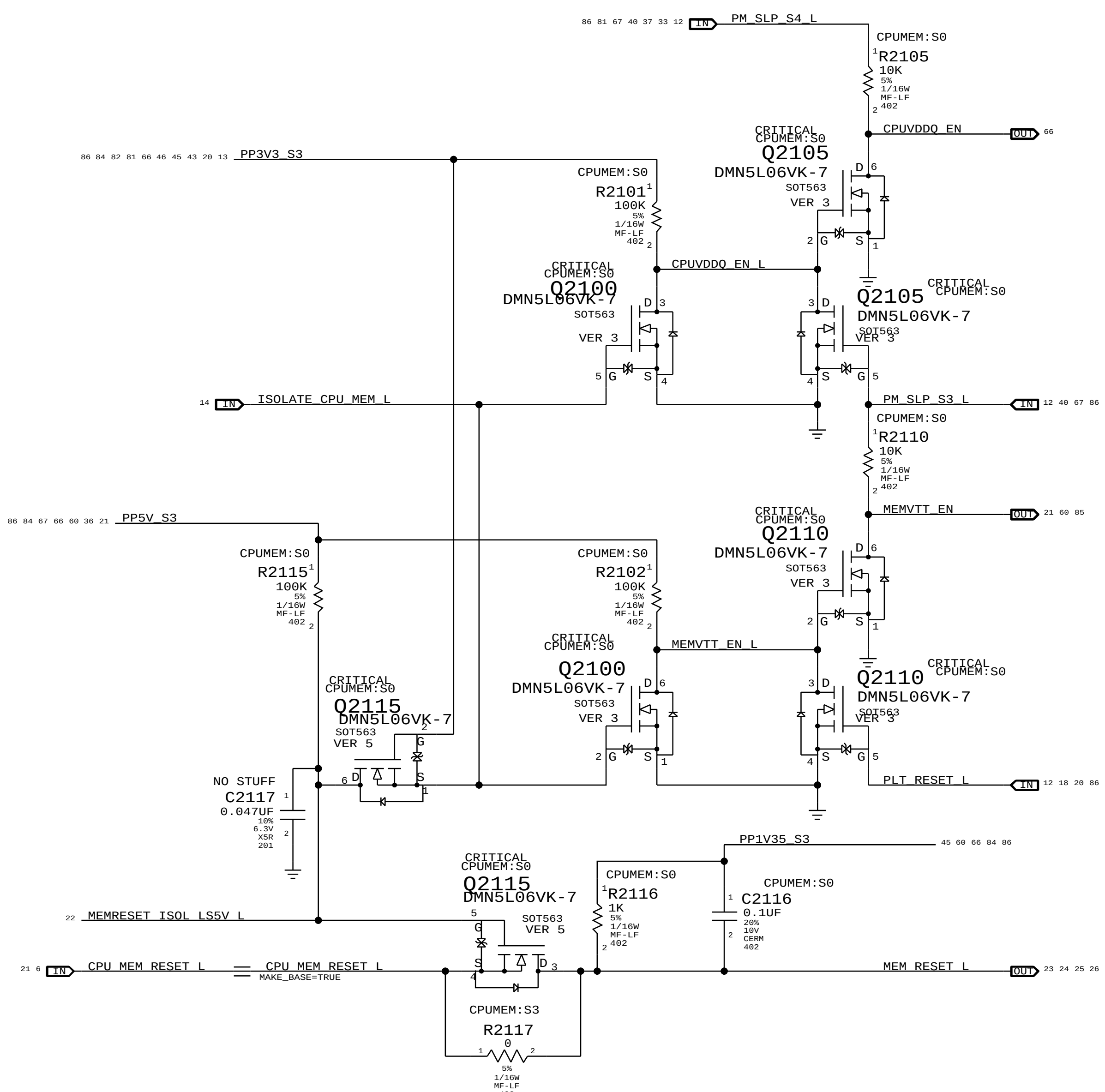
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well
as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

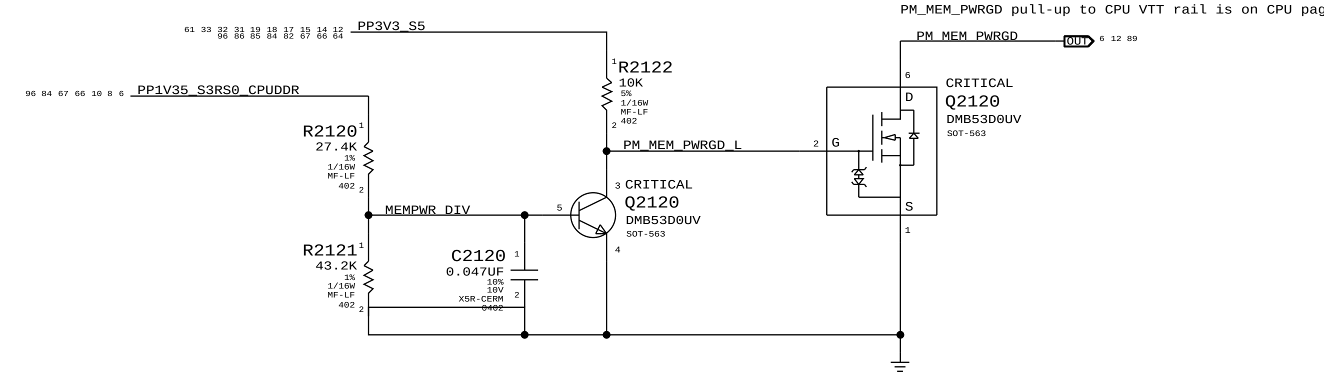
ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

CPUVDDQ_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN   = (ISOLATE_CPU_MEM_L + PLT_RST_L)   * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



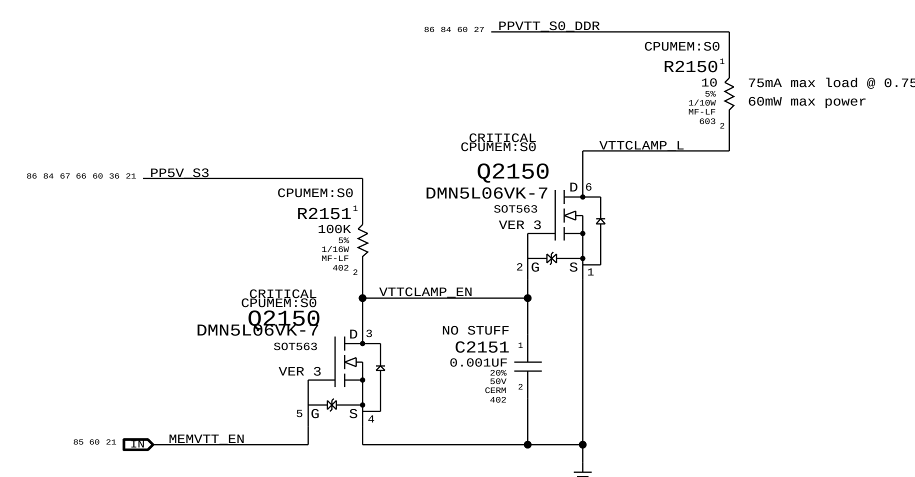
MEM S0 "PGOOD" for CPU

PM_MEM_PWRGD pull-up to CPU VTT rail is on CPU page



MEMVTT Clamp

Ensures CKE signals are held low in S3



	Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	CPUVDDQ_EN
S0	0	1	1	1	1	1	CPU_MEM_RESET_L	1	1
to	1	0	1	1	1	1	1	1	1
	2	0	0	1	1	1	1	0	1
S3	3	0	0	0	1	X	1	0	0
	4	0	0	1	1	X	1	0	1
to	5	0	1	1	1	0 (*)	1	1	1
	6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=CLEAN_MAXWELL SYNC_DATE=07/02/2014

PAGE TITLE: CPU Memory S3 Support

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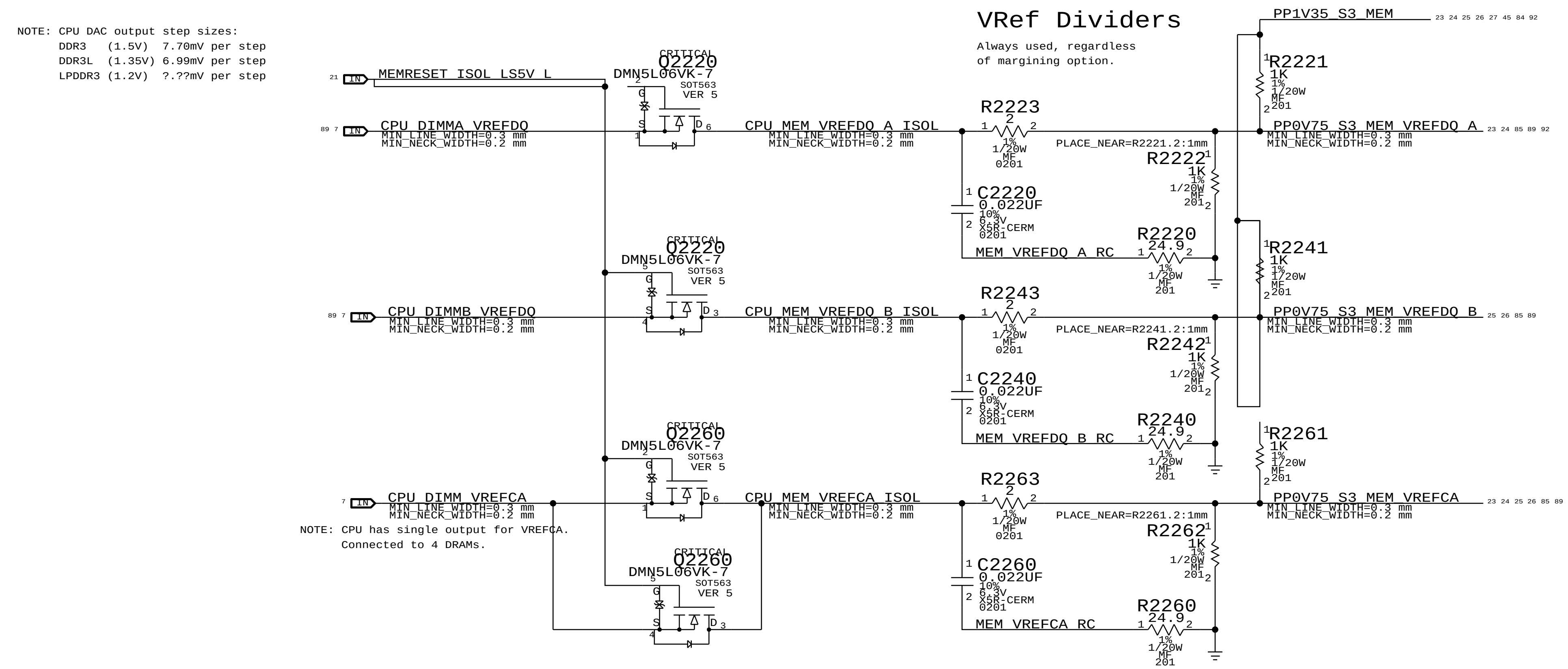
CPU-Based Margining



SYNC_MASTER=CLEAN_X425 SYNC_DATE=08/11/2014

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DDR3 VREF MARGINING

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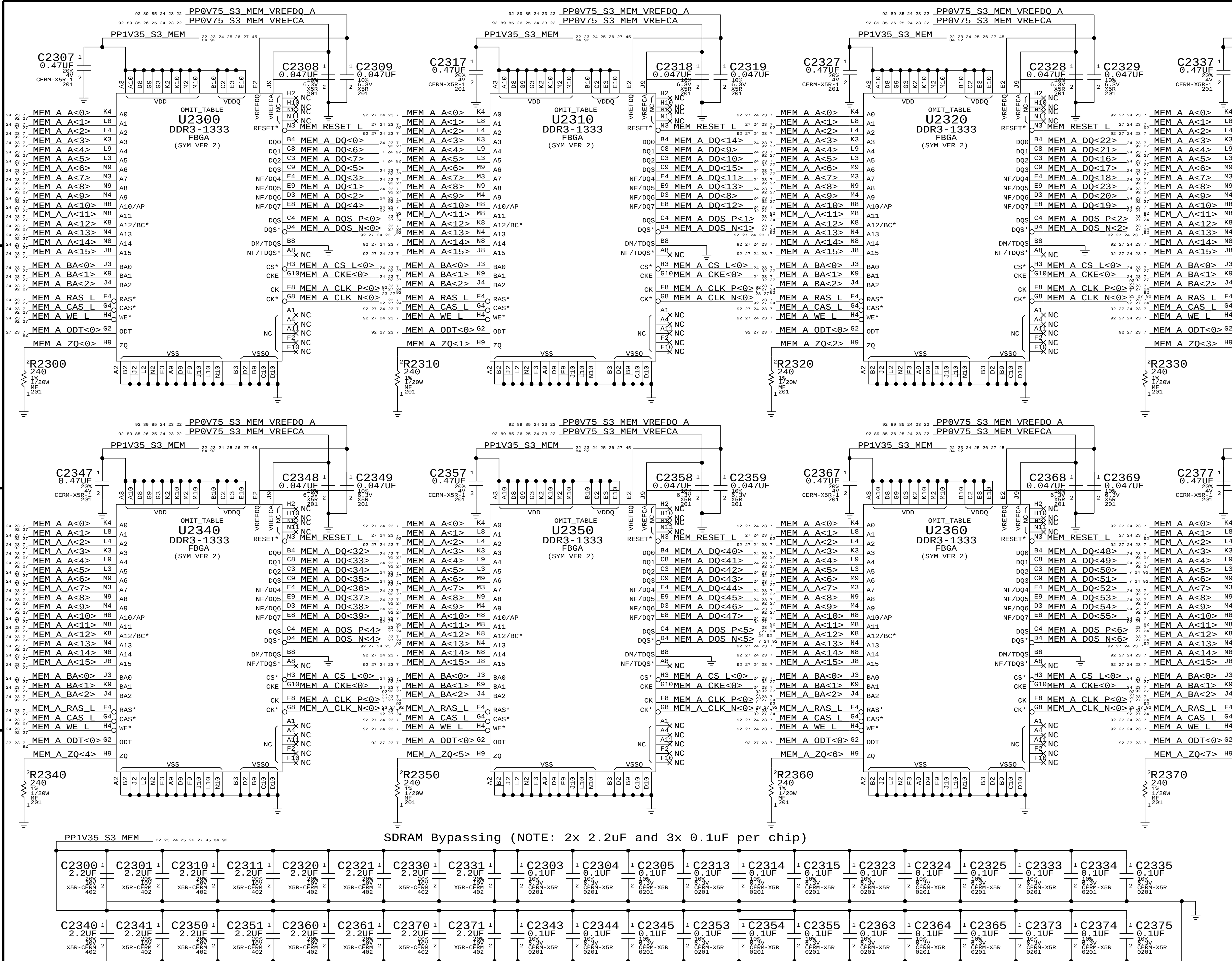

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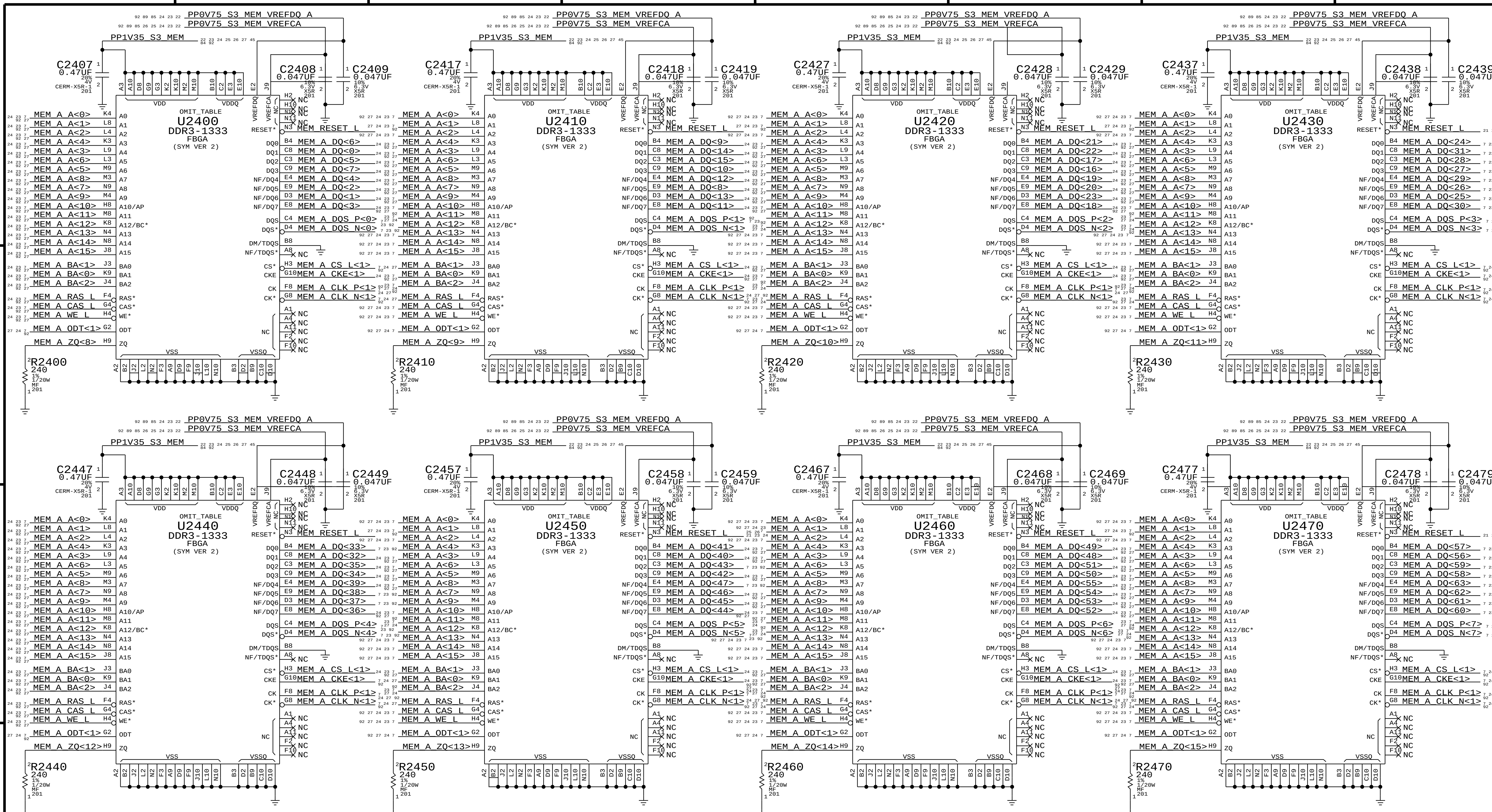
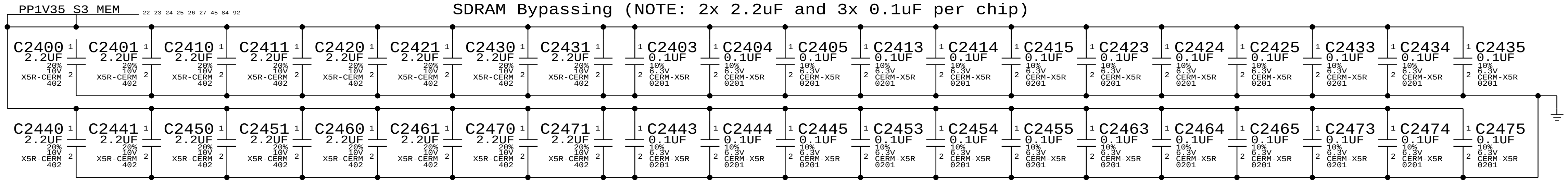
8	7	6	5	4	3	2	1



SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)



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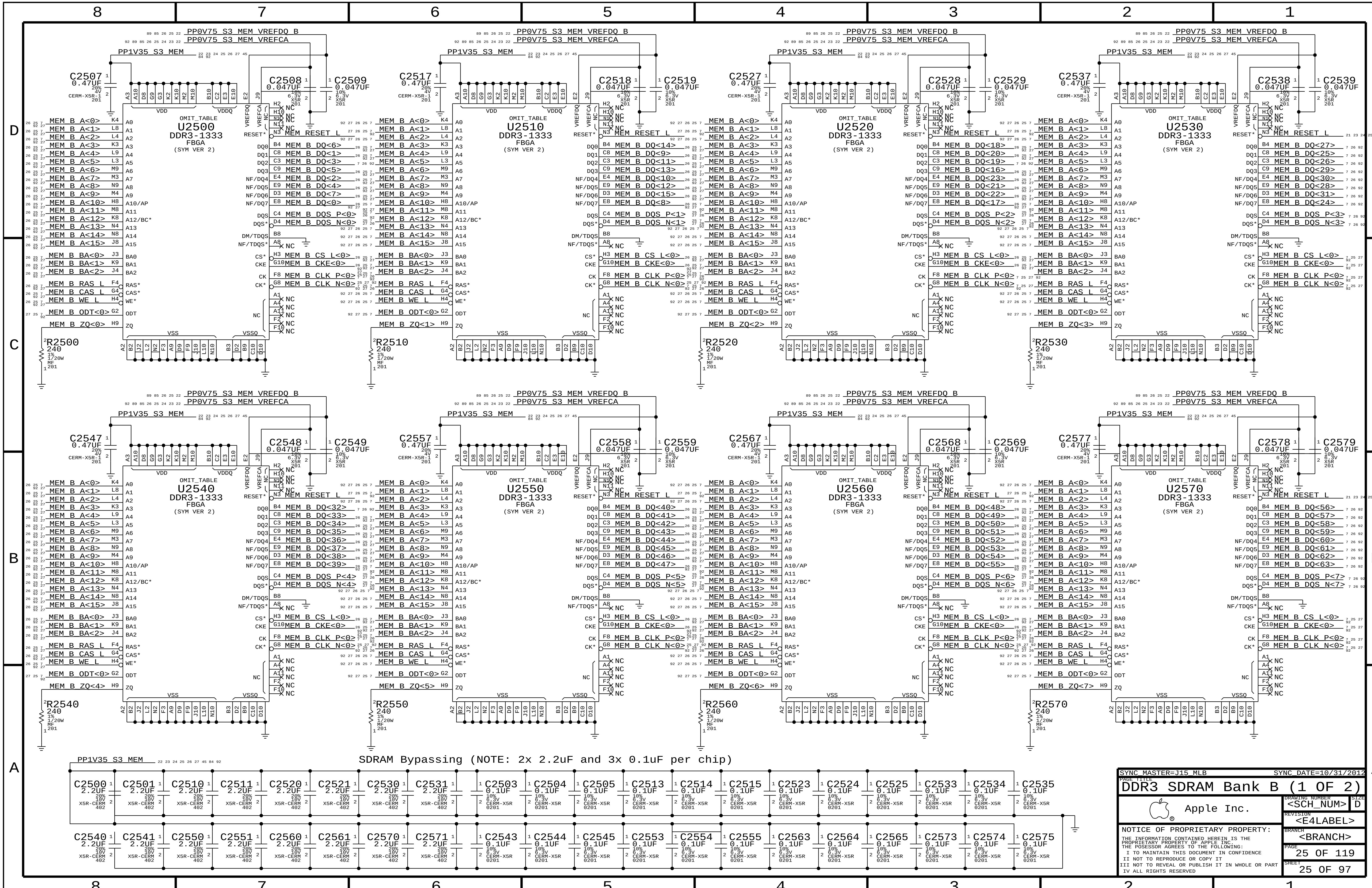
8	7	6	5	4	3	2	1



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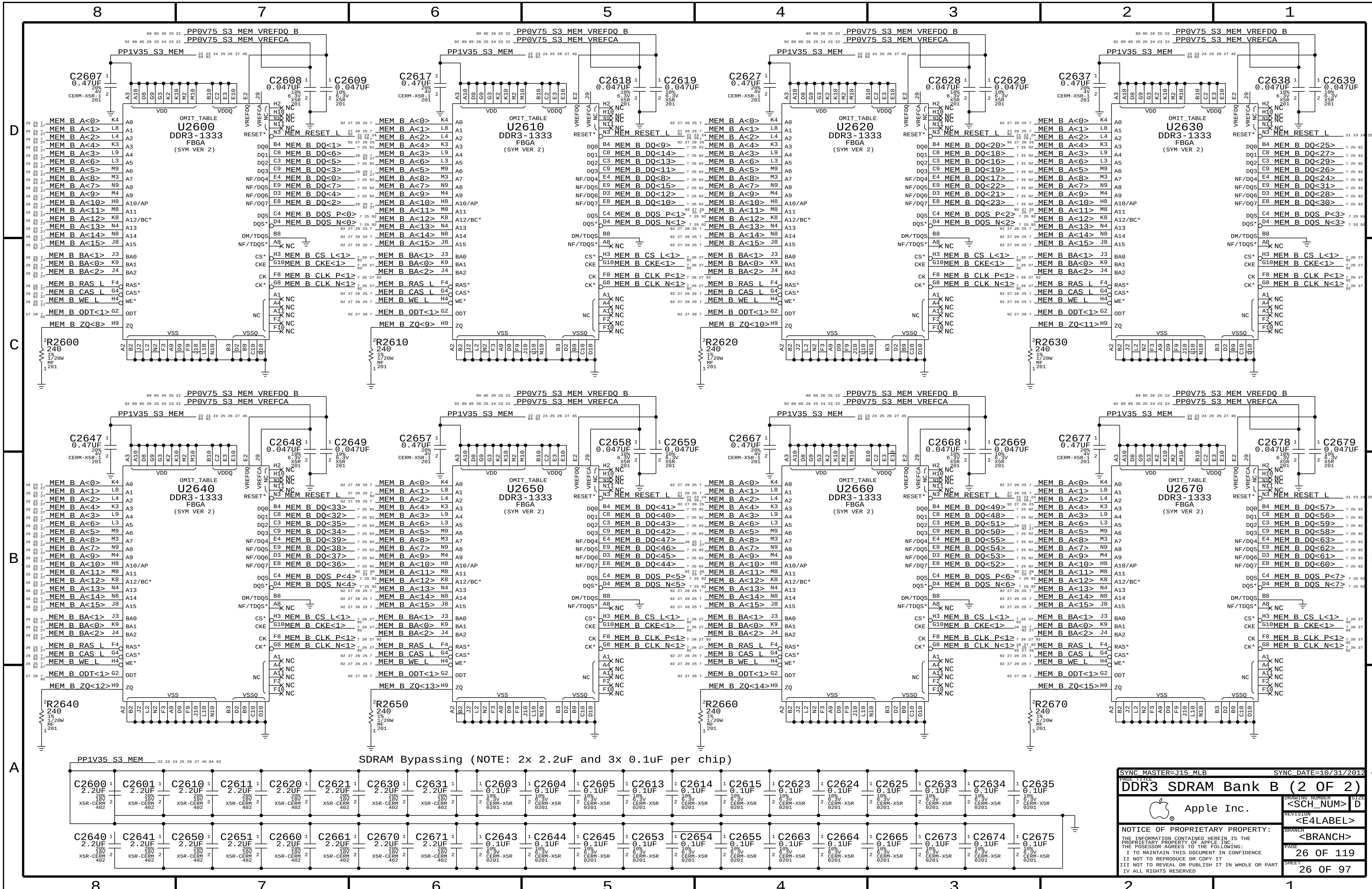

PP0V75_S3_MEM_VREFDQ_B
PP0V75_S3_MEM_VREFCA
PP1V35_S3_MEM
OMIT_TABLE
U2600
DDR3-1333
FBGA
(SYM VER 2)
U2610
U2620
U2630
U2640
U2650
U2660
U2670
R2600
R2610
R2620
R2630
R2640
R2650
R2660
R2670
240
1%
1/20W
MF
201
C2607
C2617
C2627
C2637
C2647
C2657
C2667
C2677
0.47UF
C2608
C2609
C2618
C2619
C2628
C2629
C2638
C2639
C2648
C2649
C2658
C2659
C2668
C2669
C2678
C2679
0.047UF
MEM_B_RESET_L
MEM_B_CS_L<1>
MEM_B_CKE<1>
MEM_B_CLK_P<1>
MEM_B_CLK_N<1>
MEM_B_RAS_L
MEM_B_CAS_L
MEM_B_WE_L
MEM_B_ODT<1>
VDD
VDDQ
VSS
VSSQ
SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)
C2600
C2601
C2610
C2611
C2620
C2621
C2630
C2631
C2640
C2641
C2650
C2651
C2660
C2661
C2670
C2671
2.2UF
C2603
C2604
C2605
C2613
C2614
C2615
C2623
C2624
C2625
C2633
C2634
C2635
C2643
C2644
C2645
C2653
C2654
C2655
C2663
C2664
C2665
C2673
C2674
C2675
0.1UF
SYNC_MASTER=J15_MLB
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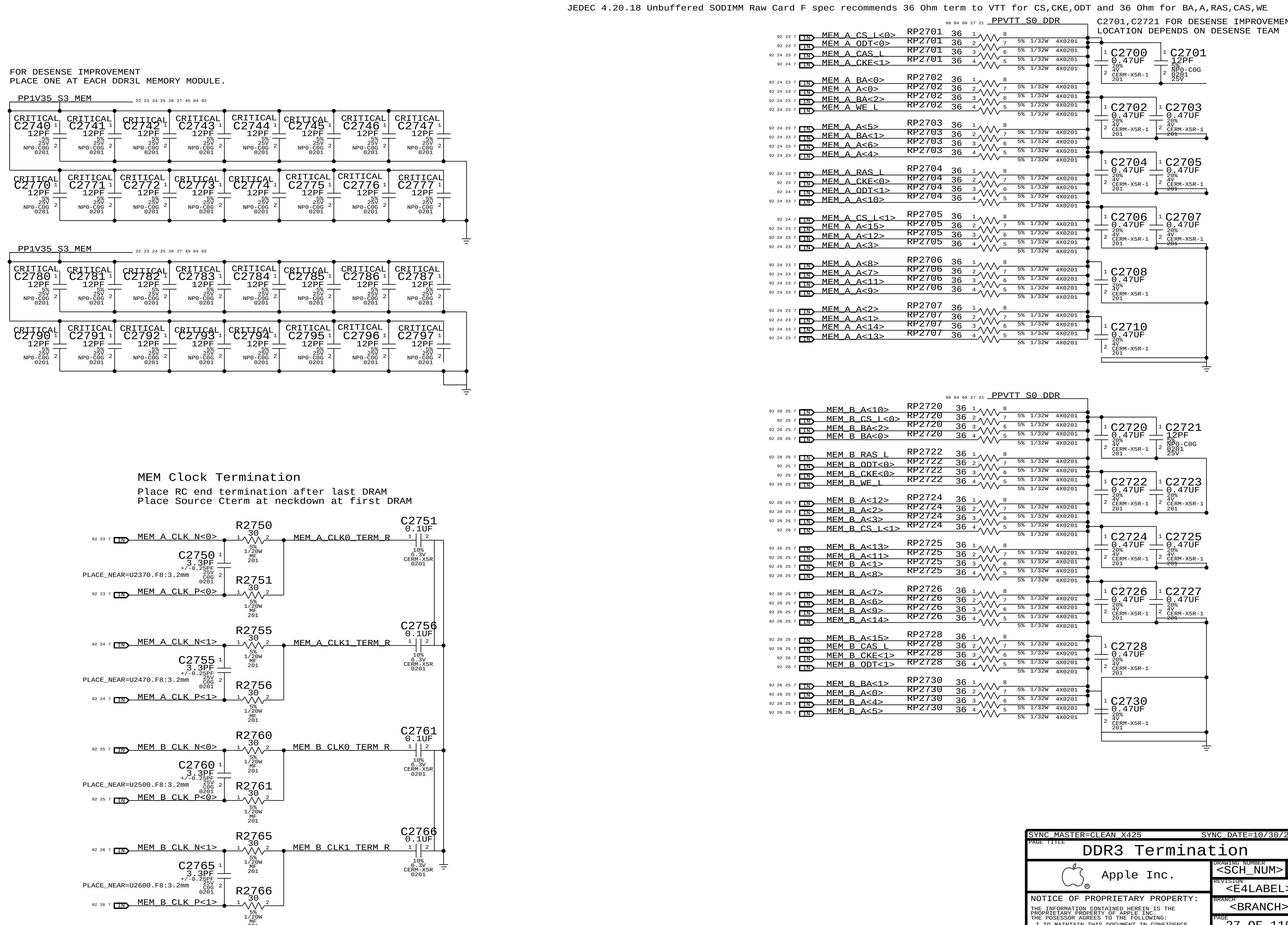

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DDR3 Termination

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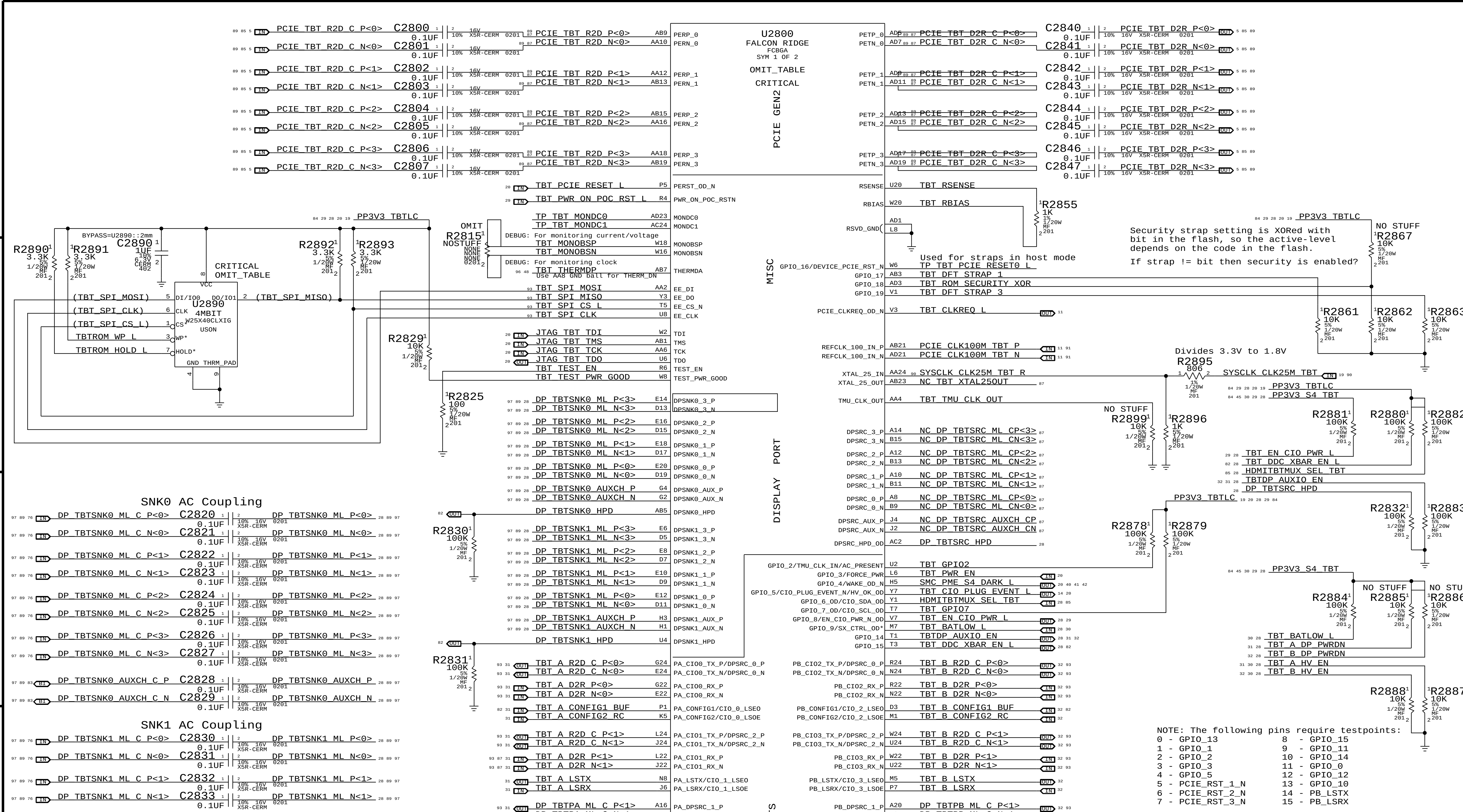

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Thunderbolt Host (1 of 2)

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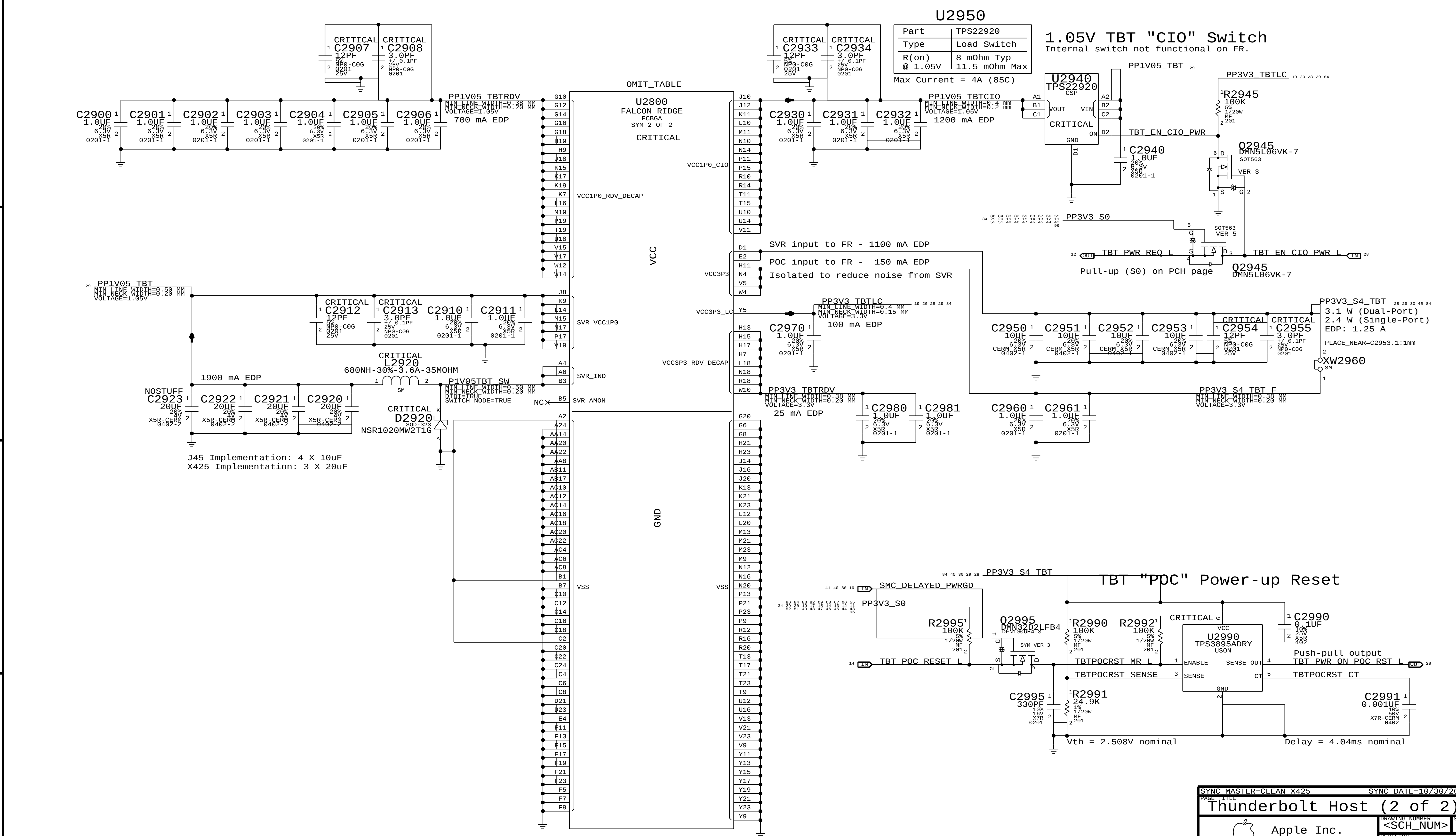

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE

Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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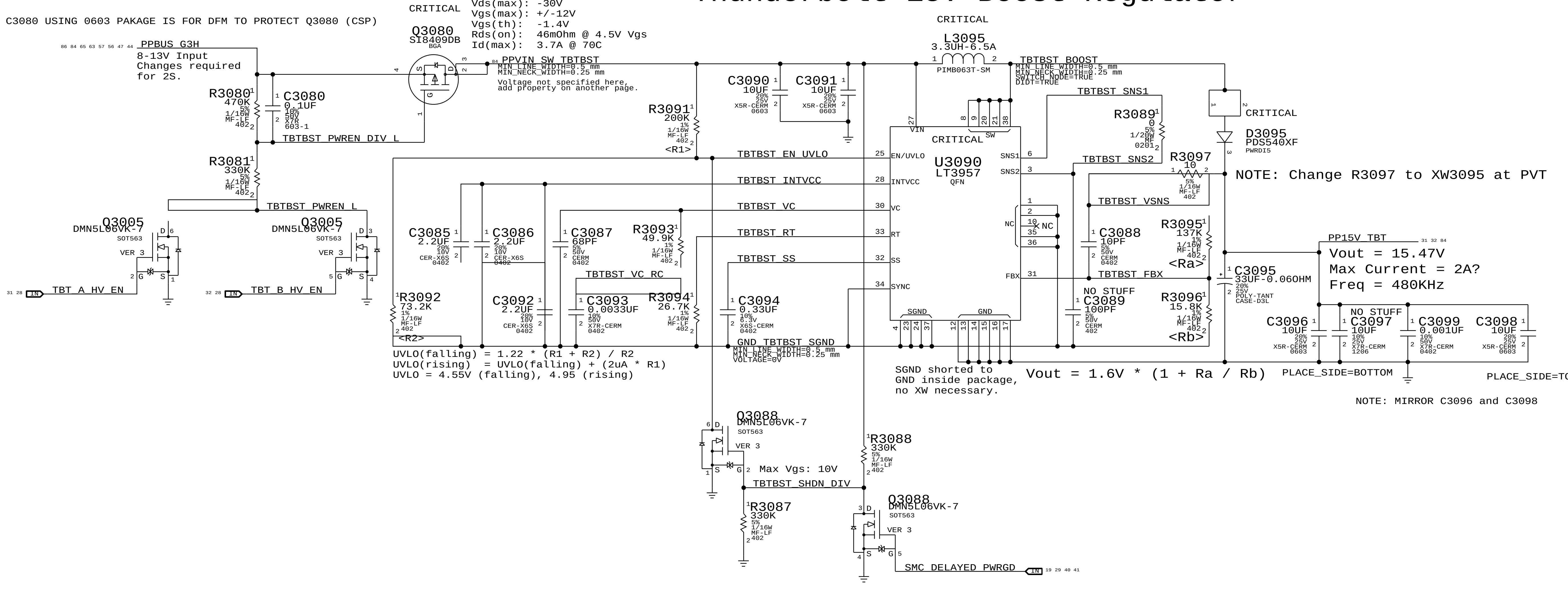
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)

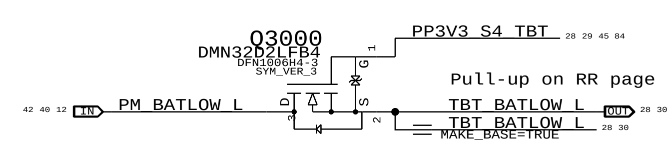
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Thunderbolt 15V Boost Regulator



BATLOW# Isolation



SYNC_MASTER=CLEAN_X305 SYNC_DATE=06/24/2014

PAGE TITLE: Thunderbolt Mobile Support

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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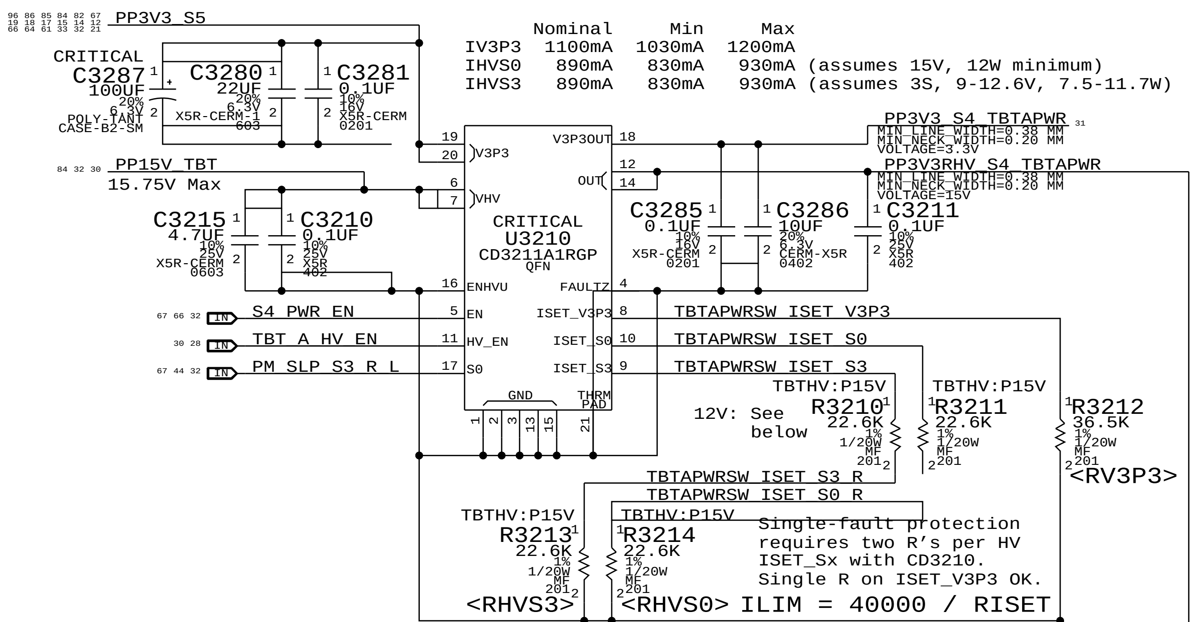
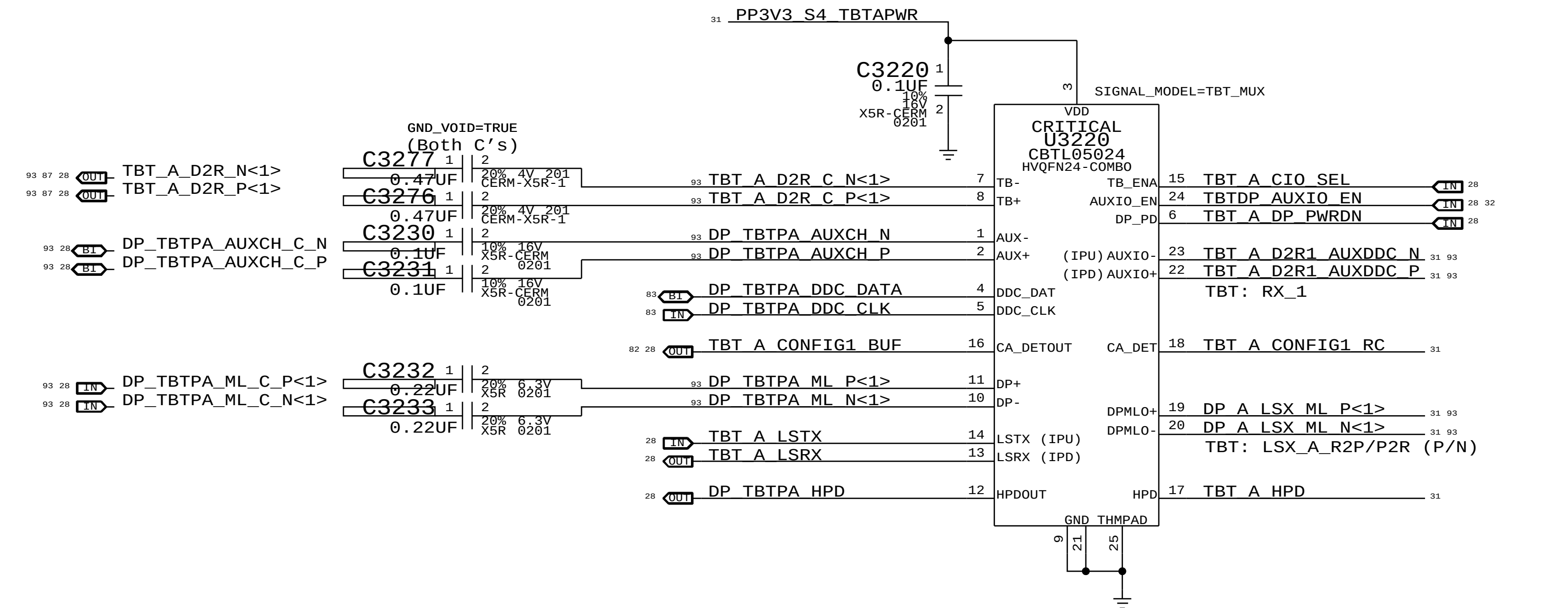
3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)



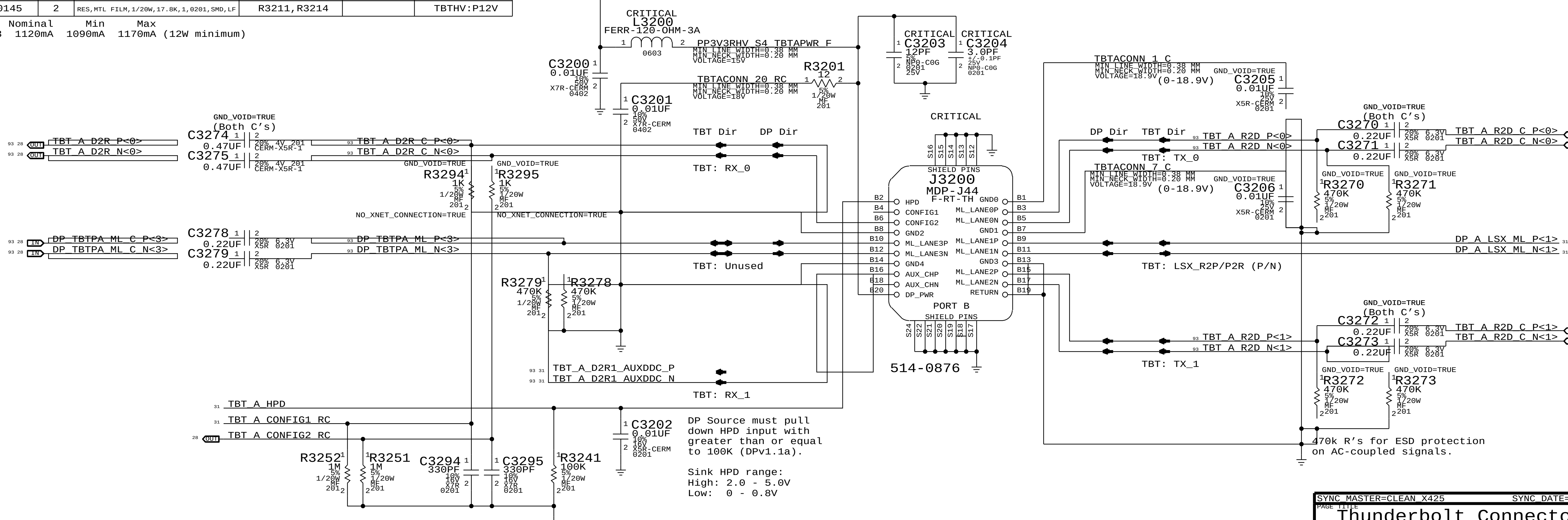


For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3210,R3213		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3211,R3214		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

Thunderbolt Connector A



DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

470k R's for ESD protection on AC-coupled signals.

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

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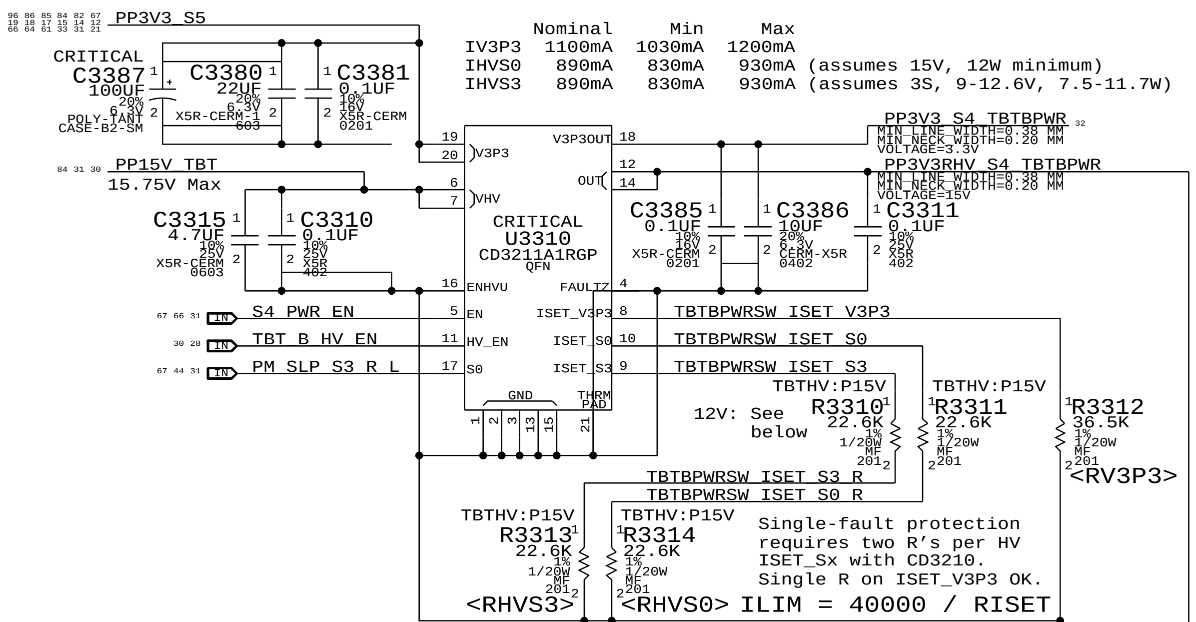
3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)



Single-fault protection requires two R's per HV ISET_Sx with CD3210. Single R on ISET_V3P3 OK.

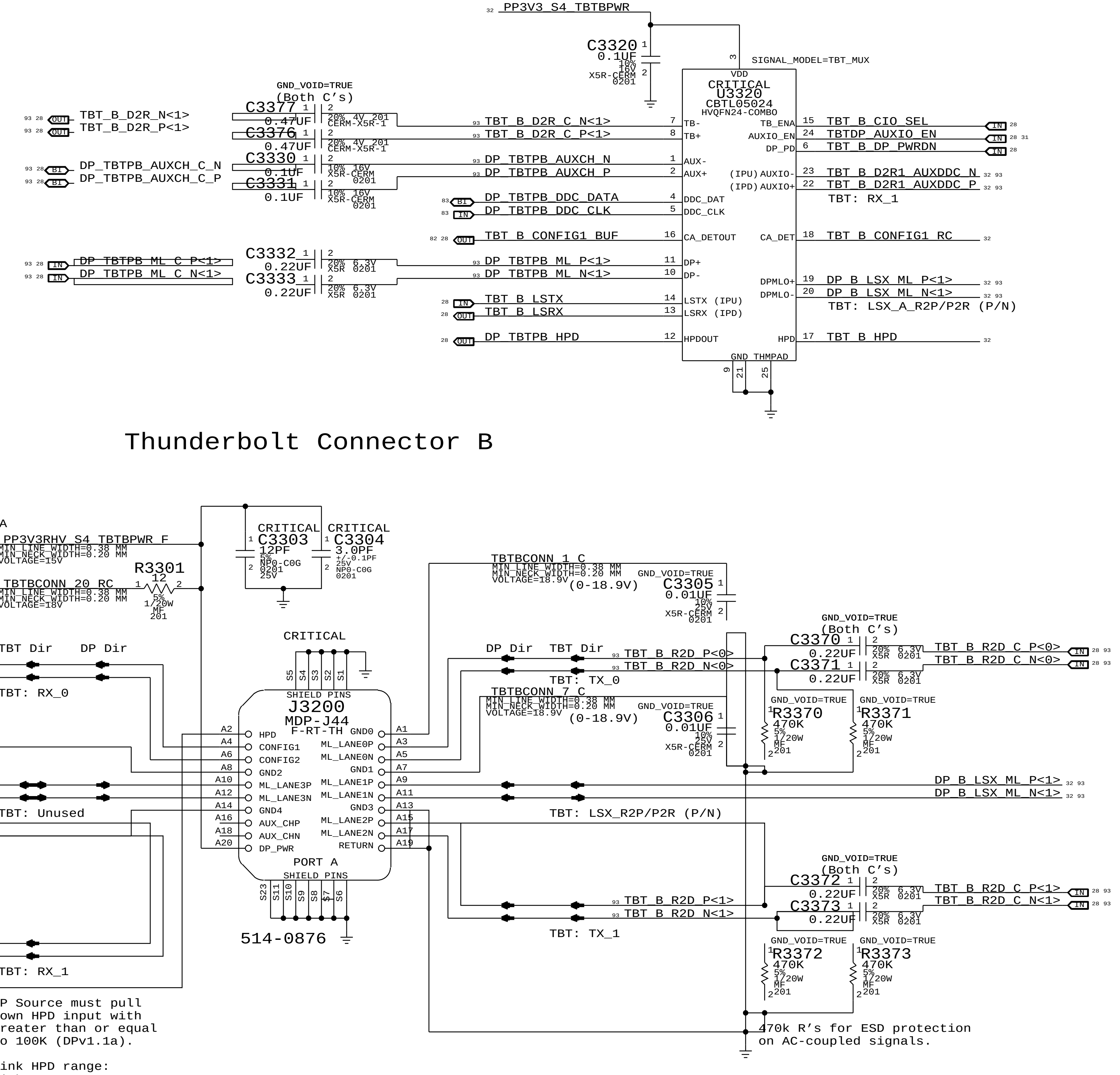
ILIM = 40000 / RISET

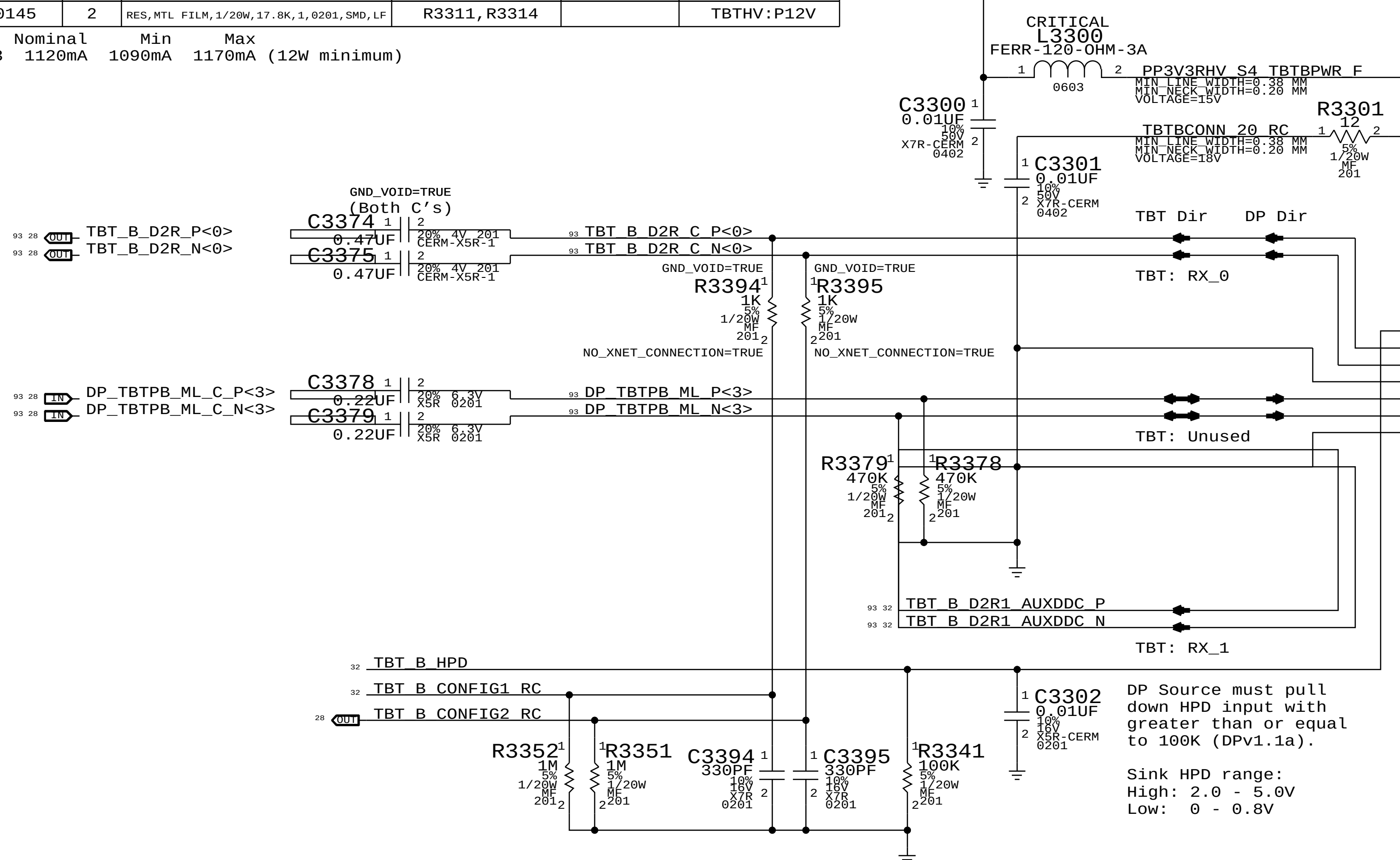
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3310,R3313		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3311,R3314		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

Thunderbolt Connector B





DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

470k R's for ESD protection on AC-coupled signals.

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

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REVISION: <E4LABEL>

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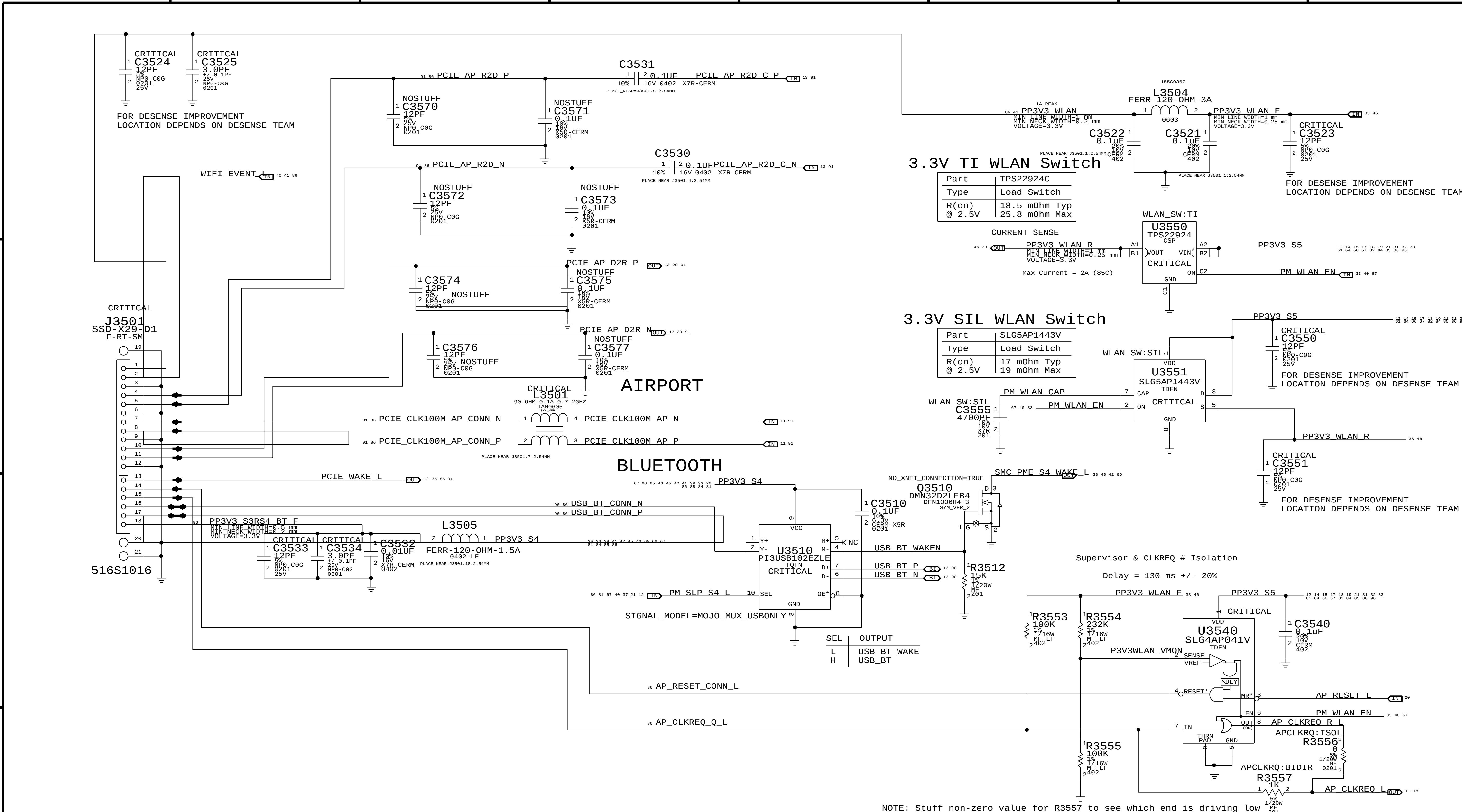

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE: X87 CONNECTOR

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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PAGE: 35 OF 119

SHEET: 33 OF 97

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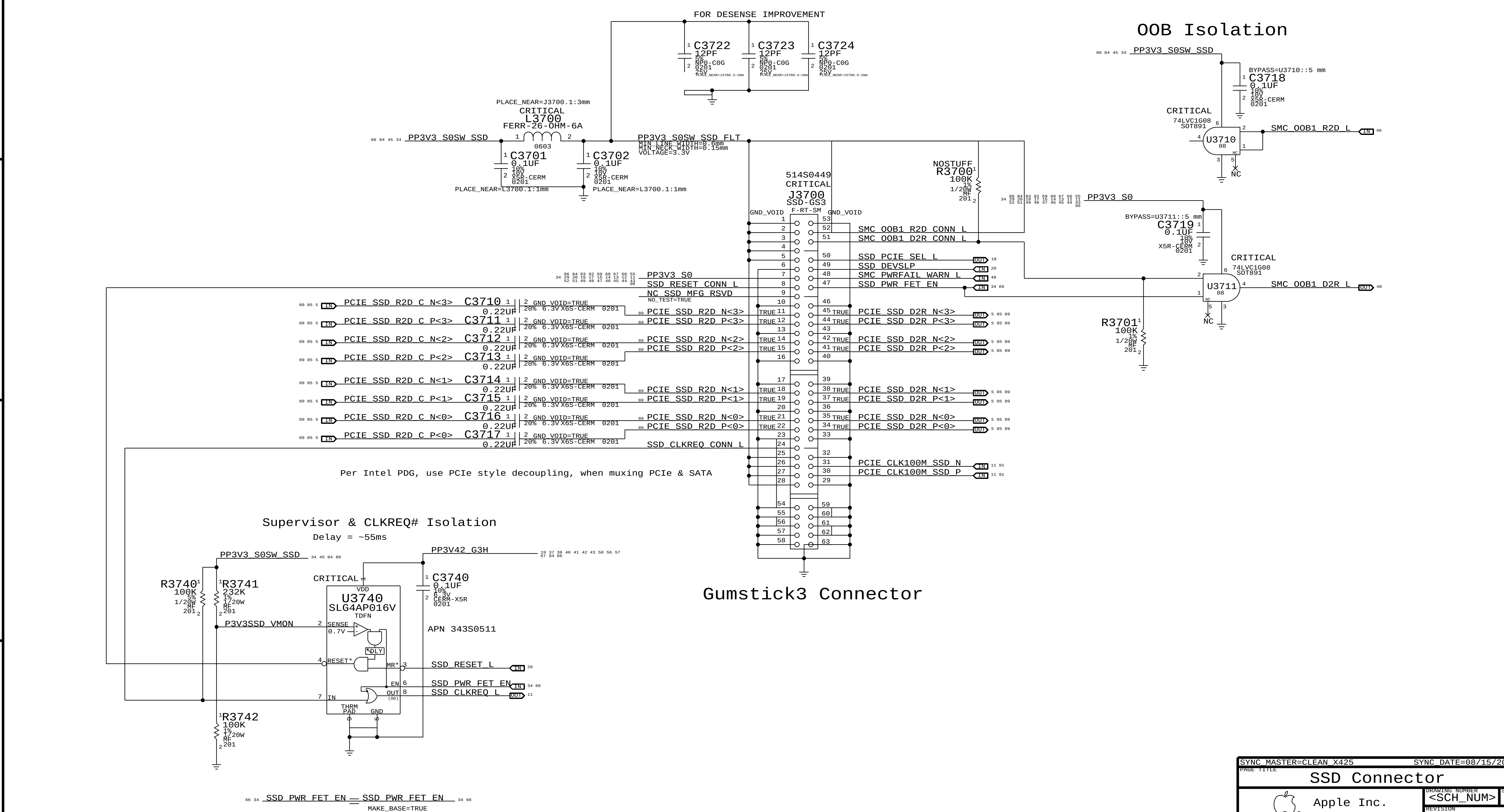

SYNC_MASTER=CLEAN_X425 SYNC_DATE=08/15/2014

PAGE TITLE

SSD Connector

Apple Inc.

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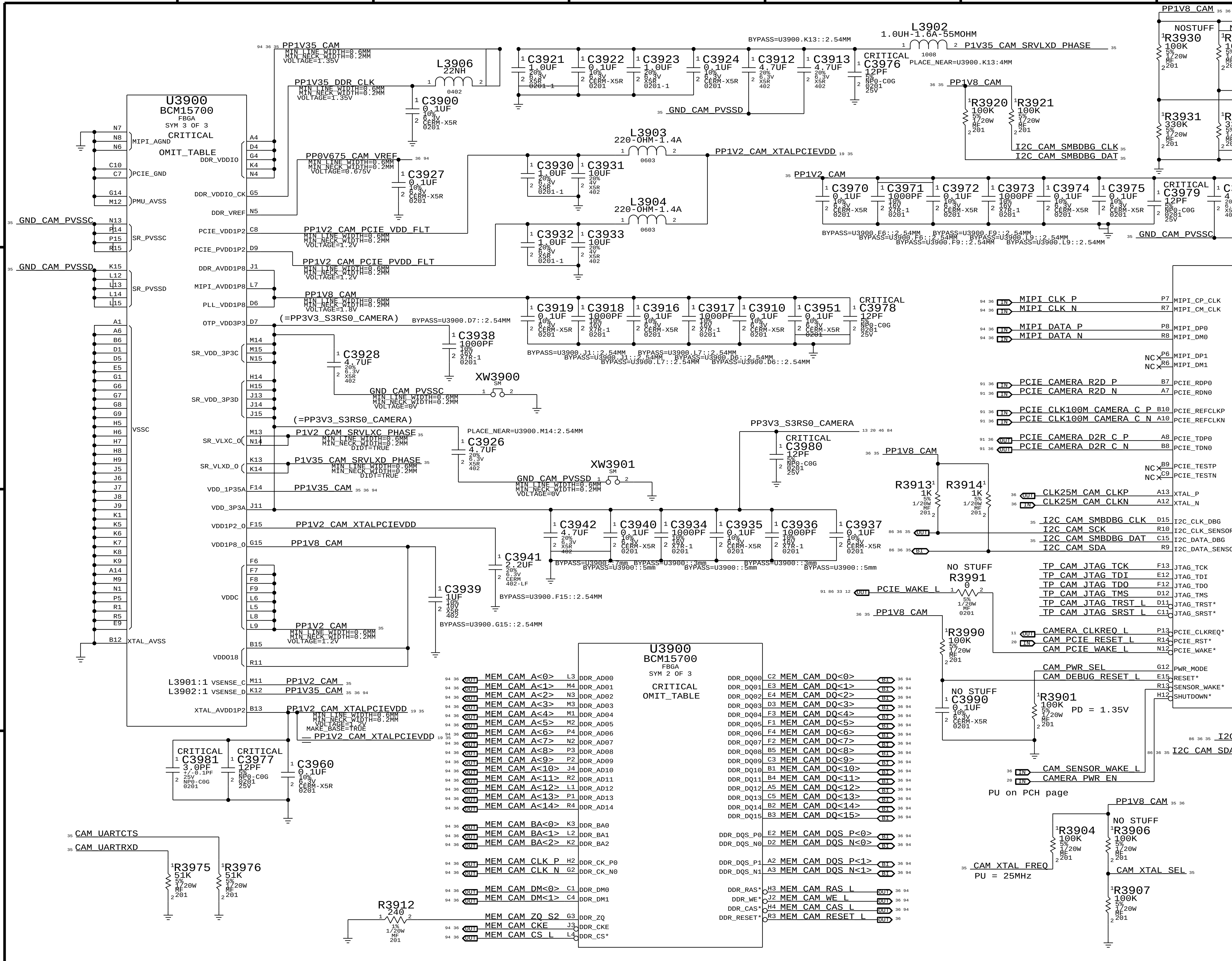

SYNC_MASTER=CLEAN_X425

SYNC_DATE=10/30/2014

PAGE TITLE

Camera 1 of 2

Apple Inc.

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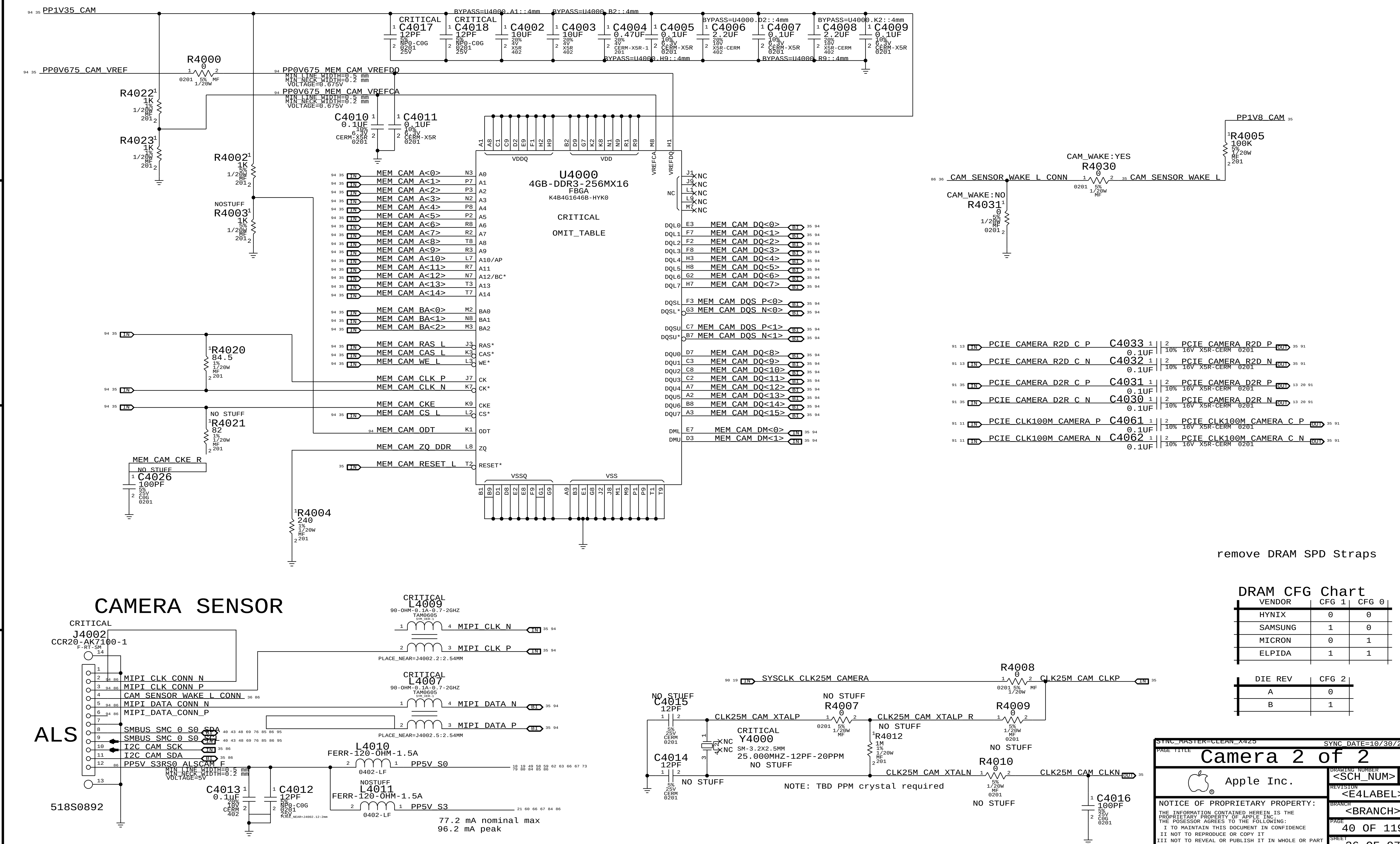

DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

DIE REV	CFG 2
A	0
B	1

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE

Camera 2 of 2

Apple Inc.

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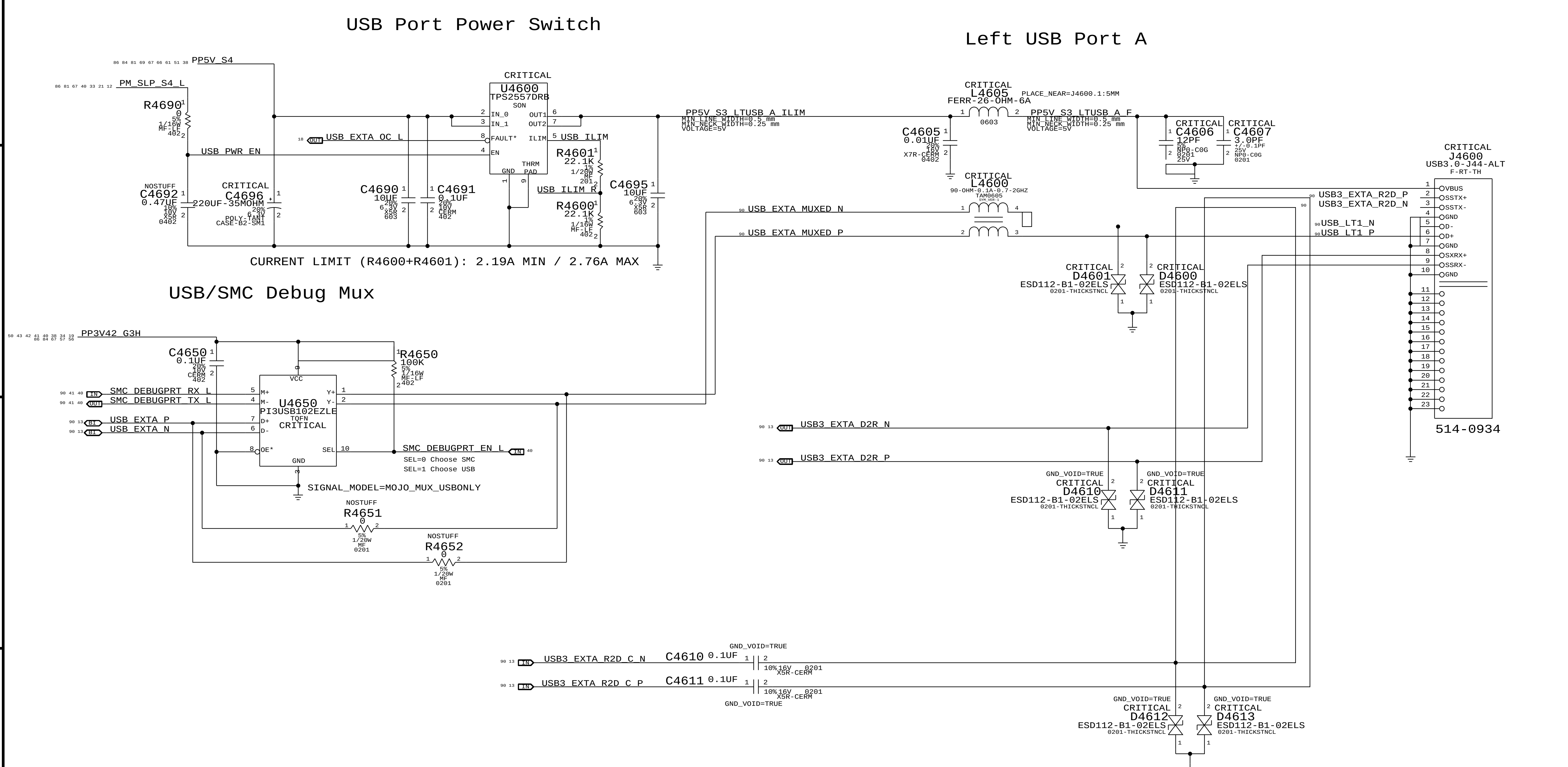

SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE

USB 3.0 CONNECTORS

Apple Inc.

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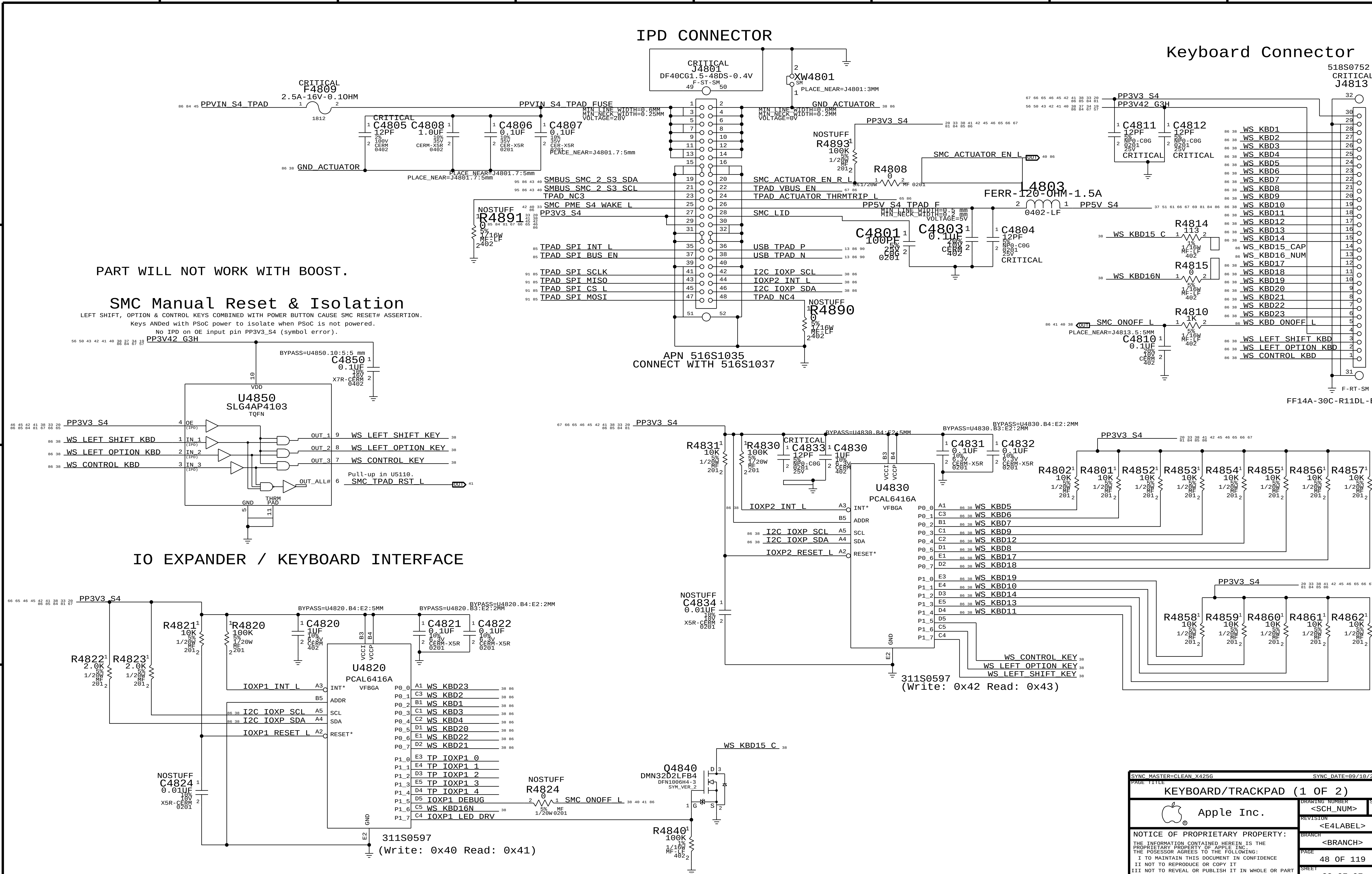

SYNC_MASTER=CLEAN_X425G SYNC_DATE=09/10/2014

PAGE TITLE: KEYBOARD/TRACKPAD (1 OF 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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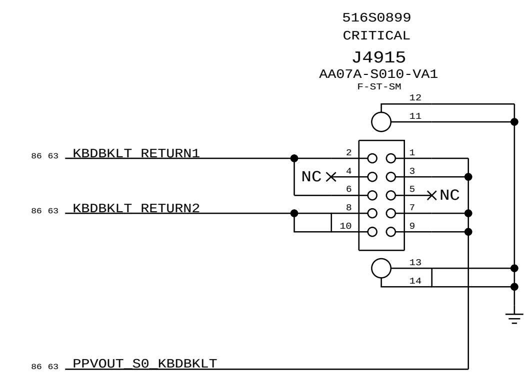
Keyboard Backlight Connector



SYNC_MASTER=CLEAN_MAXWELL SYNC_DATE=07/02/2014

PAGE TITLE
KEYBOARD/TRACKPAD (2 OF 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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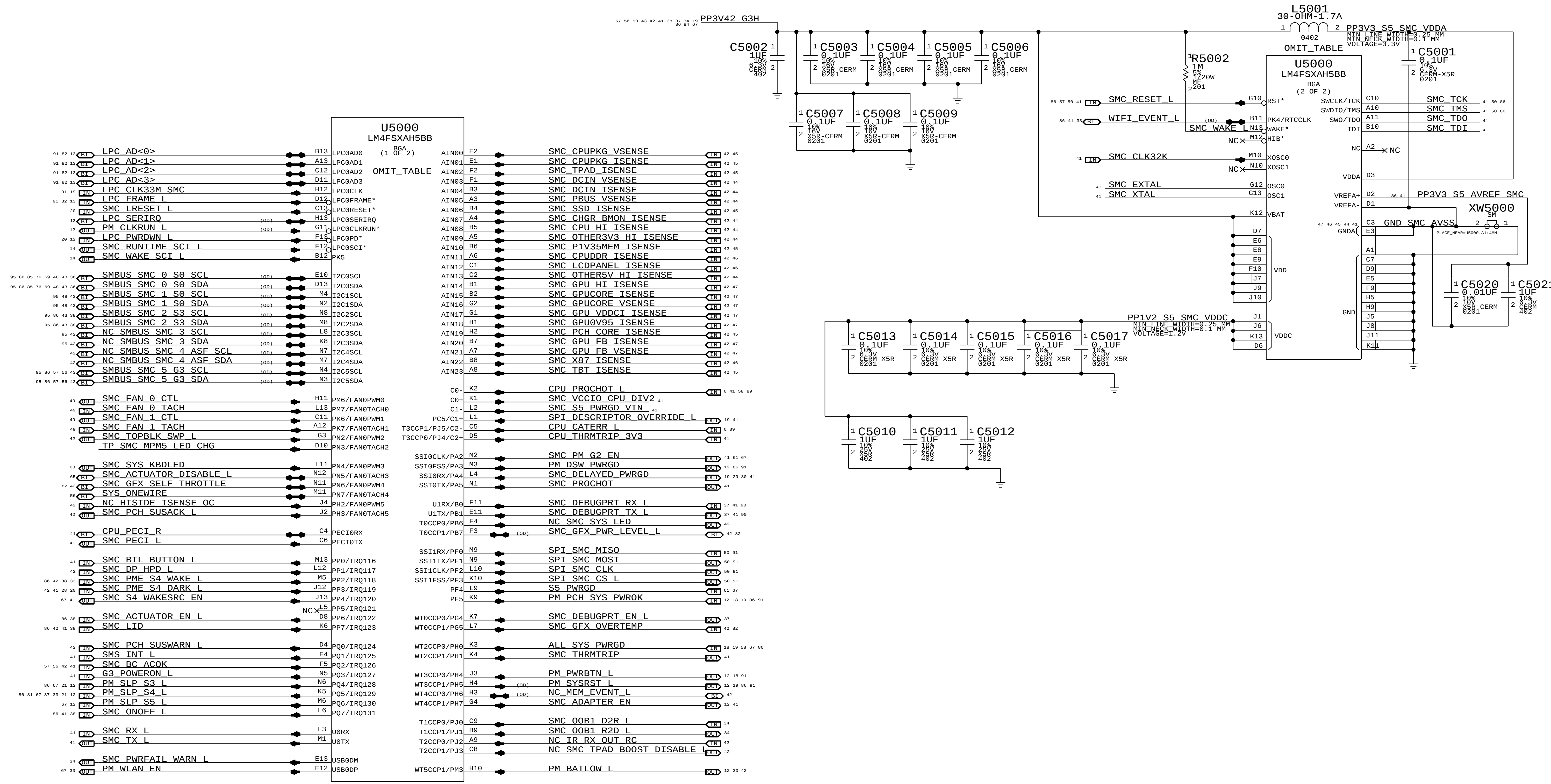
NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.



NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

SYNC_MASTER=CLEAN_X305 SYNC_DATE=01/15/2014
PAGE TITLE
SMC
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 50 OF 119
SHEET 40 OF 97
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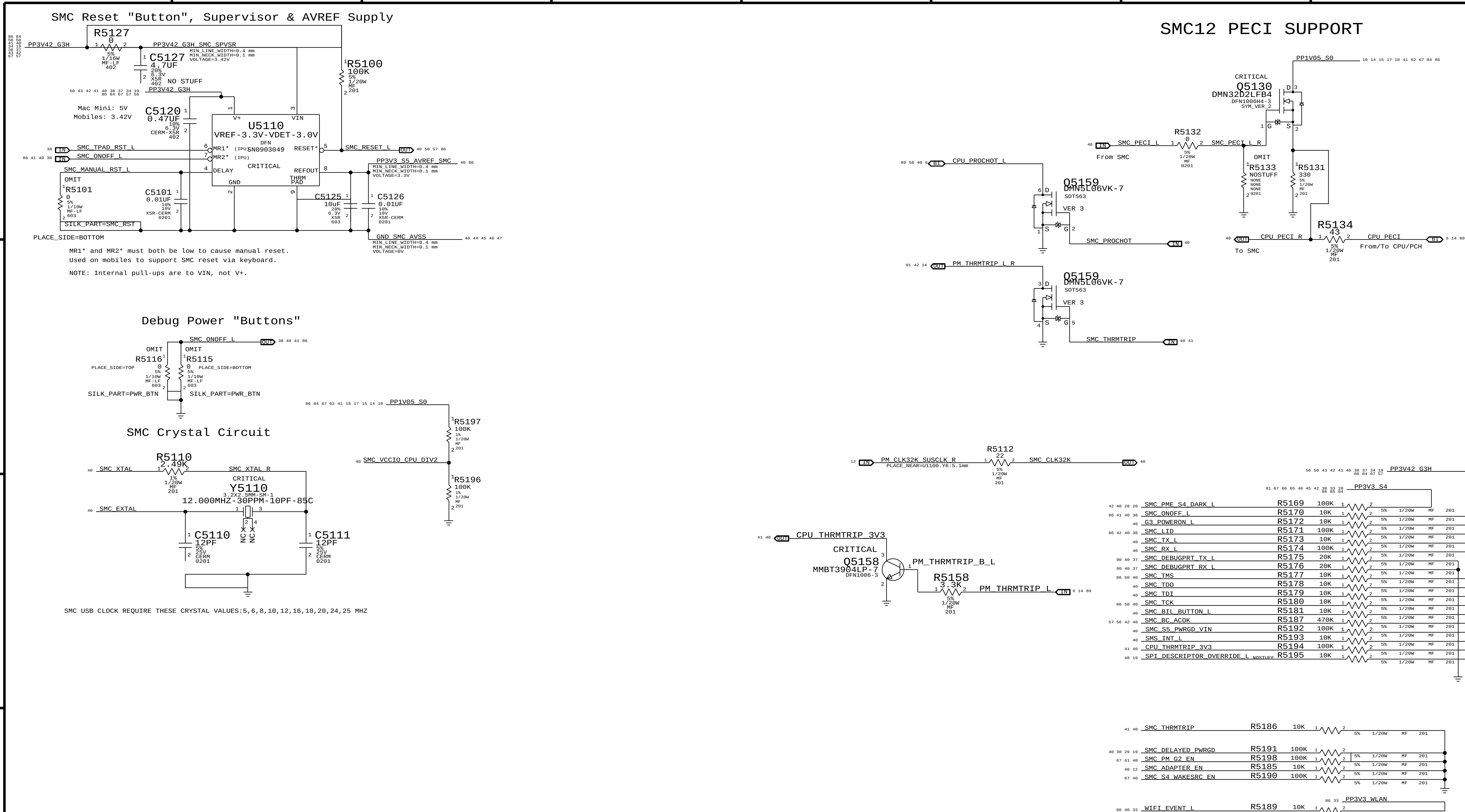

SYNC_MASTER=CLEAN_X305 SYNC_DATE=06/24/2014

PAGE TITLE

SMC Shared Support

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6811	1	SUBASSY,PCBA HALL EFFECT,K99	J5250	CRITICAL	

PAGE TITLE
SMC Project Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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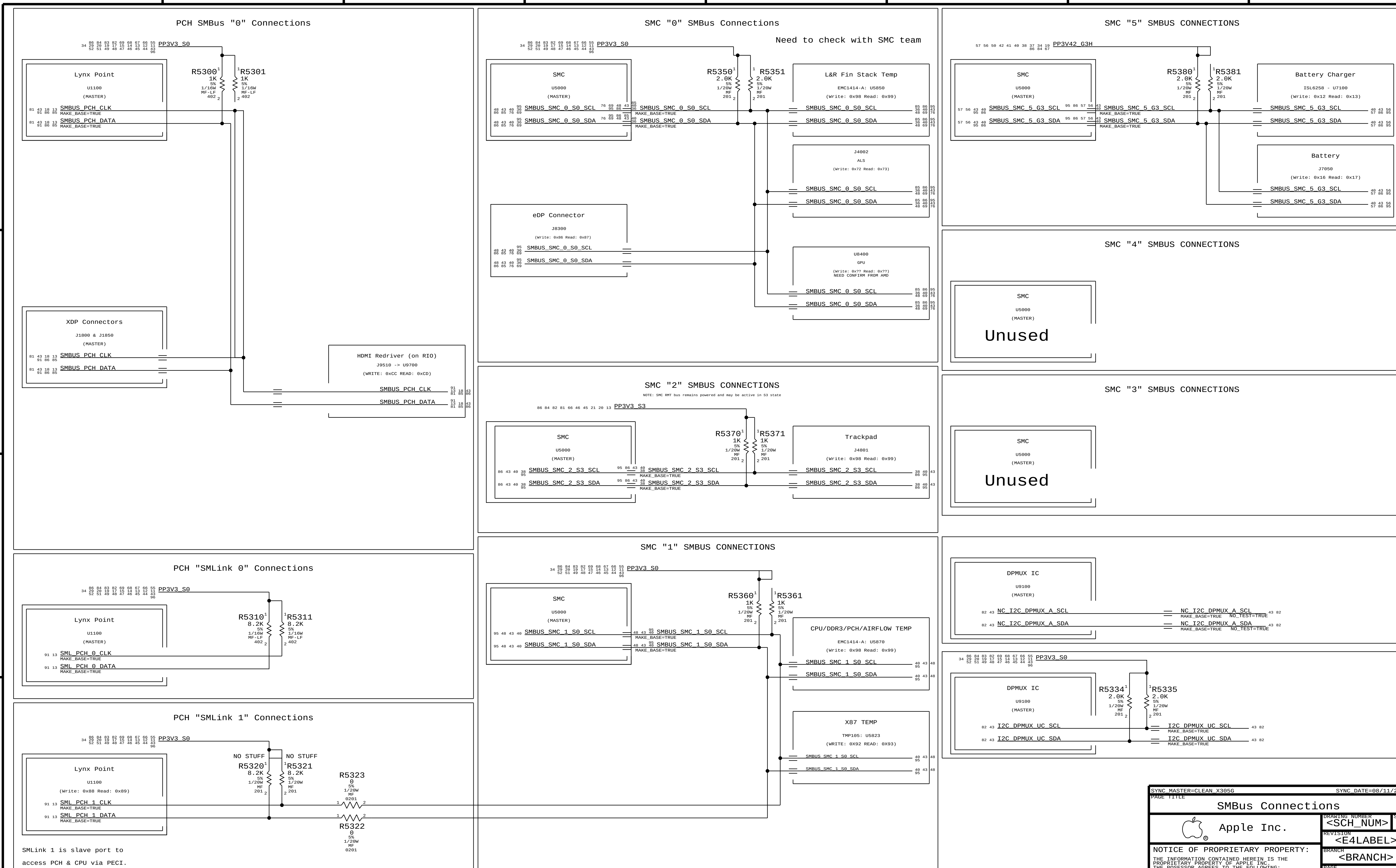

SYNC_MASTER=CLEAN_X305G SYNC_DATE=08/11/2014

PAGE TITLE

SMBus Connections

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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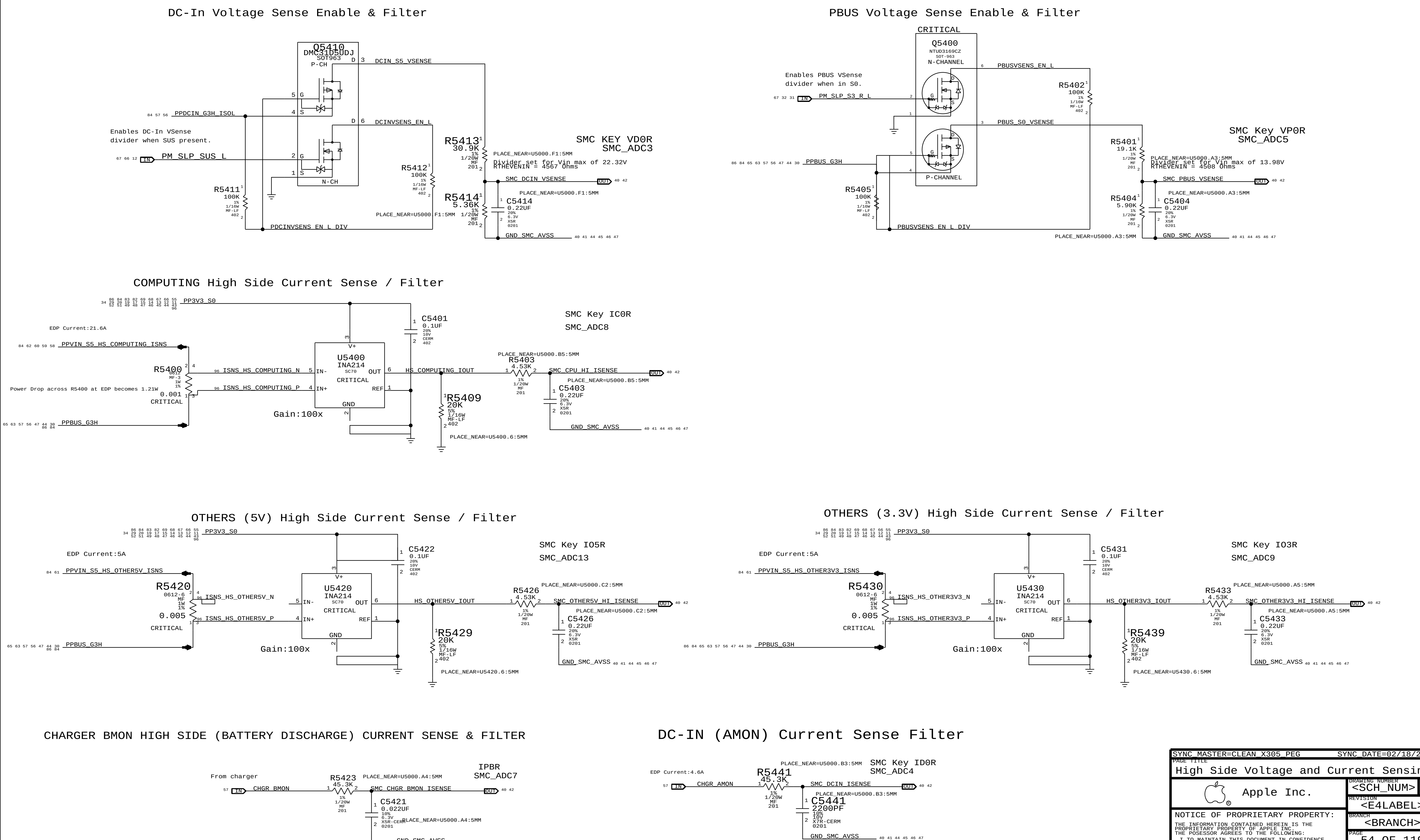

SYNC_MASTER=CLEAN_X305_PEG SYNC_DATE=02/18/2014

PAGE TITLE: High Side Voltage and Current Sensing

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

PAGE: 54 OF 119

SHEET: 44 OF 97

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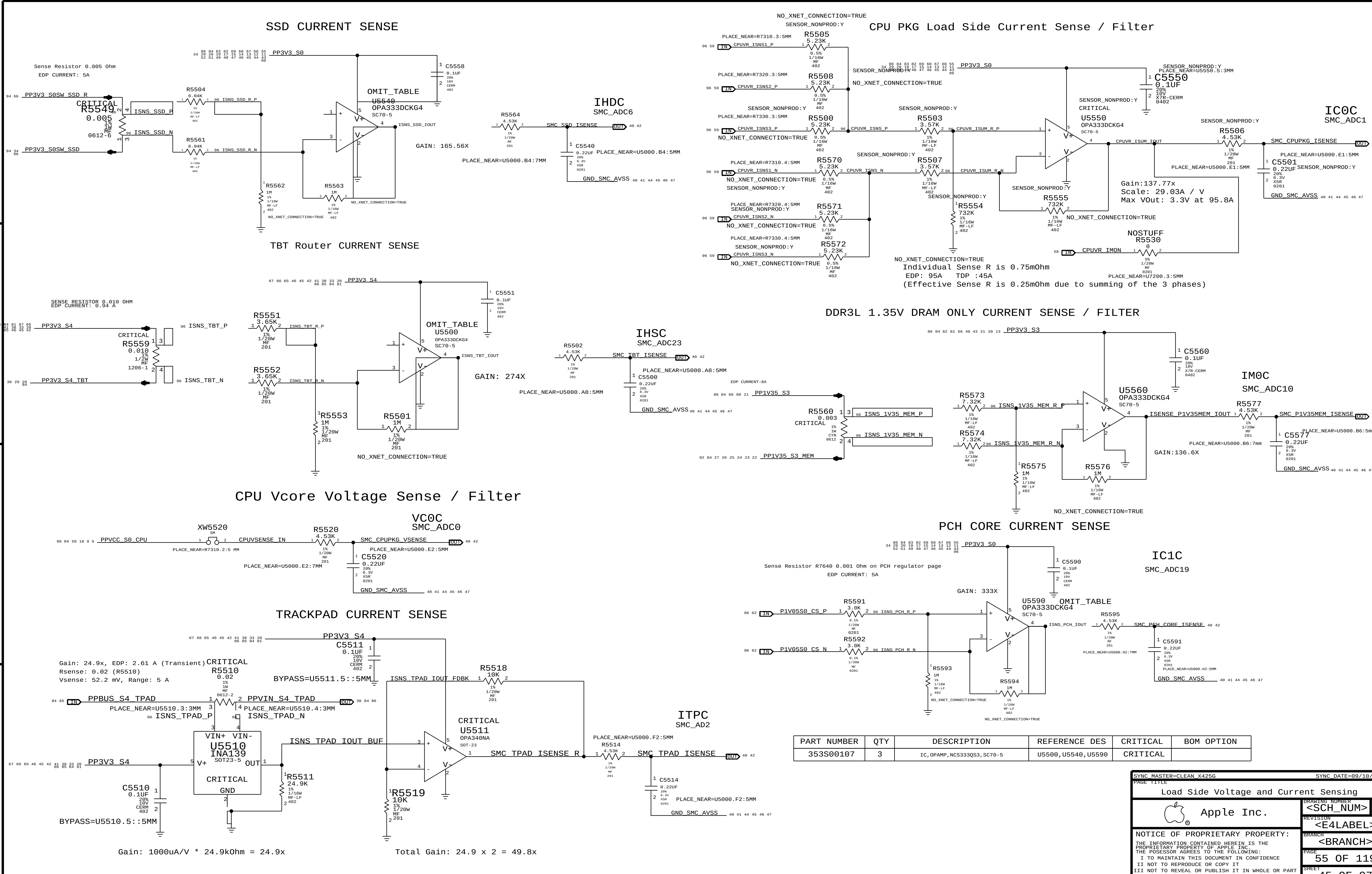

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00107	3	IC,OPAMP,NCS333QS3,SC70-5	U5500,U5540,U5590	CRITICAL	

SYNC_MASTER=CLEAN_X425G SYNC_DATE=09/10/2014

PAGE TITLE

Load Side Voltage and Current Sensing

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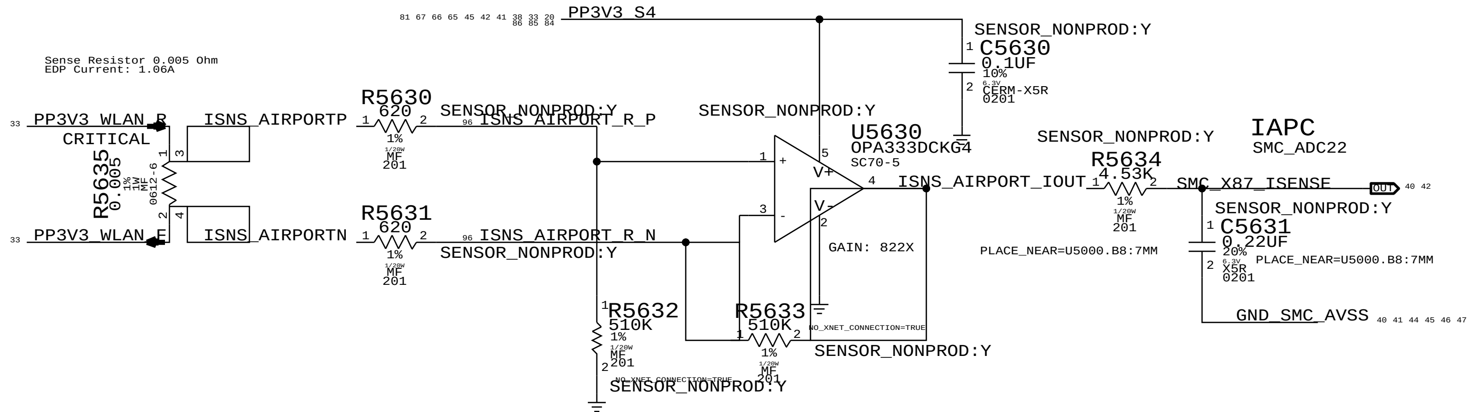
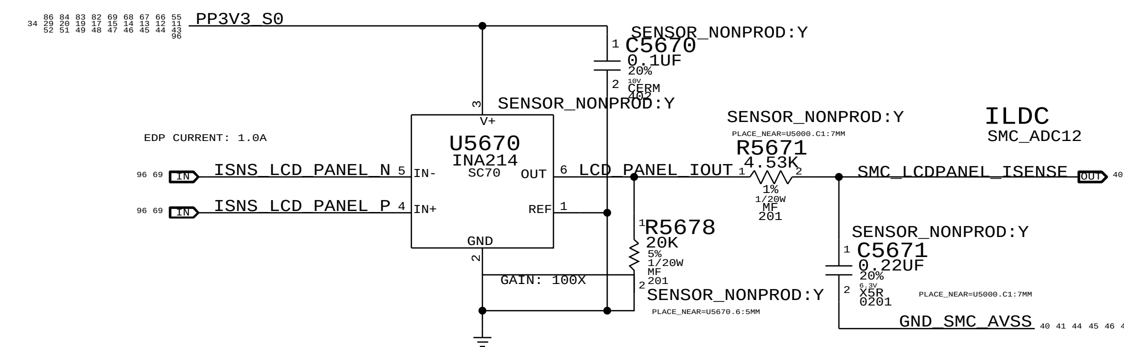
X87 AIRPORT CURRENT SENSE

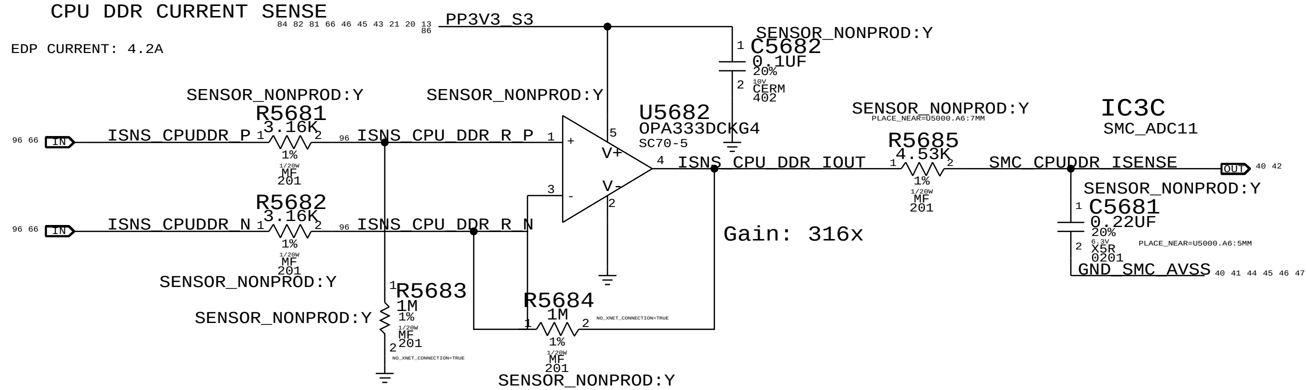


LCD PANEL CURRENT SENSE



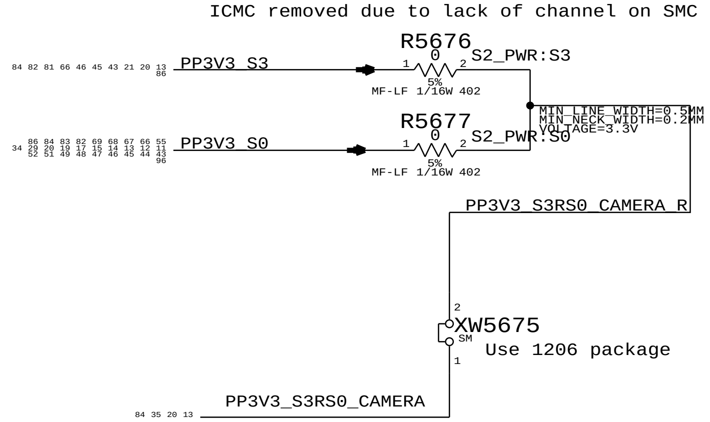
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	5	RES,MTL FILM,100K,5,1/20W,0201,SMD,LF	C5601,C5631,C5600,C5681,C5671		SENSOR_NONPROD:N

CPU DDR CURRENT SENSE



S2 CAMERA CONTROLLER CURRENT SENSE

ICMC removed due to lack of channel on SMC



SENSE RESISTOR 0.003 OHM
EDP CURRENT: 5A

SYNC_MASTER=CLEAN_X305 SYNC_DATE=01/14/2014

PAGE TITLE: Debug Sensors

DRAWING NUMBER: <SCH_NUM> SIZE: D

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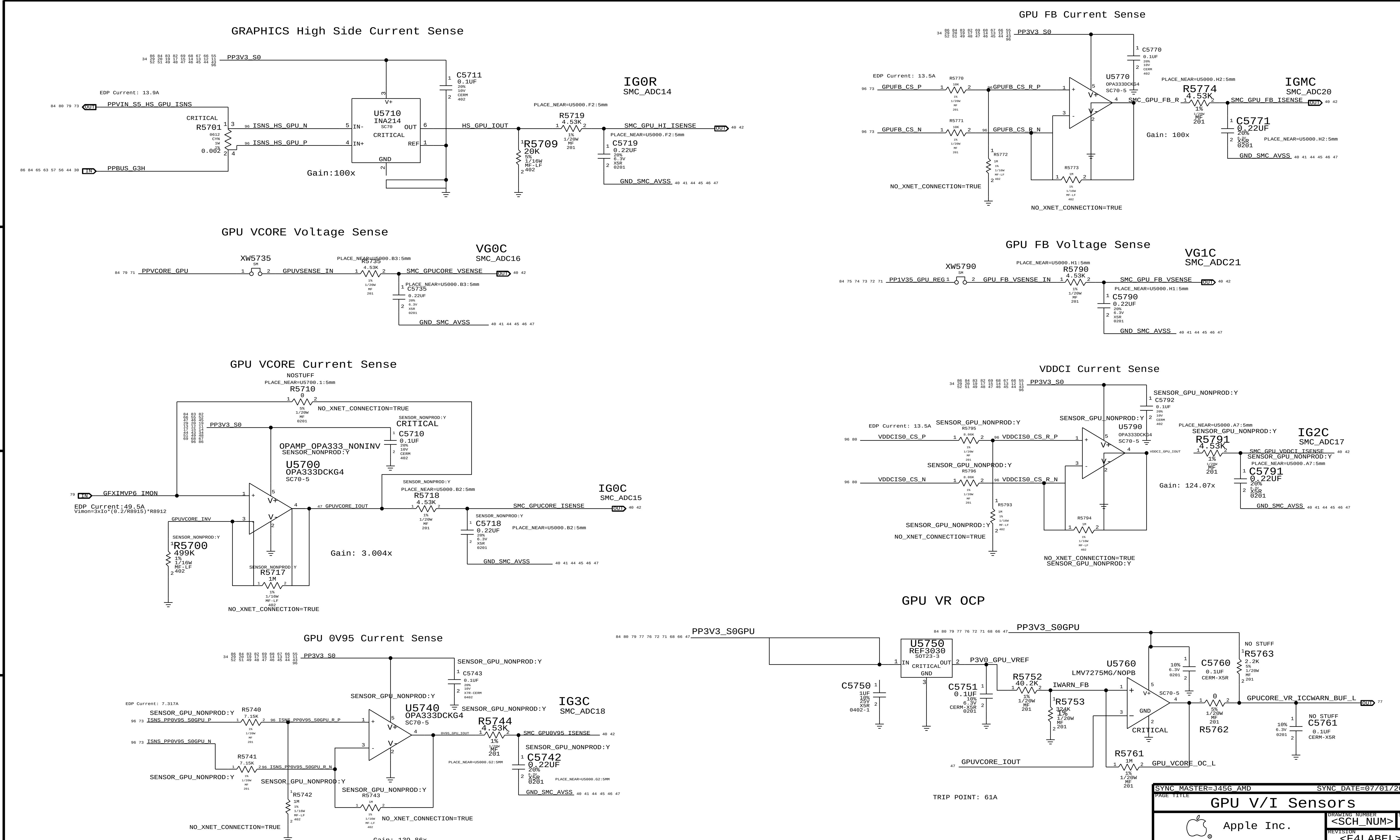

SYNC_MASTER=J45G_AMD SYNC_DATE=07/01/2014

PAGE TITLE

GPU V/I Sensors

DRAWING NUMBER <SCH_NUM> SIZE D

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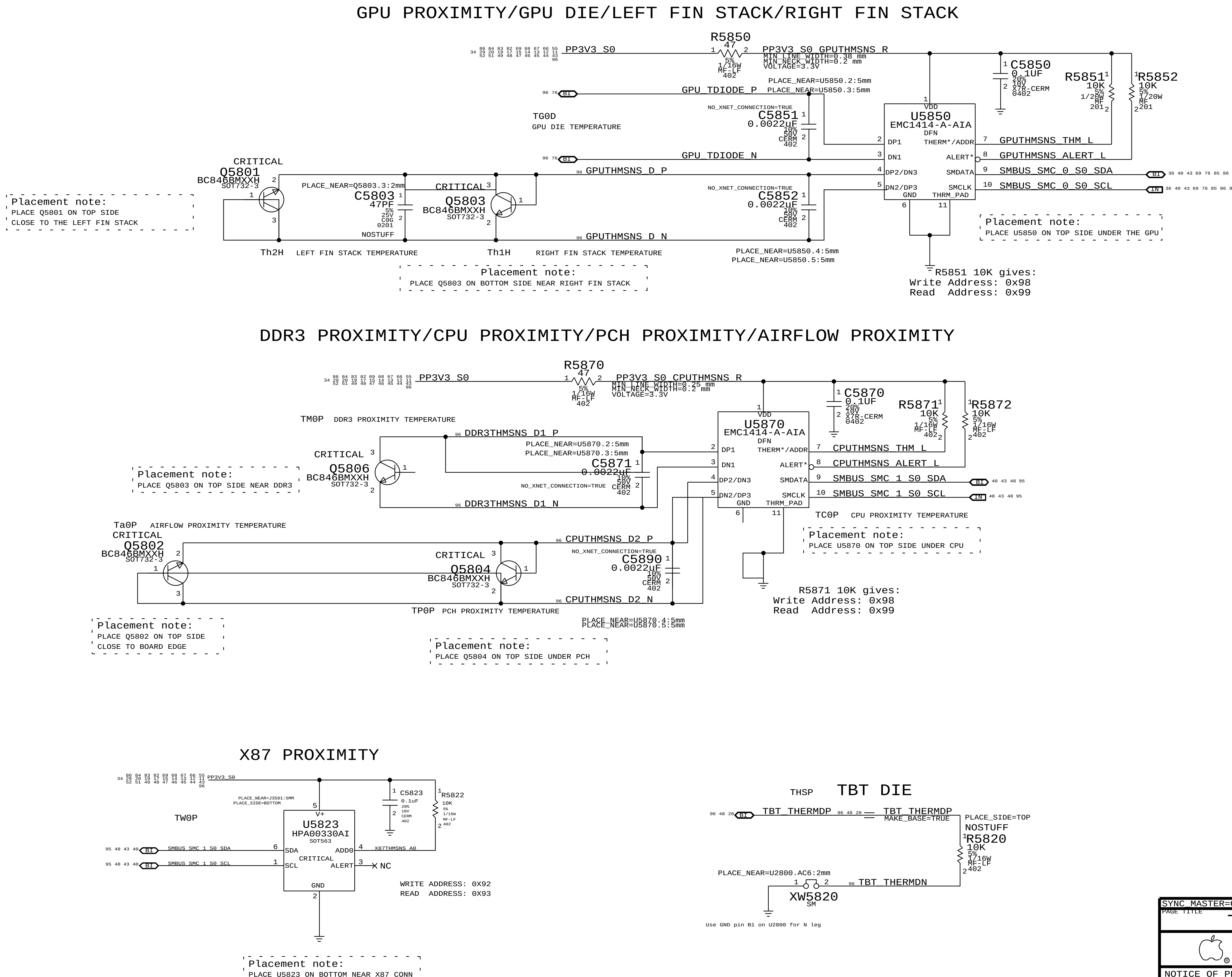

SYNC_MASTER=CHANG_J45 SYNC_DATE=11/26/2012

PAGE TITLE

Thermal Sensors

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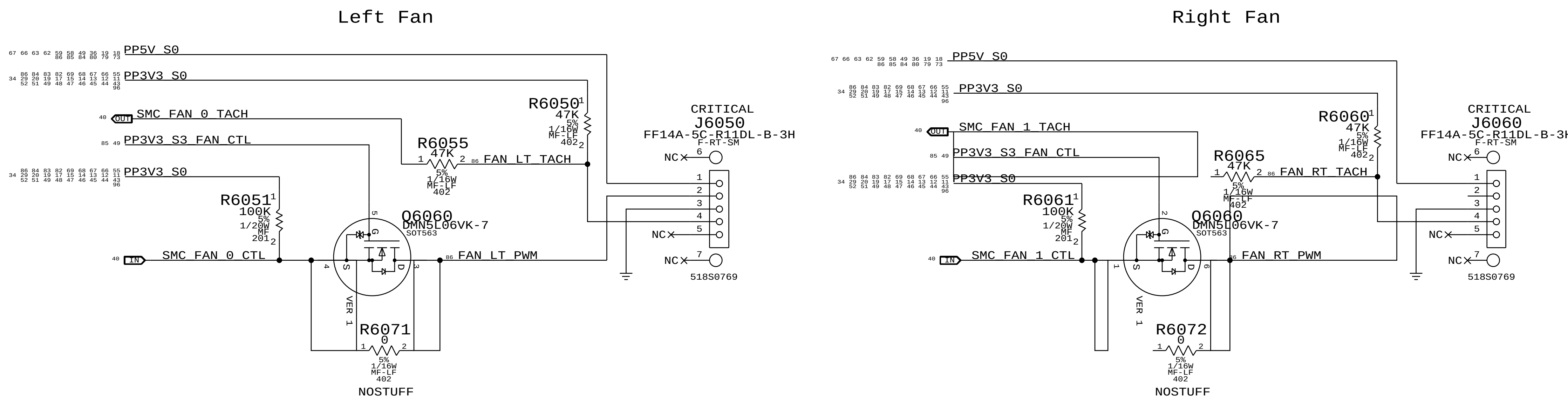

SYNC_MASTER=J15_MLB SYNC_DATE=10/31/2012

PAGE TITLE

Fan Connectors

Apple Inc.

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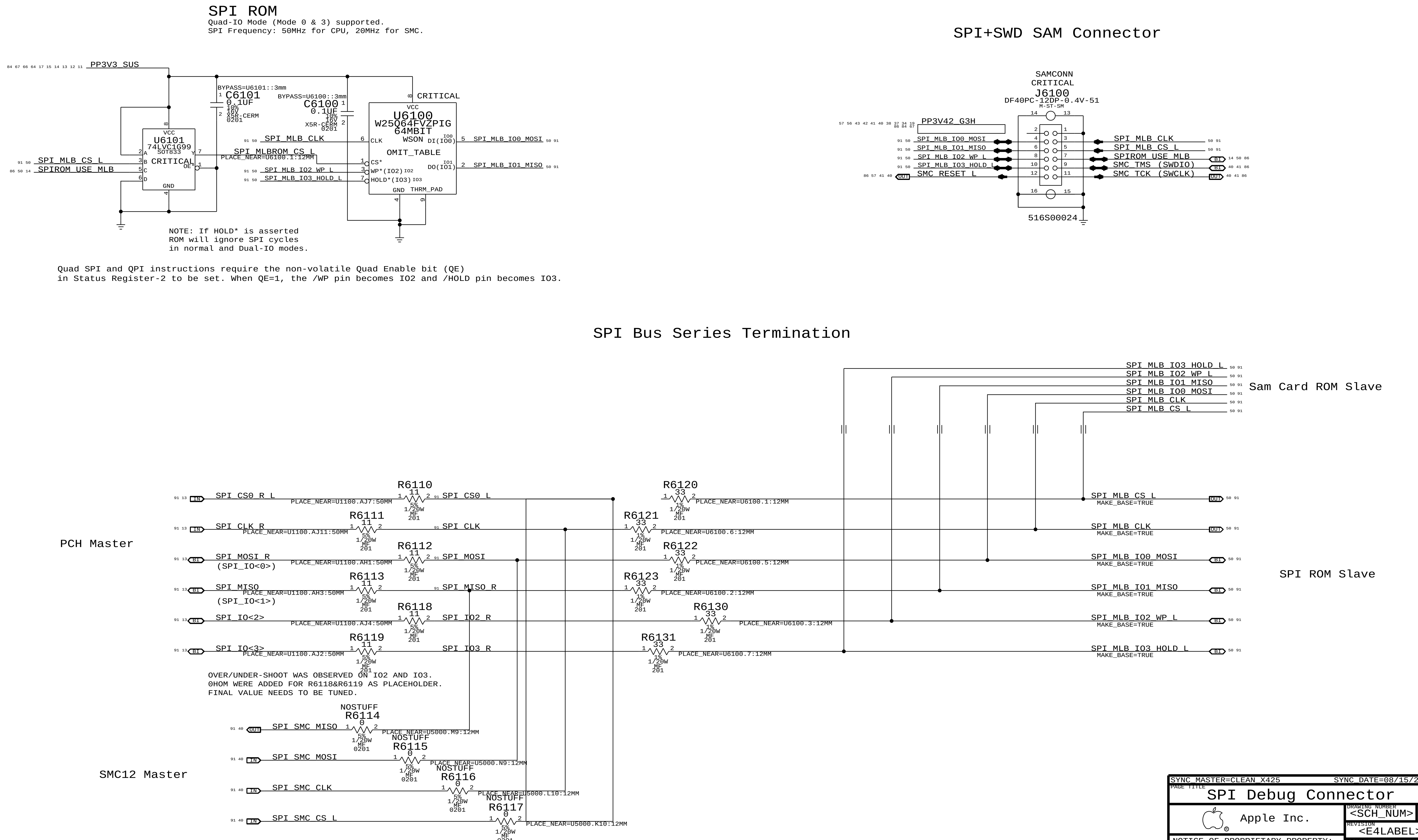

SYNC_MASTER=CLEAN_X425 SYNC_DATE=08/15/2014

PAGE TITLE

SPI Debug Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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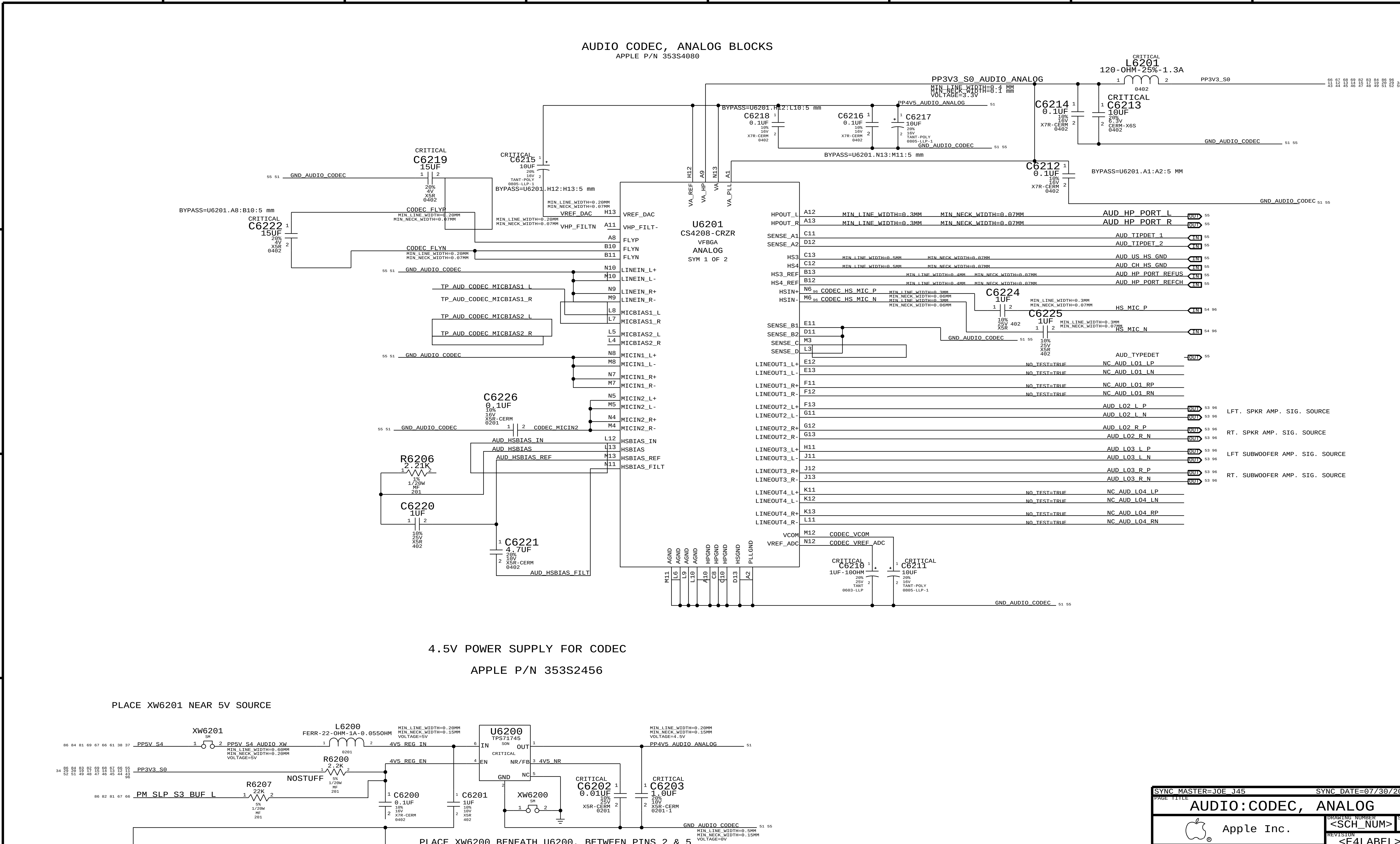

SYNC_MASTER=JOE_J45 SYNC_DATE=07/30/2013

PAGE TITLE

AUDIO:CODEC, ANALOG

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DRAWING NUMBER <SCH_NUM> SIZE D

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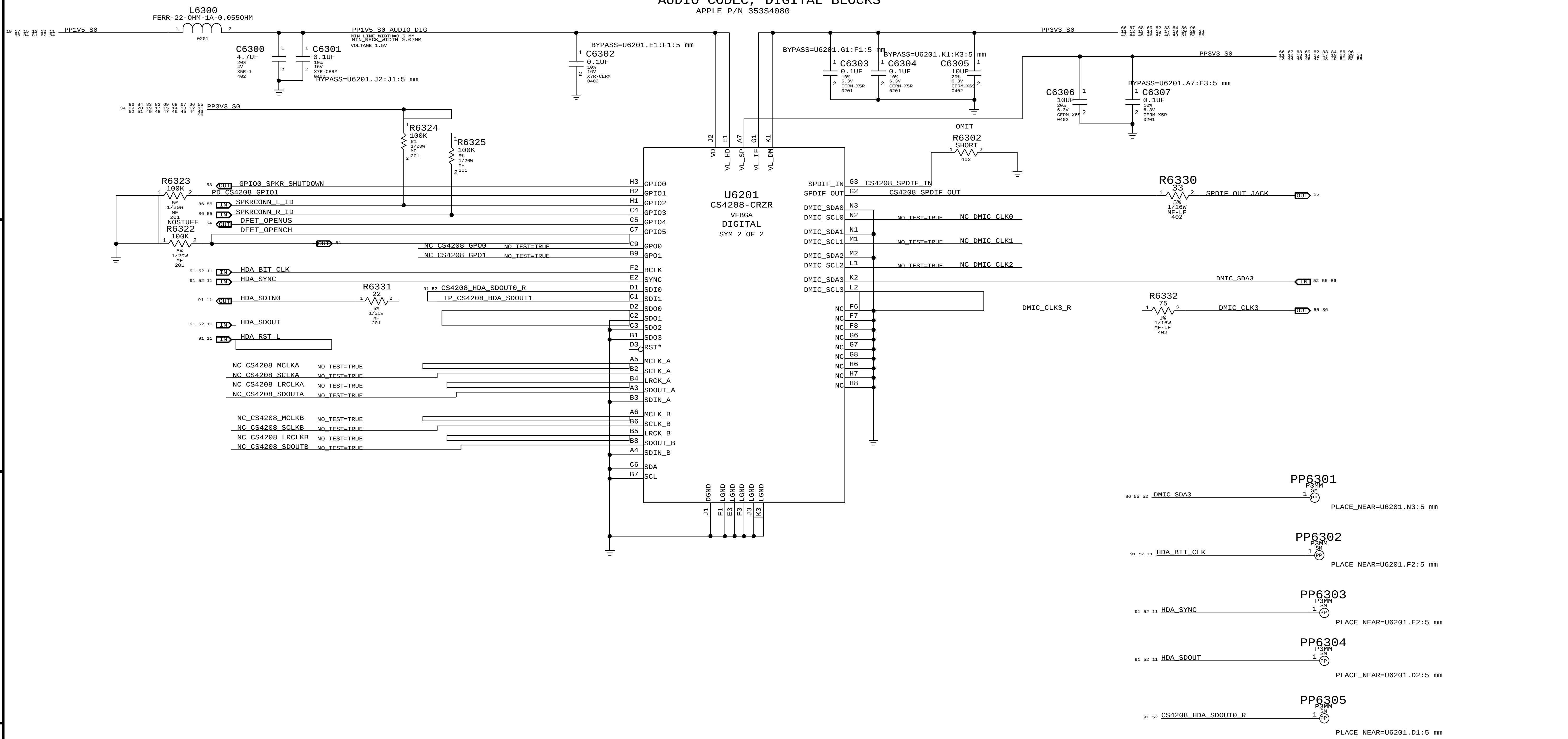
AUDIO CODEC, DIGITAL BLOCKS

APPLE P/N 353S4080



SYNC_MASTER=JOE_J45 SYNC_DATE=07/30/2013

PAGE TITLE

AUDIO:CODEC, DIGITAL

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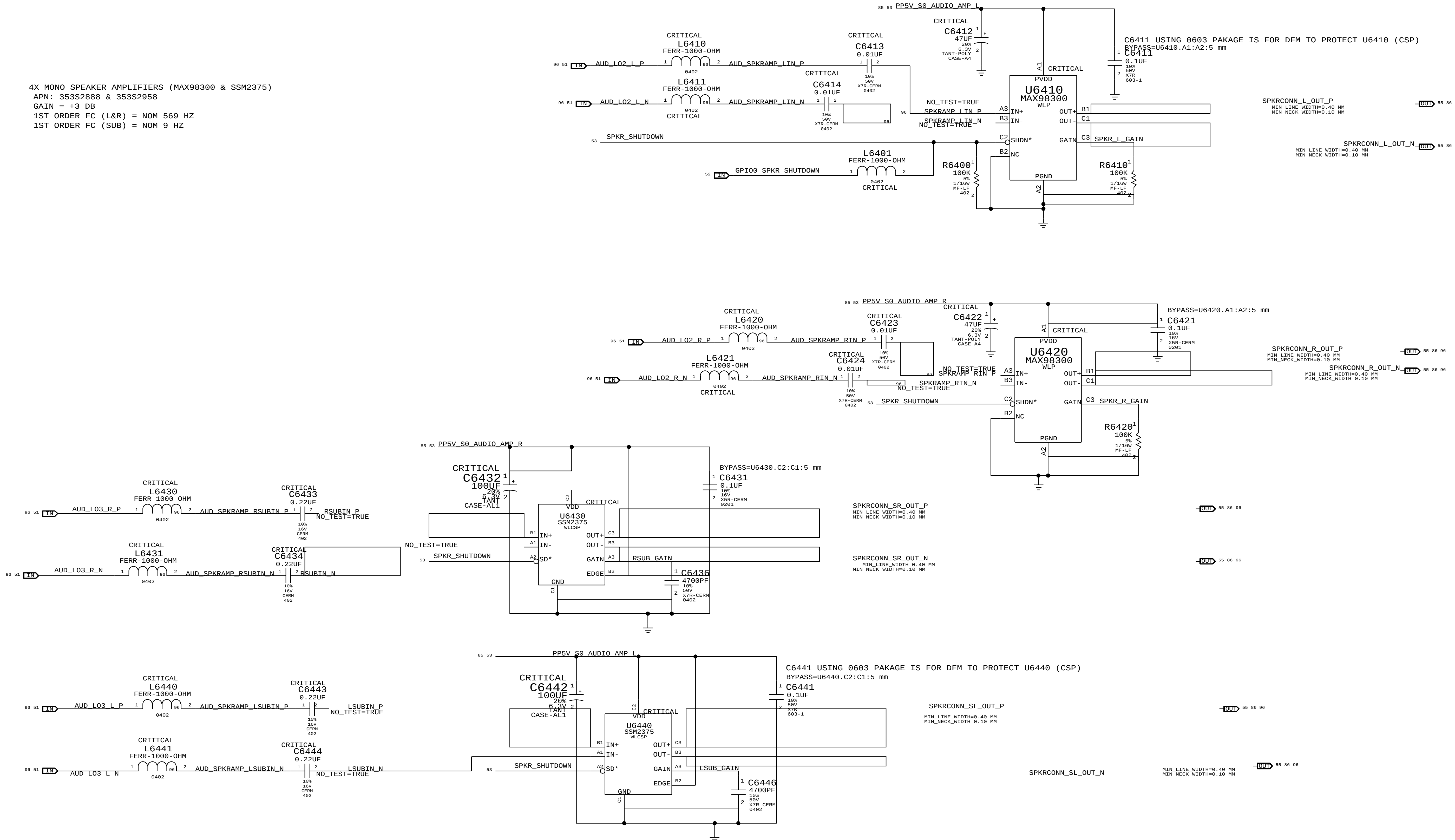
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



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PAGE TITLE
AUDIO: SPEAKER AMP

Apple Inc.

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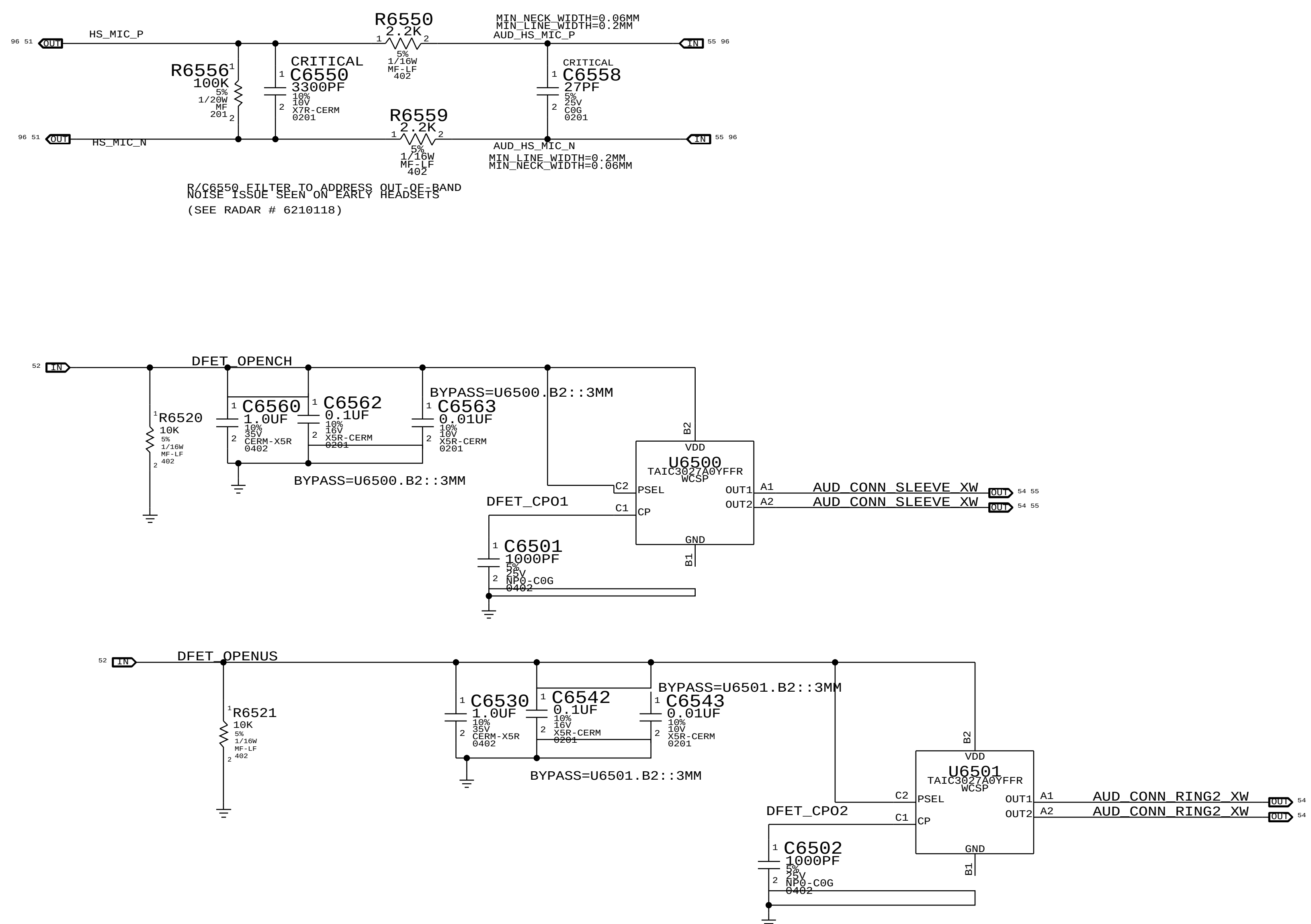

SYNC_MASTER=CLEAN_X305 SYNC_DATE=06/24/2014

PAGE TITLE

AUDIO: JACK

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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CODEC OUTPUT SIGNAL PATHS

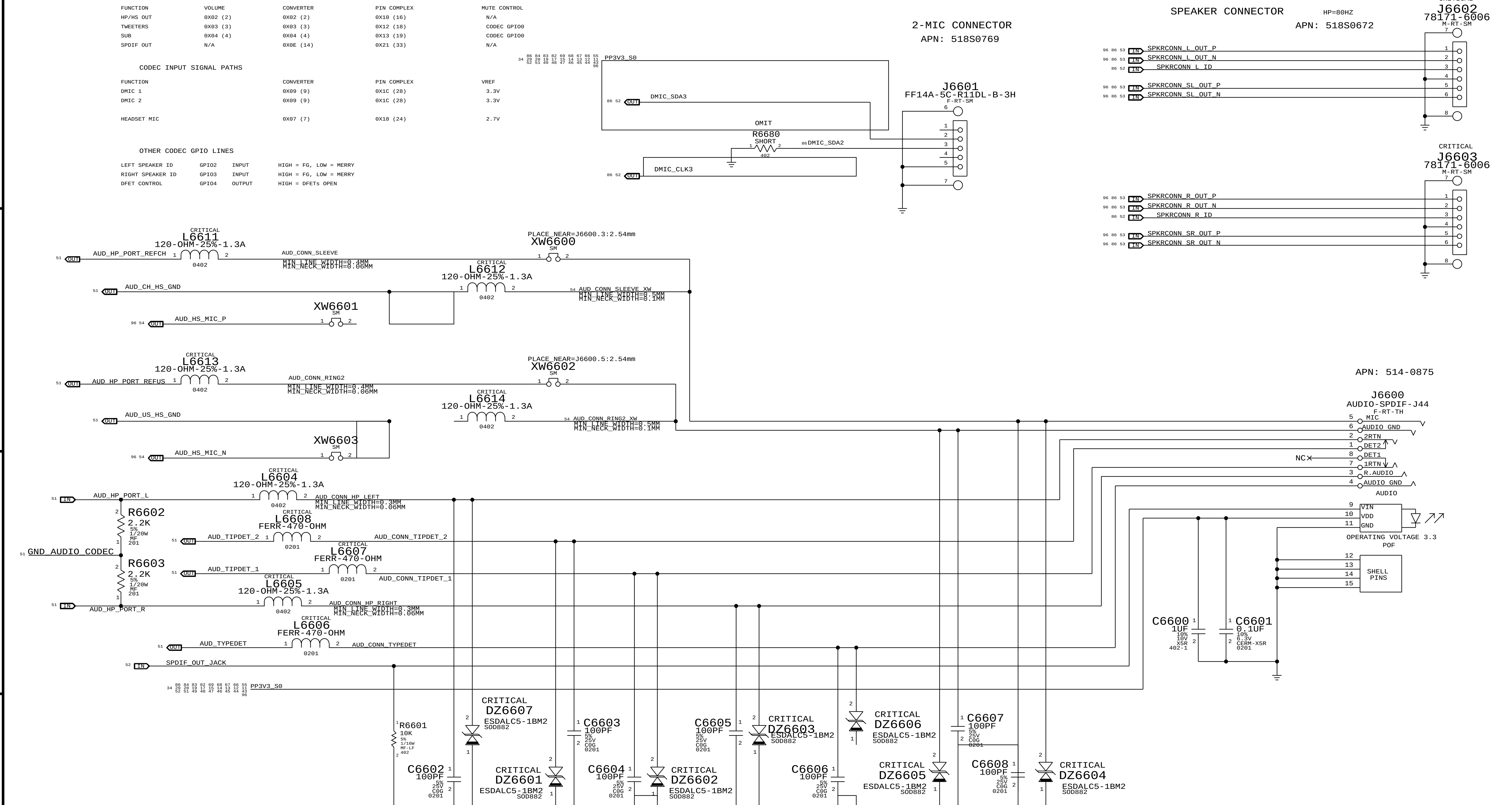
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2	INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3	INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4	OUTPUT	HIGH = DFETs OPEN



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PAGE TITLE	AUDIO: JACK TRANSLATORS
DRAWING NUMBER	<SCH_NUM>
SIZE	D
Apple Inc.	
REVISION	<E4LABEL>
BRANCH	<BRANCH>
PAGE	66 OF 119
SHEET	55 OF 97

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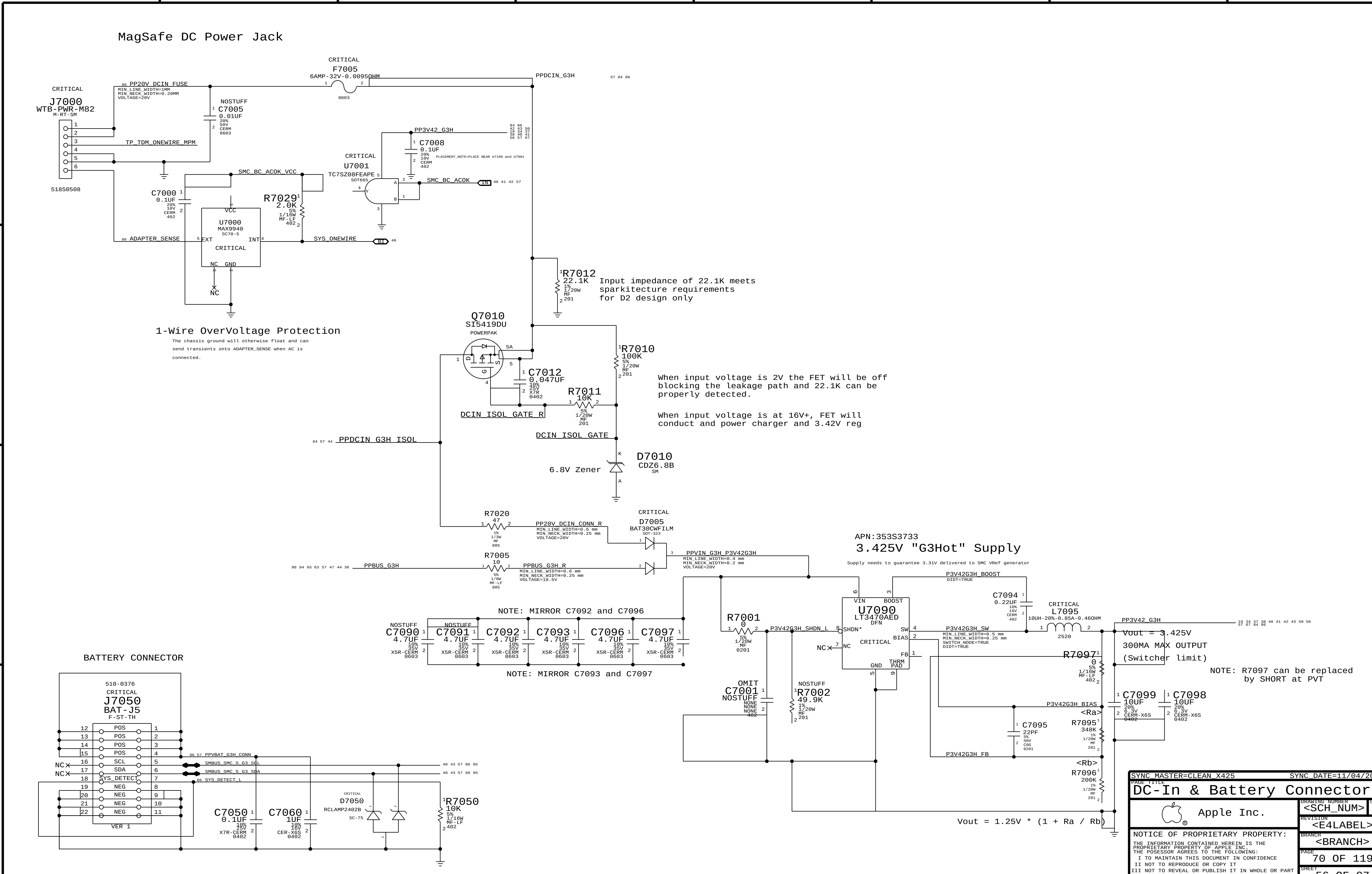

SYNC_MASTER=CLEAN_X425 SYNC_DATE=11/04/2014

PAGE TITLE

DC-In & Battery Connectors

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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PAGE 70 OF 119

SHEET 56 OF 97

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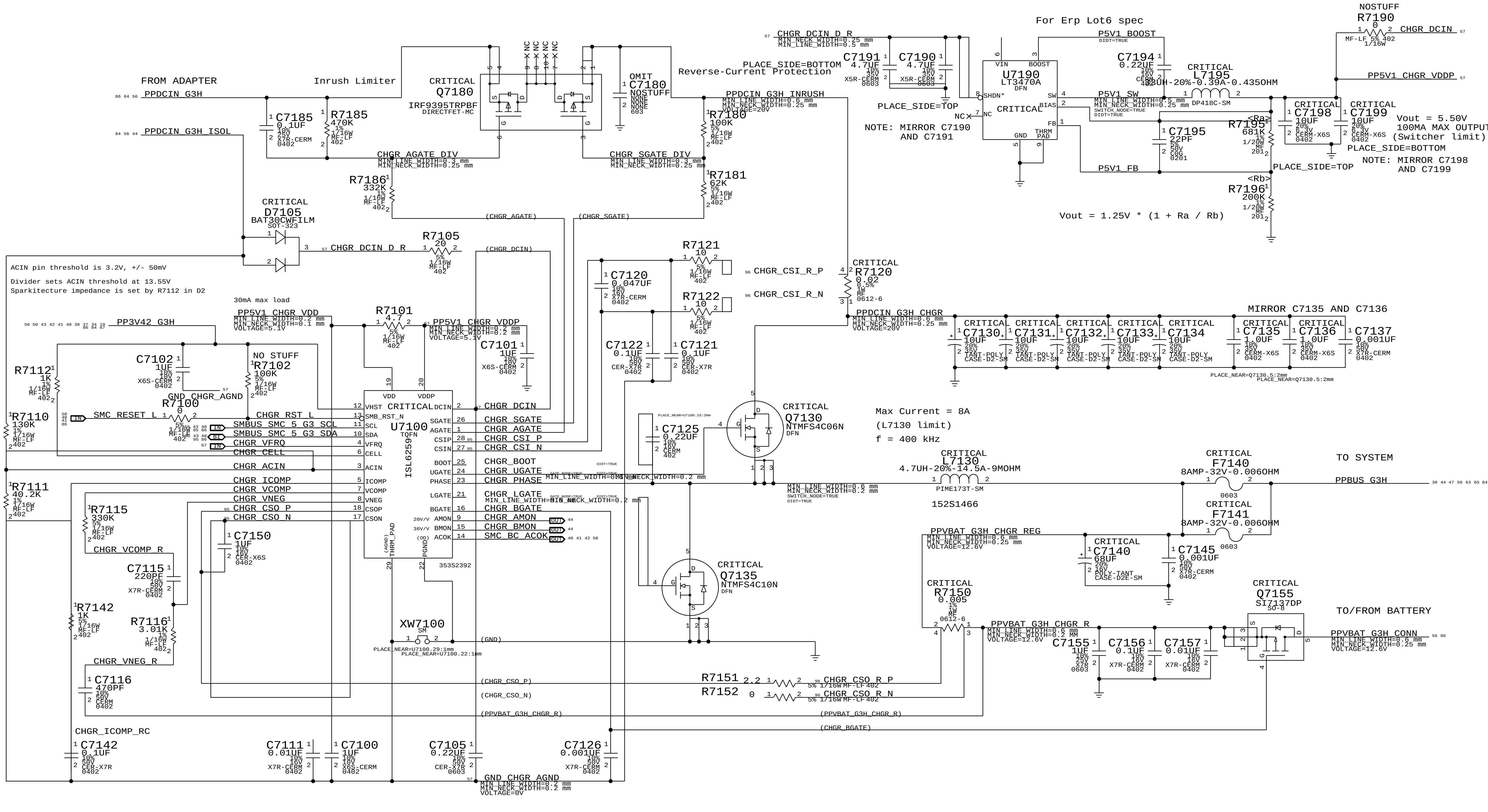

SYNC_MASTER=CLEAN_X305 SYNC_DATE=01/15/2014

PAGE TITLE

PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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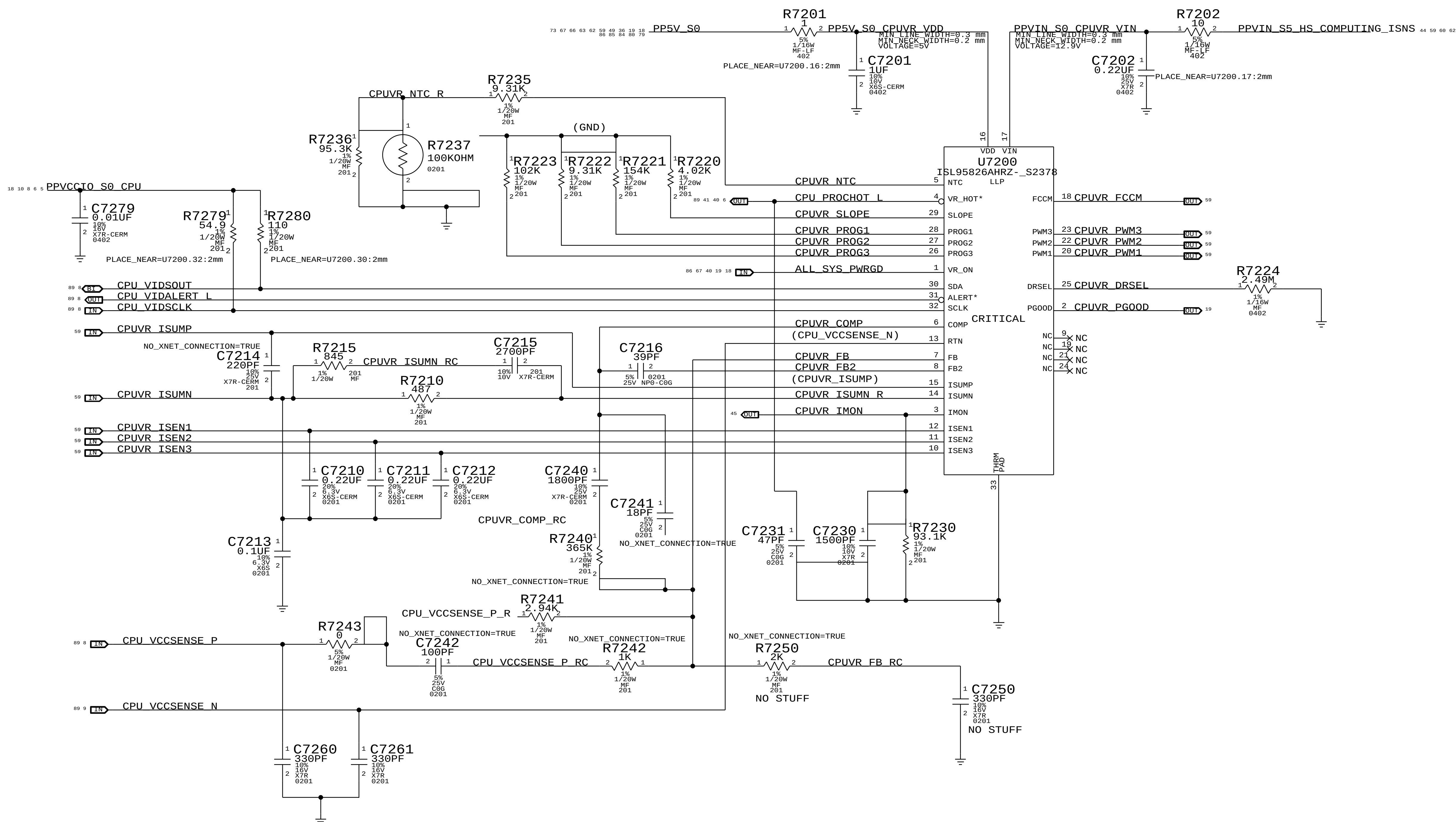

SYNC_MASTER=CLEAN_X425 SYNC_DATE=01/09/2015

PAGE TITLE

CPU VR12.5 VCC Regulator IC

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

REVISION <E4LABEL>

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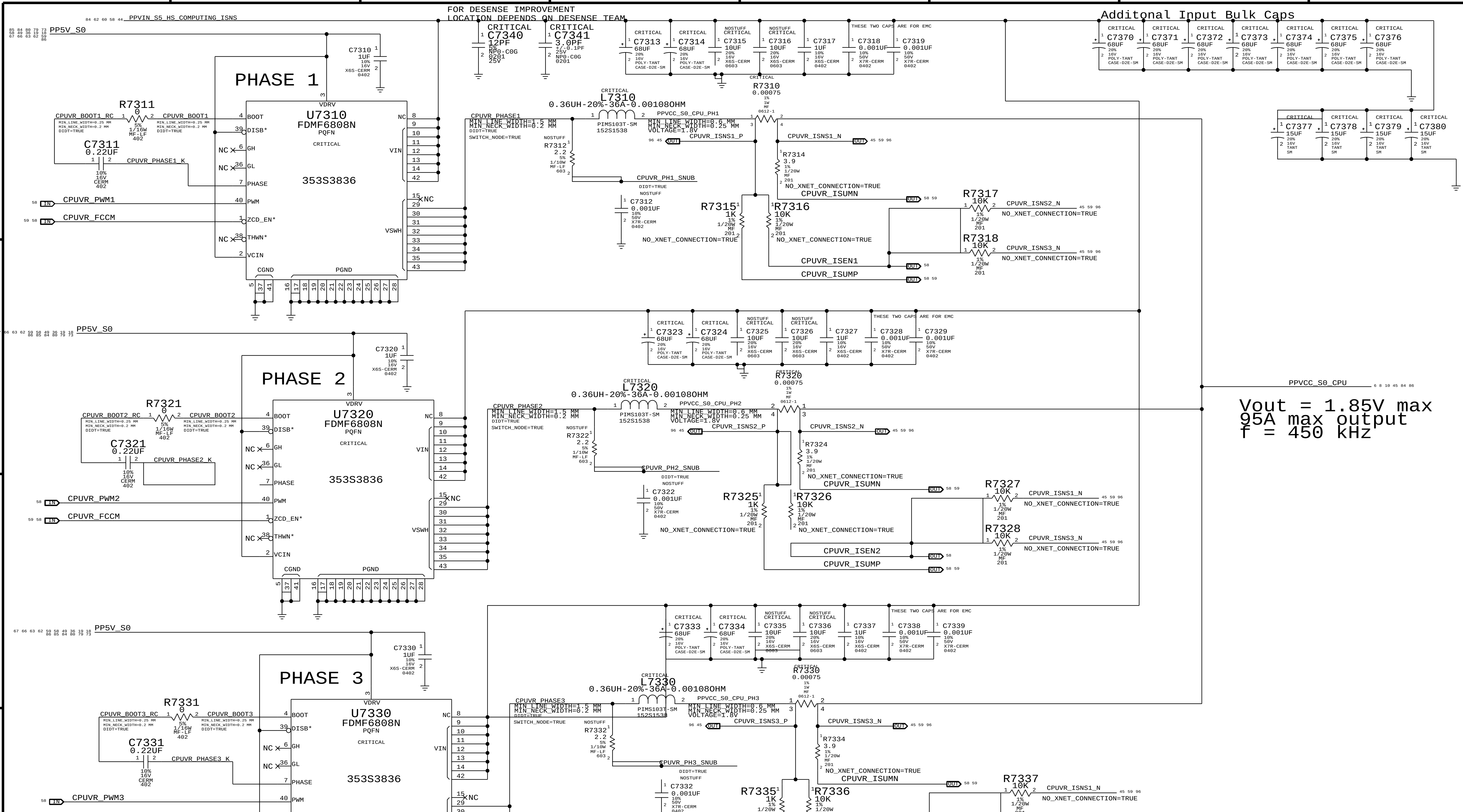

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PAGE TITLE

CPU VR12.5 VCC Power Stage

Apple Inc.

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SHEET 59 OF 97

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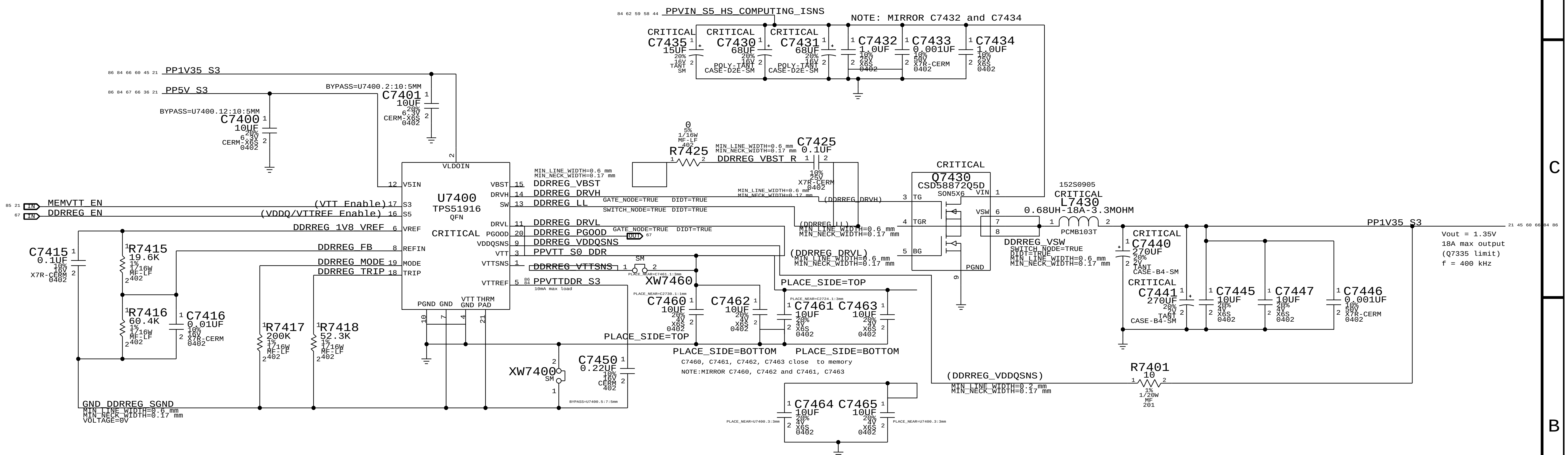
DDR3L (1V35 S3) REGULATOR



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Apple Inc.	DRAWING NUMBER: <SCH_NUM>	SIZE: D
	REVISION: <E4LABEL>	
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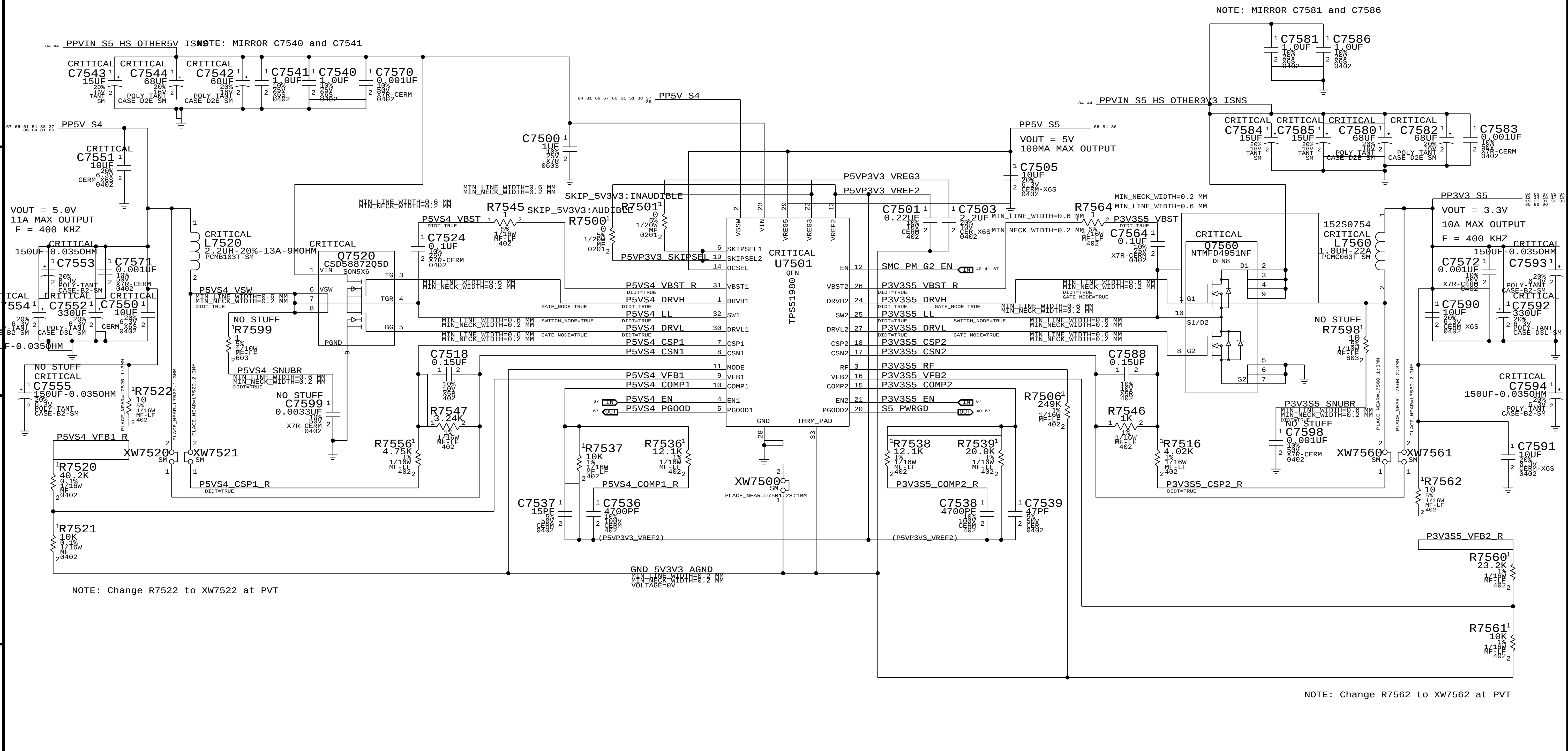
	SHEET: 60 OF 97	



SYNC_MASTER=CLEAN_X425 SYNC_DATE=11/04/2014

PAGE TITLE

5V / 3.3V Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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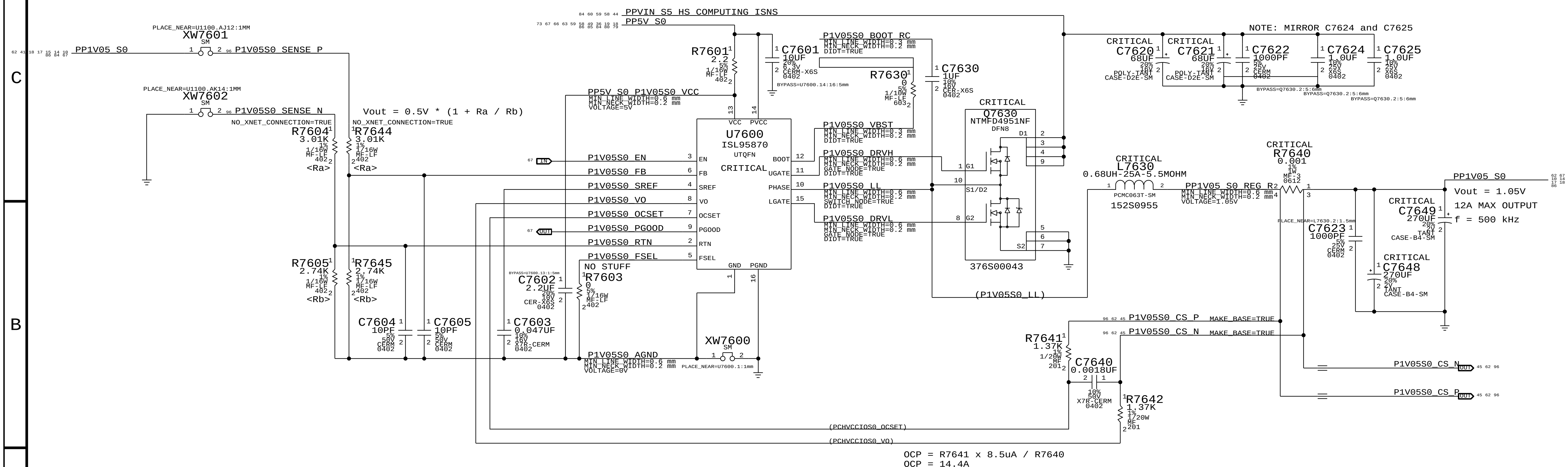
1V05 S0 REGULATOR



SYNC_MASTER=CLEAN_X305_PEG SYNC_DATE=02/18/2014

PAGE TITLE

1V05V POWER SUPPLY

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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PAGE 76 OF 119

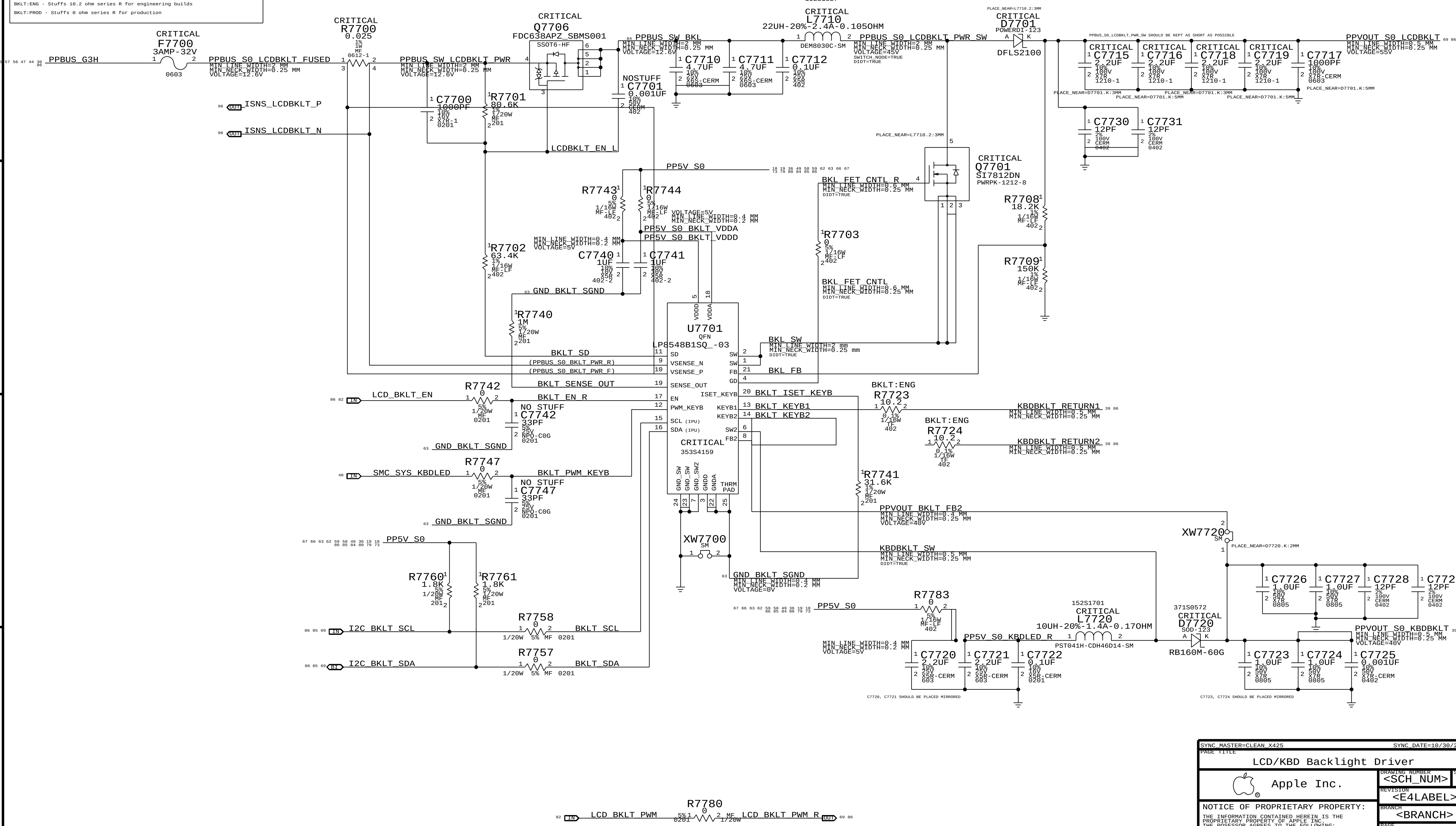
SHEET 62 OF 97

Page Notes

Power aliases required by this page:
- =PPVIN_S0_LCDBKLT (9-12.6V LCD Backlight Input)
- =PP5V_S0_BKLTCTRL (5V Backlight Driver Input)
- =PP5V_S0_KBDLED (5V Keyboard Backlight Input)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7723,R7724		BKLT:PROD



SYNC_MASTER=CLEAN_X425 SYNC_DATE=10/30/2014

PAGE TITLE: LCD/KBD Backlight Driver

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D
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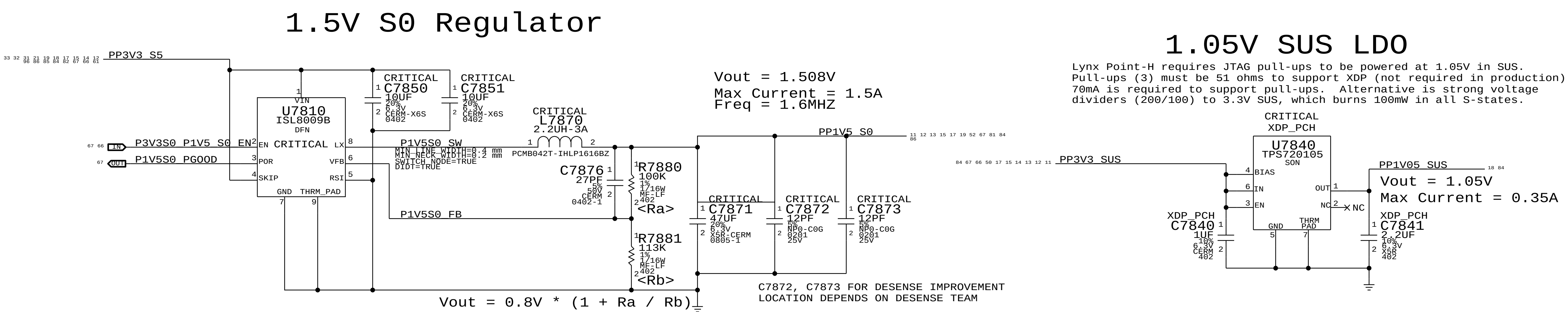

SYNC_MASTER=CLEAN_X305 SYNC_DATE=01/15/2014

PAGE TITLE

Misc Power Supplies

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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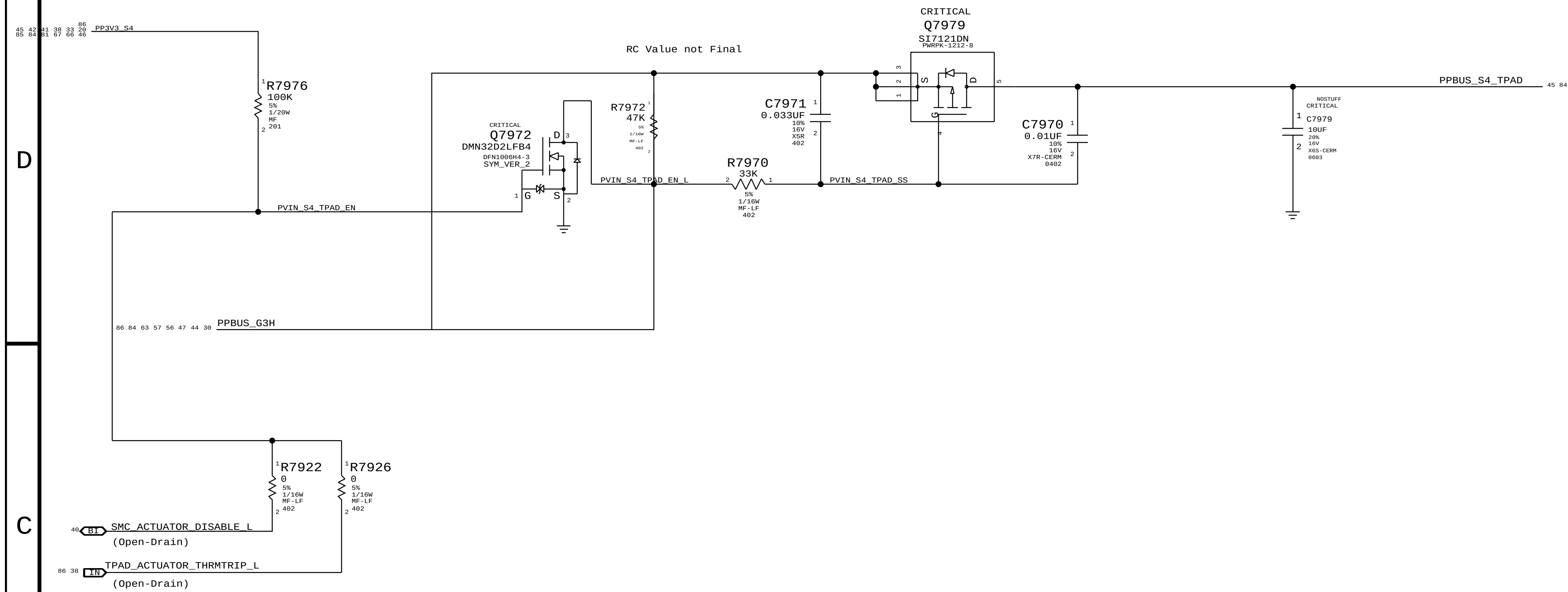

PBUS ONLY CONTROL INPUTS

SYSTEM STATE	SMC_ACTUATOR_DISABLE_L	TPAD_ACTUATOR_THRMTRIP_L	T101 POWER
G3H, S5	X	X	0 V
S4, S3, S0	Hi-Z	Hi-Z	PBUS
S4, S3, S0	0 V	X	0 V
S4, S3, S0	X	0 V	0 V

SYNC_MASTER=CLEAN_MAXWELL SYNC_DATE=07/02/2014

PAGE TITLE

X249 POWER SUPPLY

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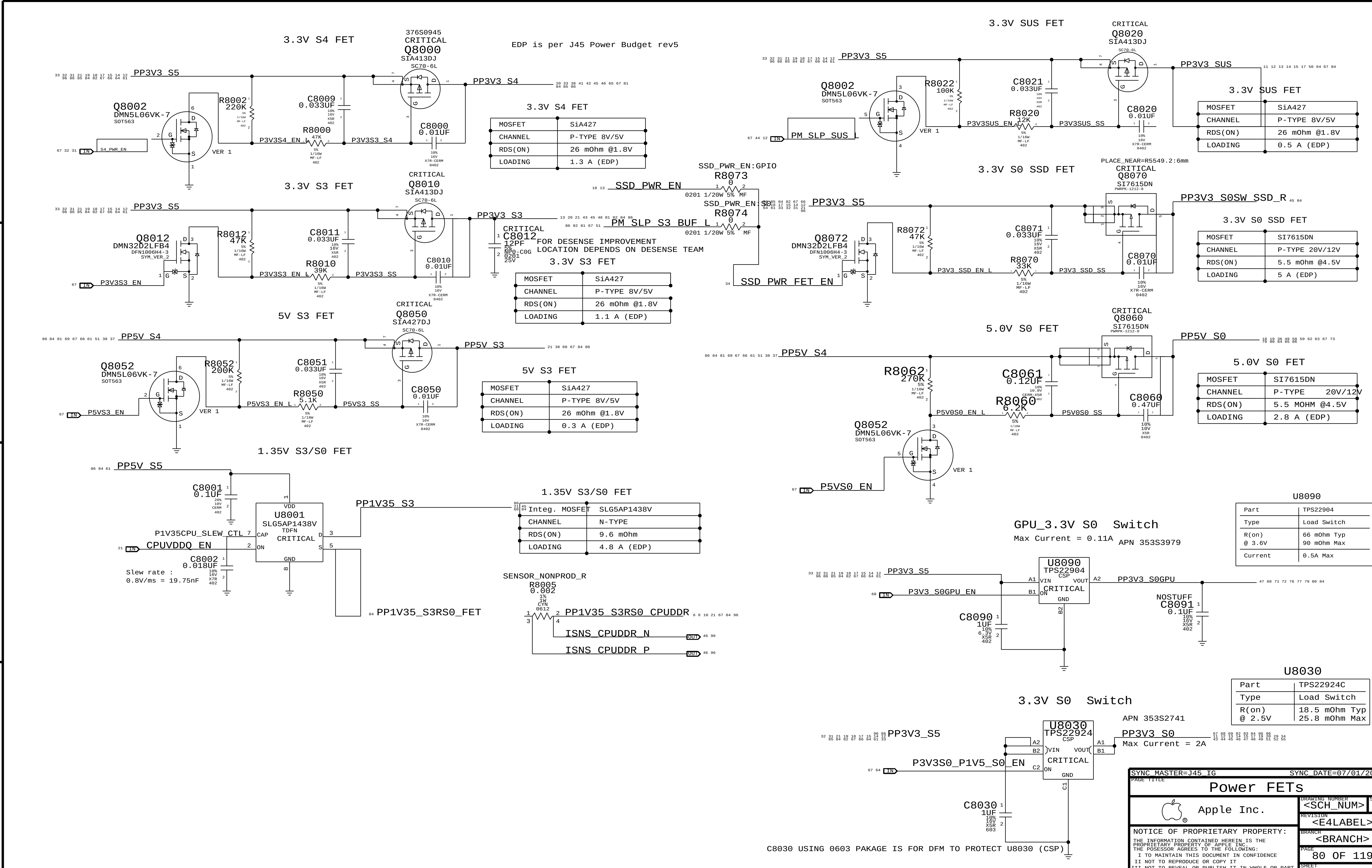

3.3V S4 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.3 A (EDP)

3.3V S3 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.1 A (EDP)

5V S3 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.3 A (EDP)

1.35V S3/S0 FET

Integ. MOSFET	SLG5AP1438V
CHANNEL	N-TYPE
RDS(ON)	9.6 mOhm
LOADING	4.8 A (EDP)

3.3V SUS FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.5 A (EDP)

3.3V S0 SSD FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 mOhm @4.5V
LOADING	5 A (EDP)

5.0V S0 FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 MOHM @4.5V
LOADING	2.8 A (EDP)

U8090

Part	TPS22904
Type	Load Switch
R(on) @ 3.6V	66 mOhm Typ 90 mOhm Max
Current	0.5A Max

U8030

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SYNC_MASTER=J45_IG SYNC_DATE=07/01/2014

PAGE TITLE

Power FETs

DRAWING NUMBER <SCH_NUM> SIZE D

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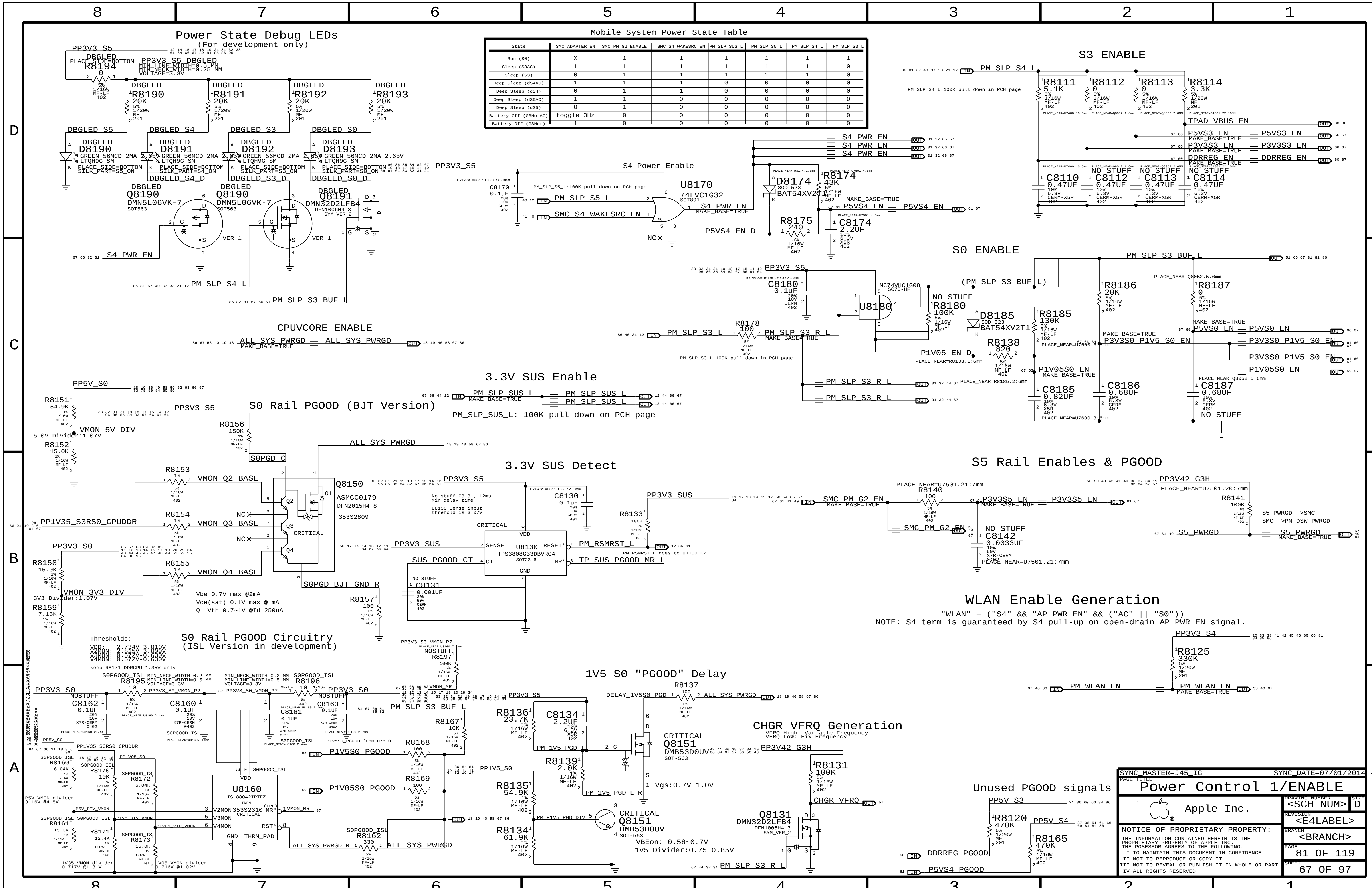

Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SLP_SUS_L	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (dS4AC)	1	1	1	0	0	0	0
Deep Sleep (dS4)	0	1	1	0	0	0	0
Deep Sleep (dS5AC)	1	1	0	0	0	0	0
Deep Sleep (dS5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC_MASTER=J45_IG SYNC_DATE=07/01/2014

PAGE TITLE: Power Control 1/ENABLE

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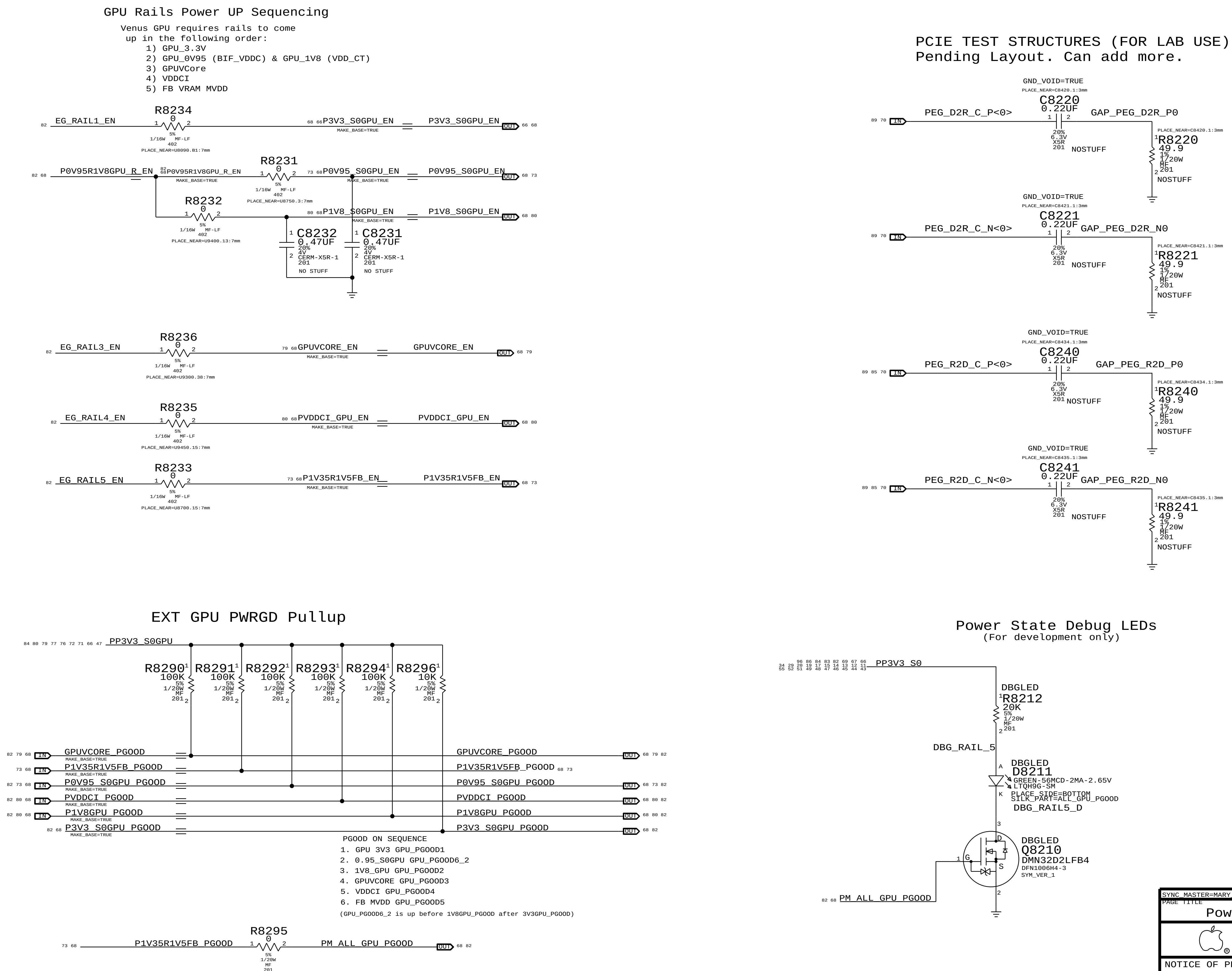

SYNC_MASTER=MARY_X425G SYNC_DATE=09/11/2014

PAGE TITLE
Power Sequencing EG/PGOOD

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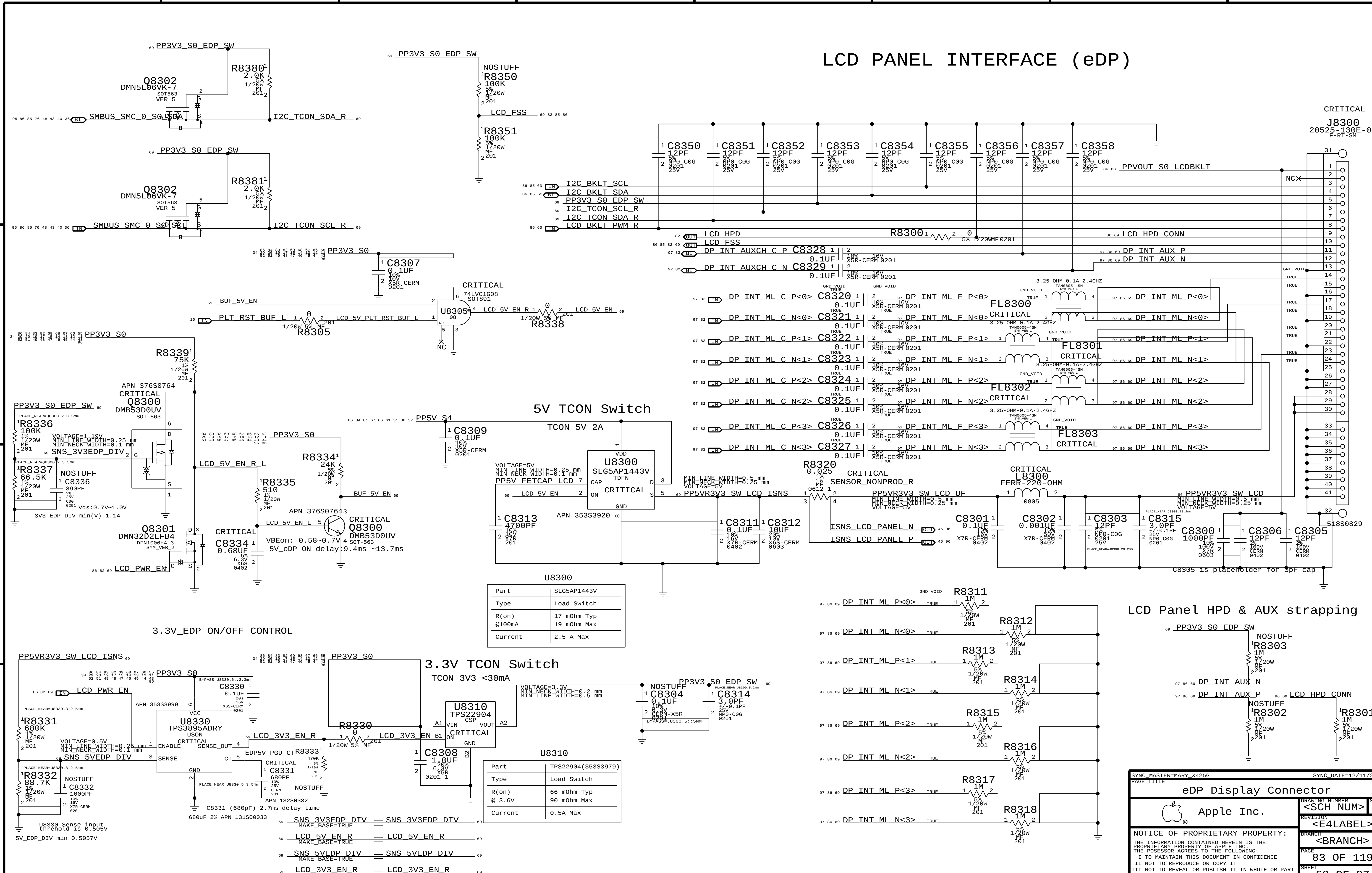
LCD PANEL INTERFACE (eDP)



U8300

Part	SLG5AP1443V
Type	Load Switch
R(on) @100mA	17 mOhm Typ 19 mOhm Max
Current	2.5 A Max

U8310

Part	TPS22904(353S3979)
Type	Load Switch
R(on) @ 3.6V	66 mOhm Typ 90 mOhm Max
Current	0.5A Max

SYNC_MASTER=MARY_X425G SYNC_DATE=12/11/2014

PAGE TITLE
eDP Display Connector

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Page Notes

Power aliases required by this page:

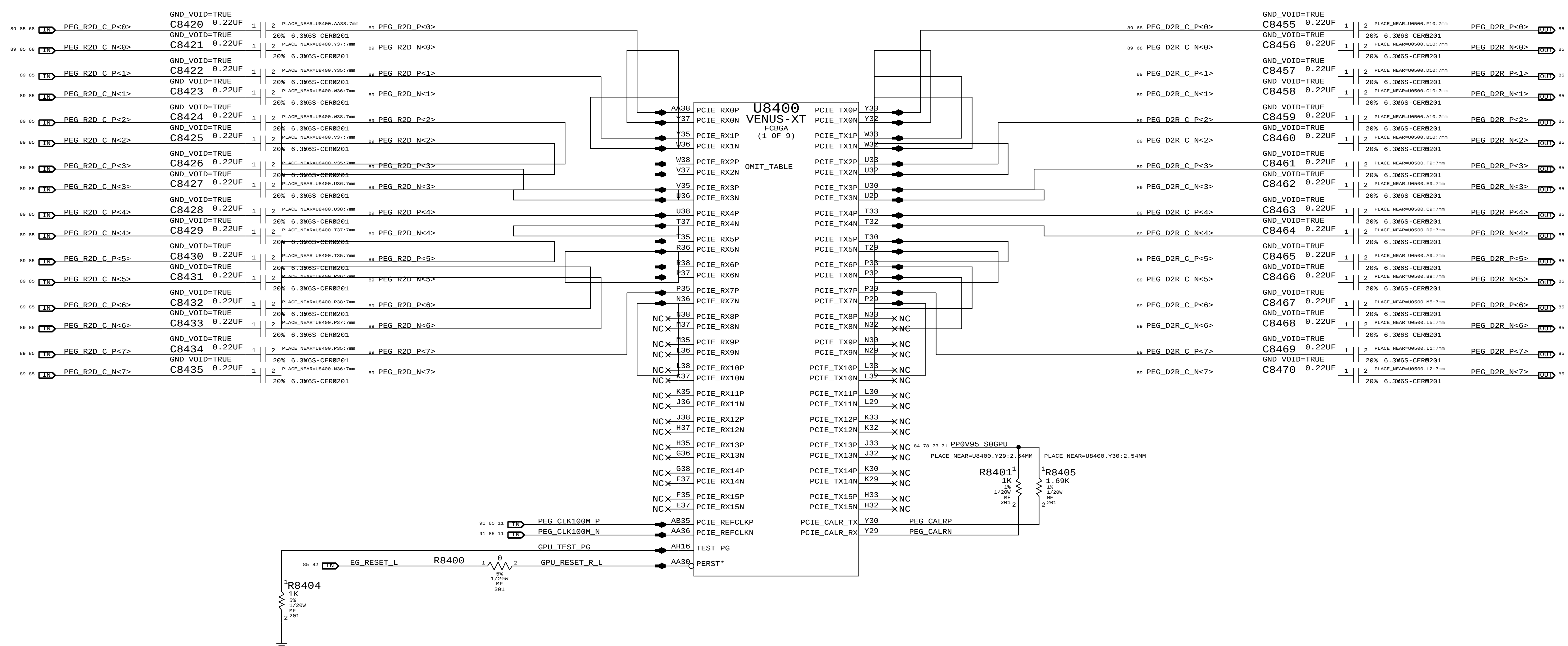
- =PPV95_GPU_PCIE

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)



SYNC_MASTER=MARY_X425G SYNC_DATE=08/22/2014

PAGE TITLE

Venus PCI-E

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Page Notes

Power aliases required by this page:

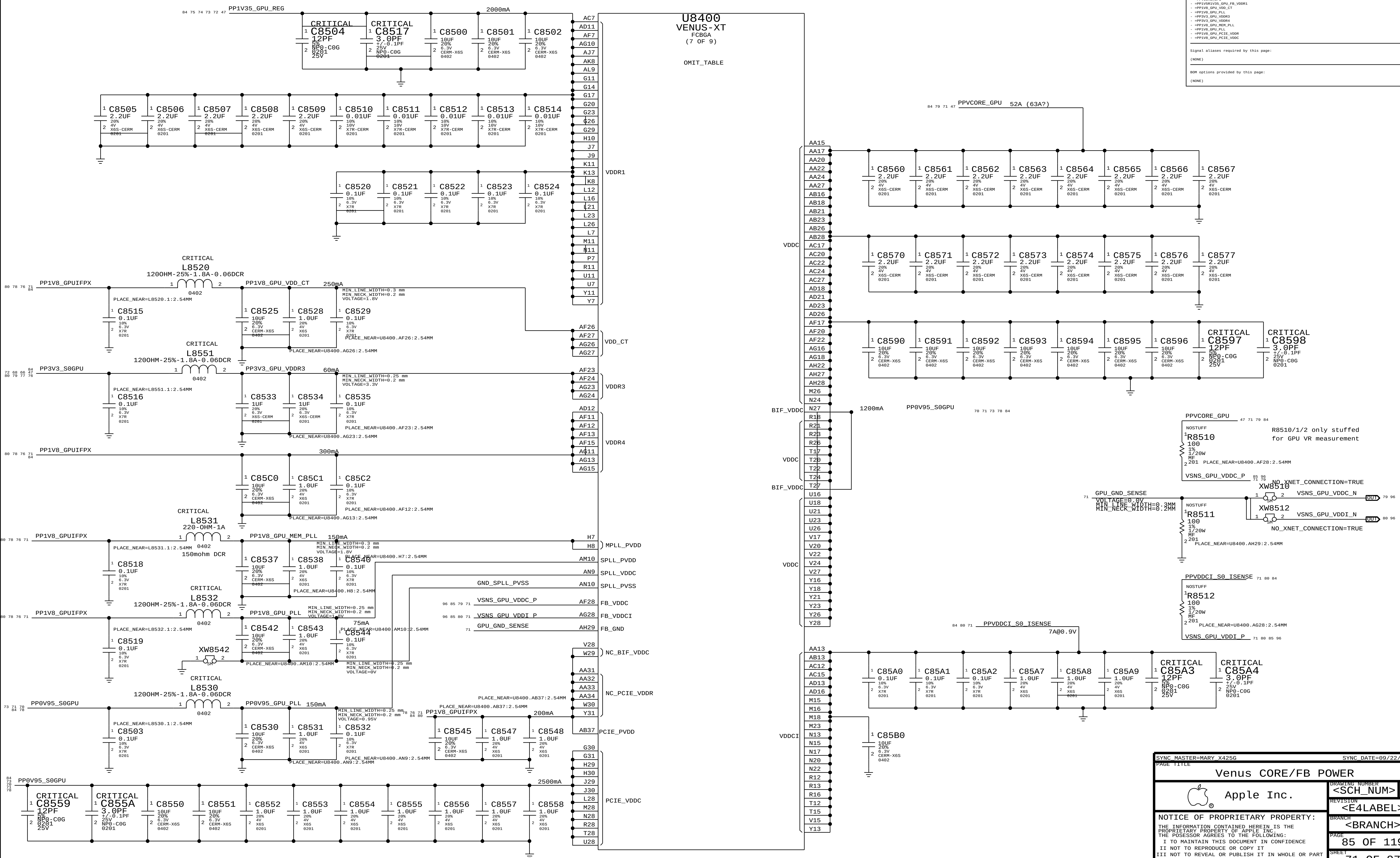
- =PPVCORE_GPU
- =PP1V5R1V35_GPU_FB_VDDR1
- =PP1V8_GPU_VDD_CT
- =PP1V0_GPU_PLL
- =PP3V3_GPU_VDDR3
- =PP3V3_GPU_VDDR4
- =PP1V8_GPU_MEM_PLL
- =PP1V8_GPU_PLL
- =PP1V8_GPU_PCIE_VDDR
- =PP1V8_GPU_PCIE_VDDC

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)



PAGE TITLE

Venus CORE/FB POWER

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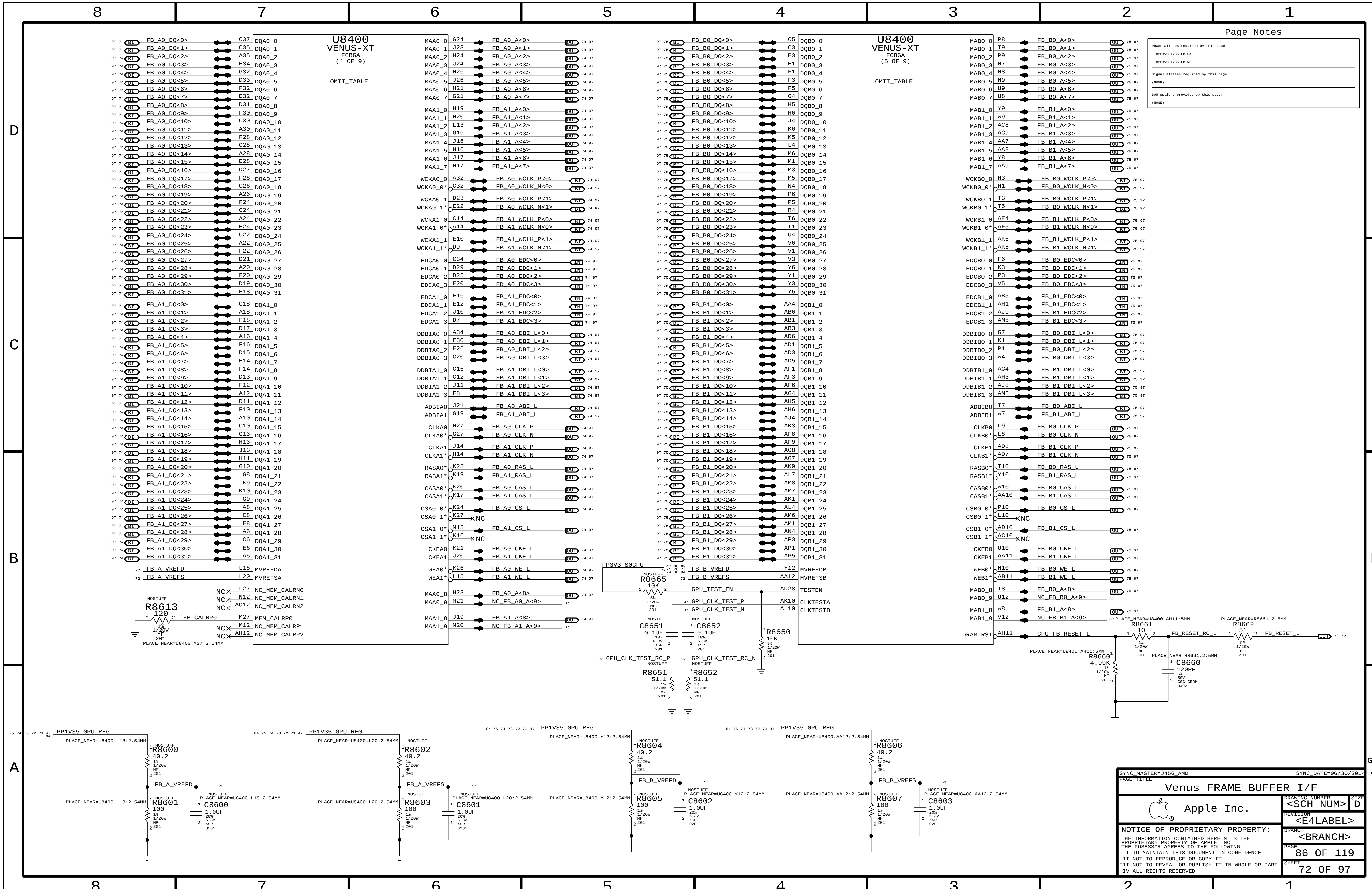
U8400
VENUS-XT
FCBGA
(4 OF 9)
OMIT_TABLE
U8400
VENUS-XT
FCBGA
(5 OF 9)
OMIT_TABLE
Page Notes
Power aliases required by this page:
- =PP1V5R1V35_FB_CAL
- =PP1V5R1V35_FB_REF
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
(NONE)
PP3V3_S0GPU
PP1V35_GPU_REG
FB_A_VREFD
FB_A_VREFS
FB_B_VREFD
FB_B_VREFS
GPU_TEST_EN
GPU_CLK_TEST_P
GPU_CLK_TEST_N
GPU_CLK_TEST_RC_P
GPU_CLK_TEST_RC_N
GPU_FB_RESET_L
FB_RESET_RC_L
FB_RESET_L
FB_CALRP0
SYNC_MASTER=J45G_AMD
SYNC_DATE=06/30/2014
PAGE TITLE
Venus FRAME BUFFER I/F
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
86 OF 119
SHEET
72 OF 97
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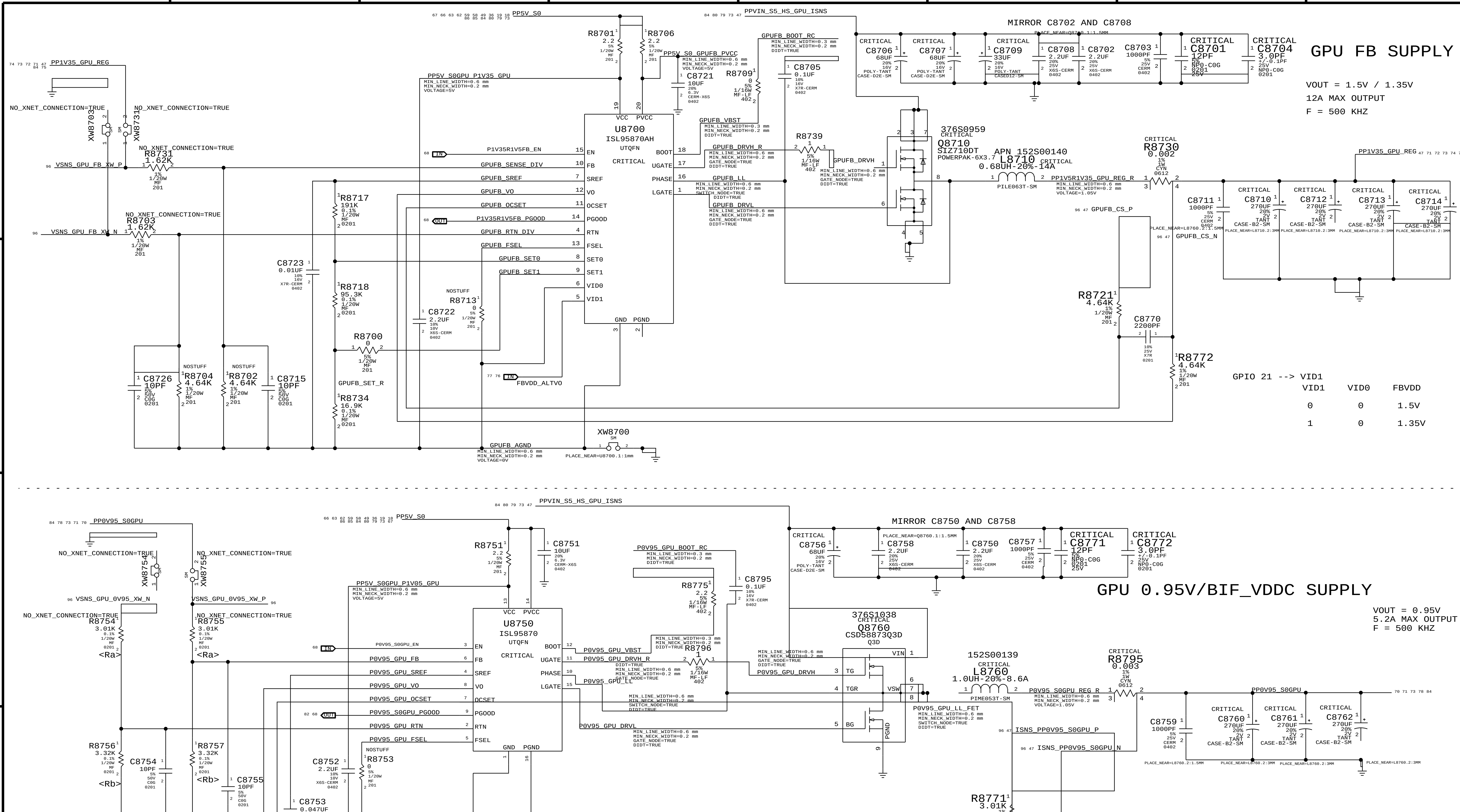
GPU



SYNC_MASTER=ADITYA_X425G SYNC_DATE=09/16/2014

PAGE TITLE

0V95 GPU / 1V35 FB Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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Page Notes

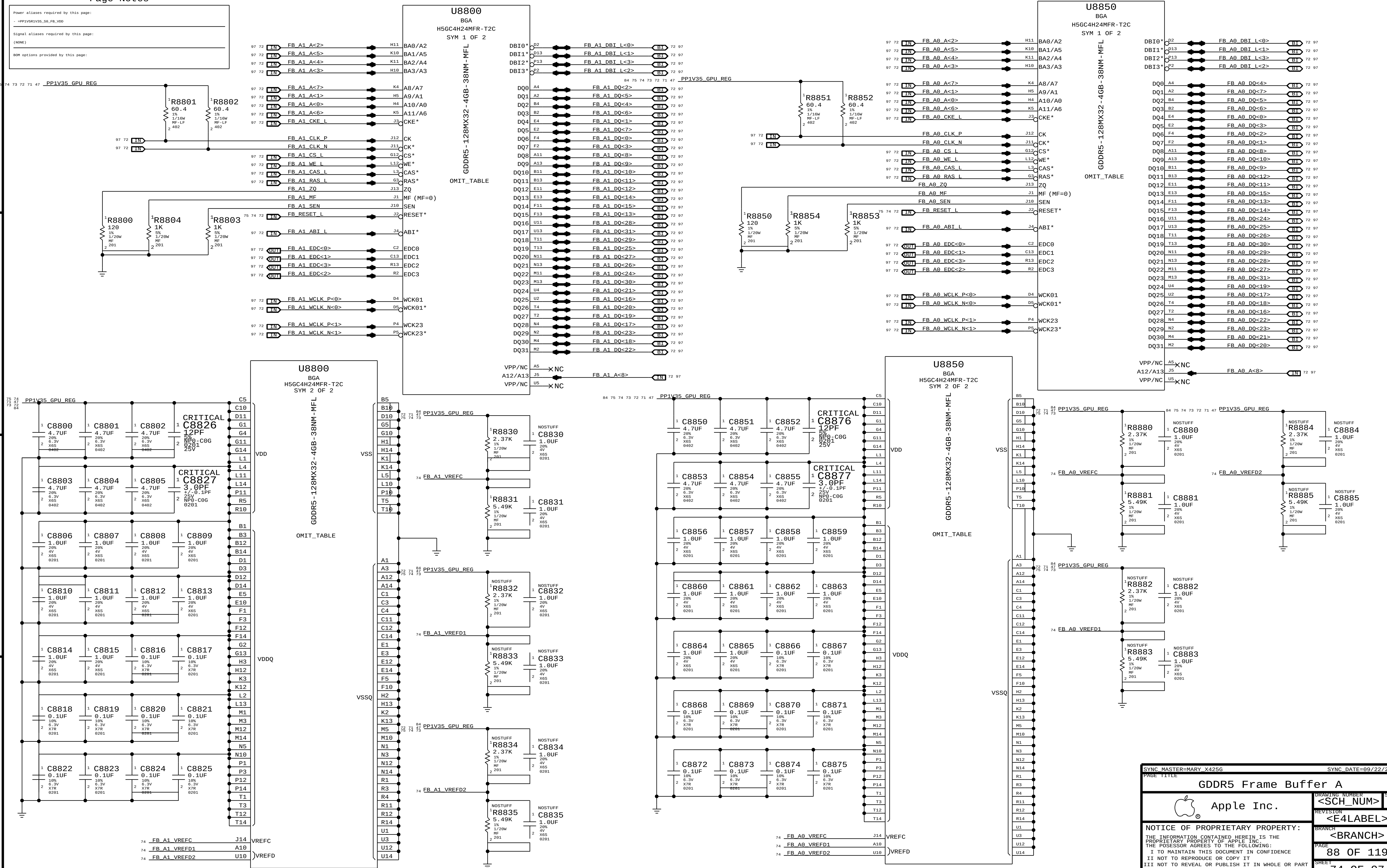
Power aliases required by this page:

- =PP1V5R1V35_S0_FB_VDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:



SYNC_MASTER=MARY_X425G

SYNC_DATE=09/22/2014

PAGE TITLE

GDDR5 Frame Buffer A

GPU

Apple Inc.

DRAWING NUMBER <SCH_NUM>

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REVISION <E4LABEL>

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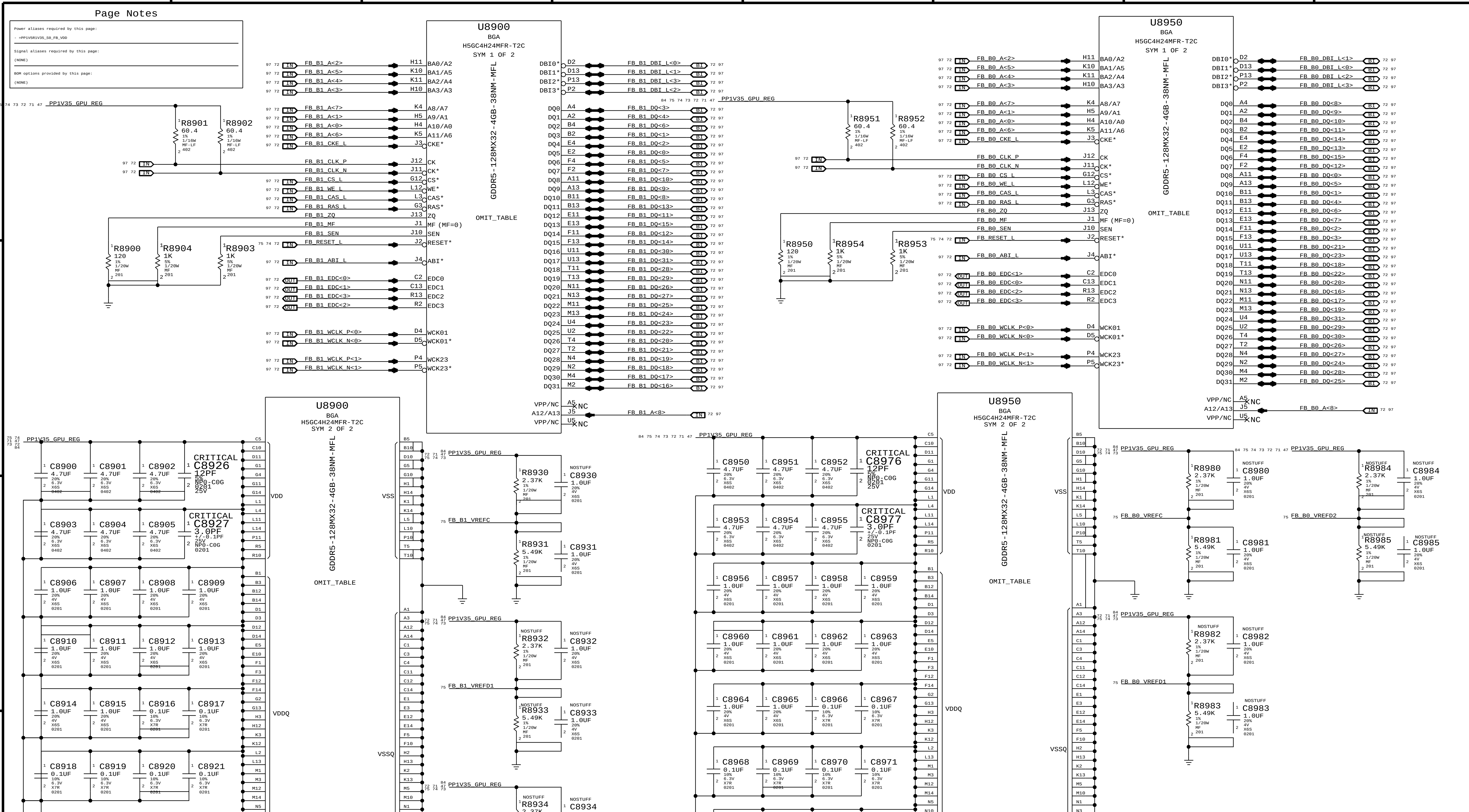

SYNC_MASTER=MARY_X425G SYNC_DATE=09/22/2014

PAGE TITLE

GDDR5 Frame Buffer B

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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GPU

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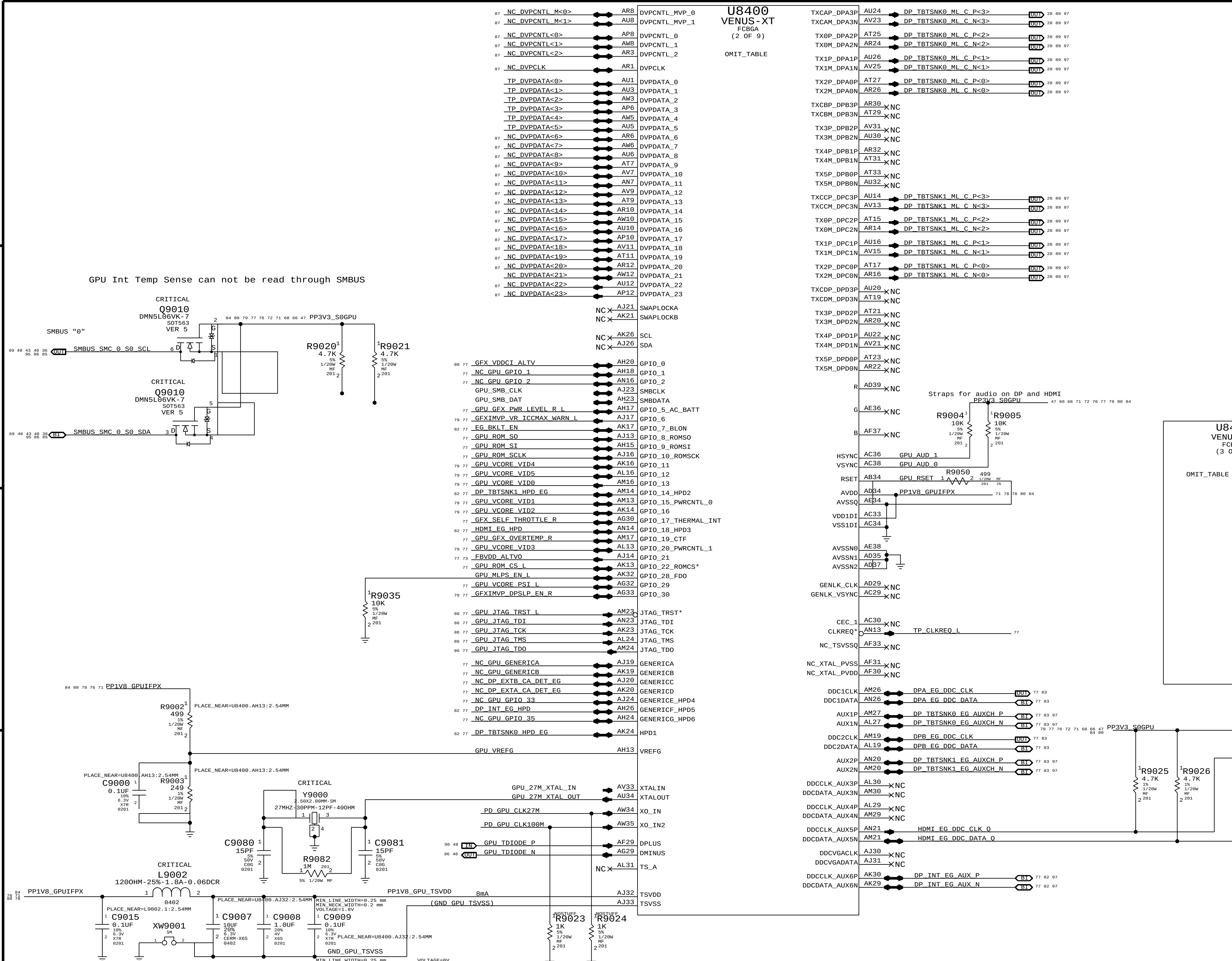

SYNC_MASTER=MARY_X425G SYNC_DATE=09/22/2014

PAGE TITLE

Venus HDMI/DP/GPIO

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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SHEET 76 OF 97

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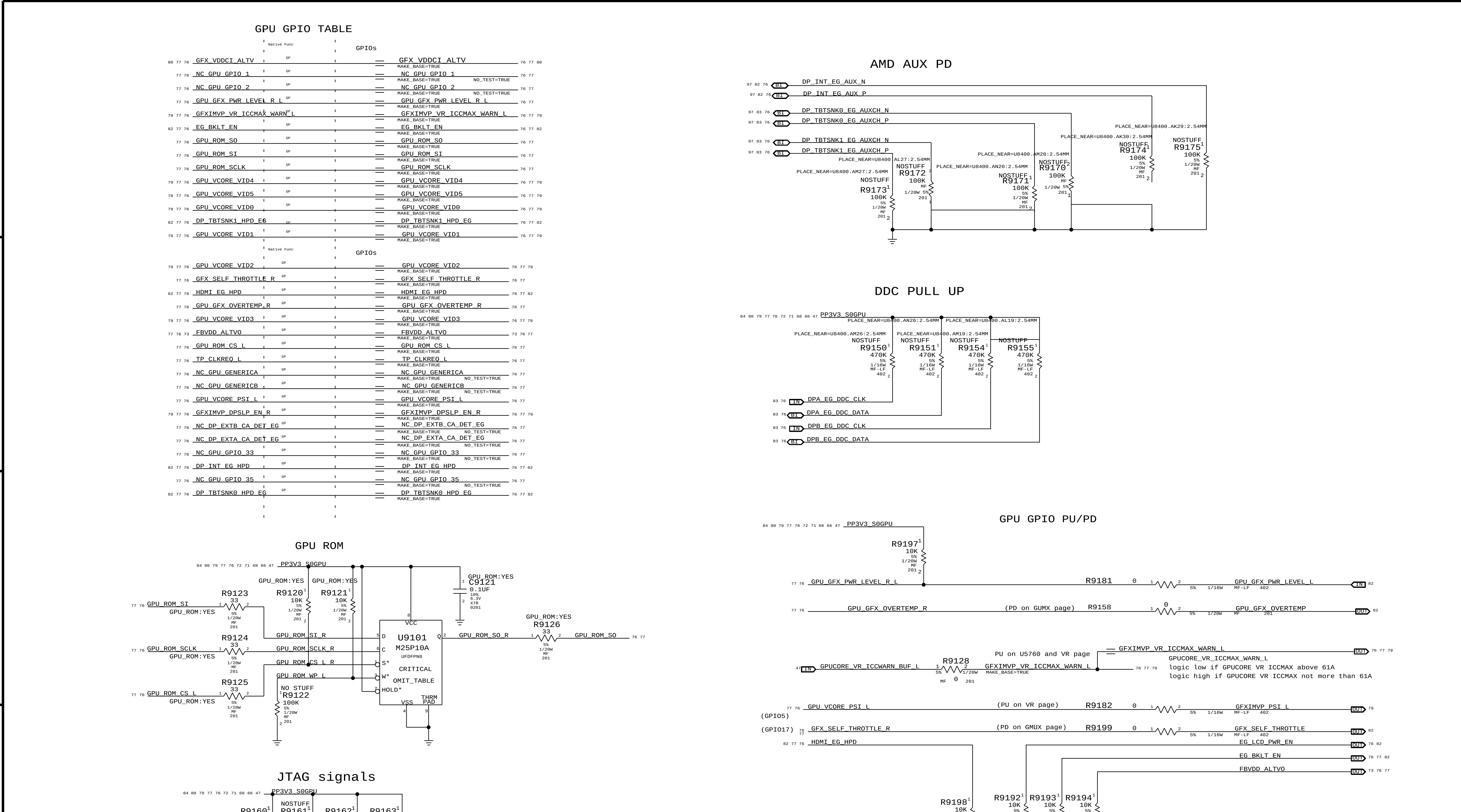

SYNC_MASTER=MARY_X425G SYNC_DATE=11/07/2014

PAGE TITLE: Venus GPIOs & STRAPs

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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Page Notes

Power aliases required by this page:

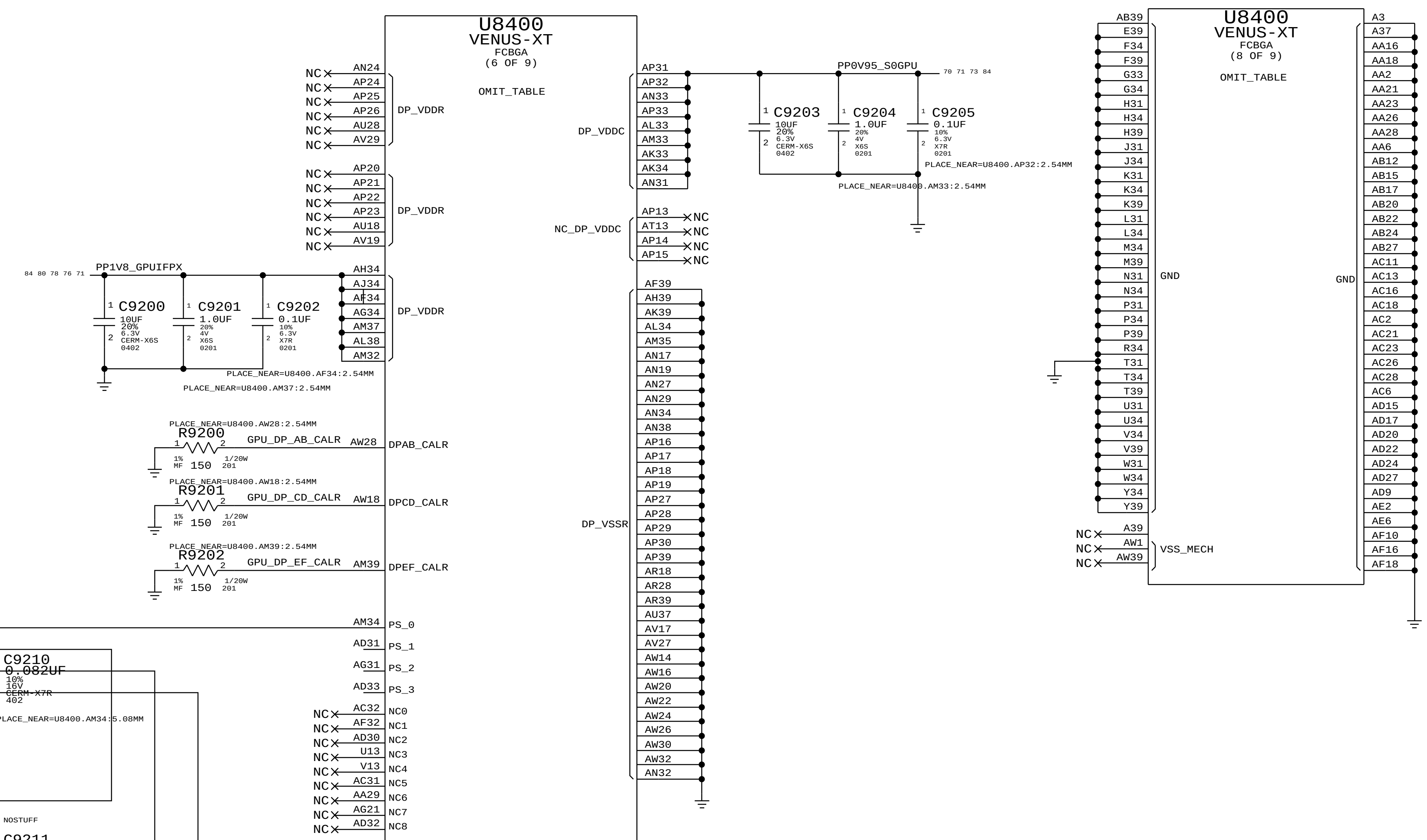
- =PP1V8_GPU_DP_AB
- =PP1V8_GPU_DP_CD
- =PP1V8_GPU_DP_EF
- =PP1V0_GPU_DP_AB
- =PP1V0_GPU_DP_CD
- =PP1V0_GPU_DP_EF

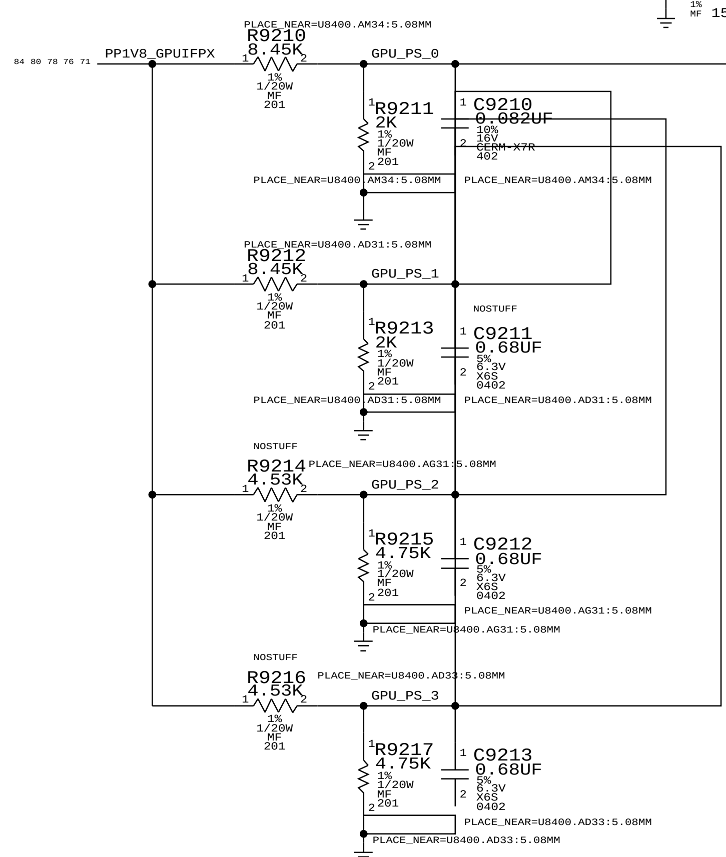
Signal aliases required by this page:

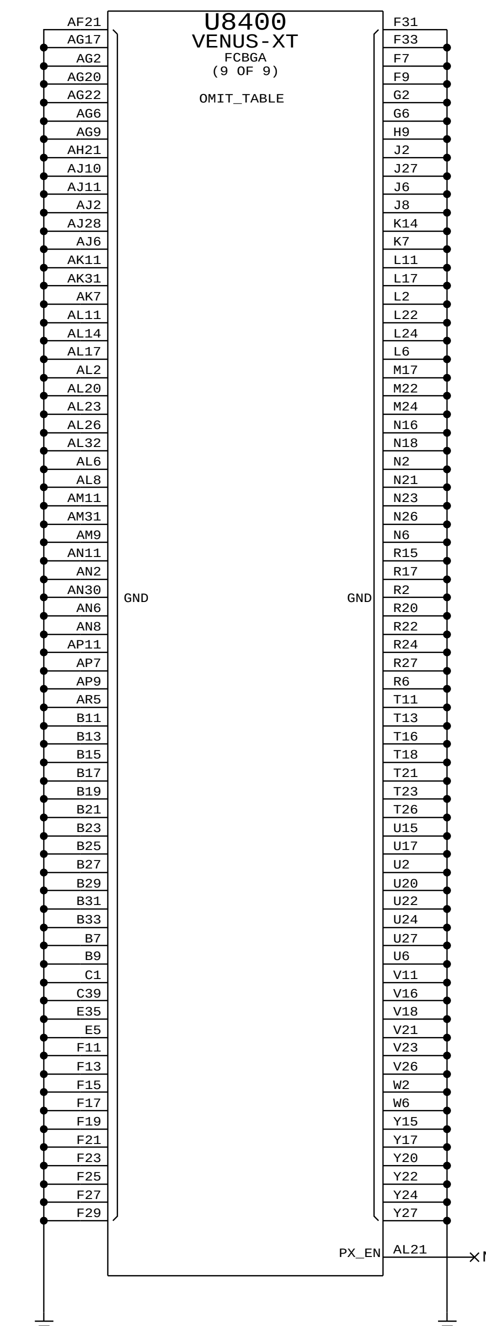
(NONE)

BOM options provided by this page:

(NONE)







CONFIG STRAPS - MLPS

PCIe Gen3 Enabled, Full-Swing, TX De-Emp Enabled

VBIOS Disabled, Boot from EFI

VGA Enabled

All Ports Audio-Capable Display Output

256MB FB Aperture Size

EFI BOOT (NO ROM)

PS_0: R9120=8.45K,R9211=2K,C9210=0.082uF

PS_1: R9122=8.45K,R9213=2K,C9211NOSTUFF

PS_2: R9214 NOSTUFF,R9215=4.75K,C9213=0.68uF

PS_2: R9216 NOSTUFF,R9217=4.75K,C9213=0.68uF

GPIO_28 pull low at power up

PS_0	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	8.45K	2K	0.082
ROM BOOT	3.24K	5.62K	0.082

PS_1	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	8.45K	2K	NC
ROM BOOT	8.45K	2K	NC

PS_2	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	NC	4.75K	0.68
ROM BOOT	4.53K	4.99K	0.68

PS_3	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	NC	4.75K	0.68
ROM BOOT	NC	4.75K	0.68

GPU

SYNC_MASTER=MARY_X425G SYNC_DATE=09/22/2014

PAGE TITLE

Venus DP PWR/GNDs

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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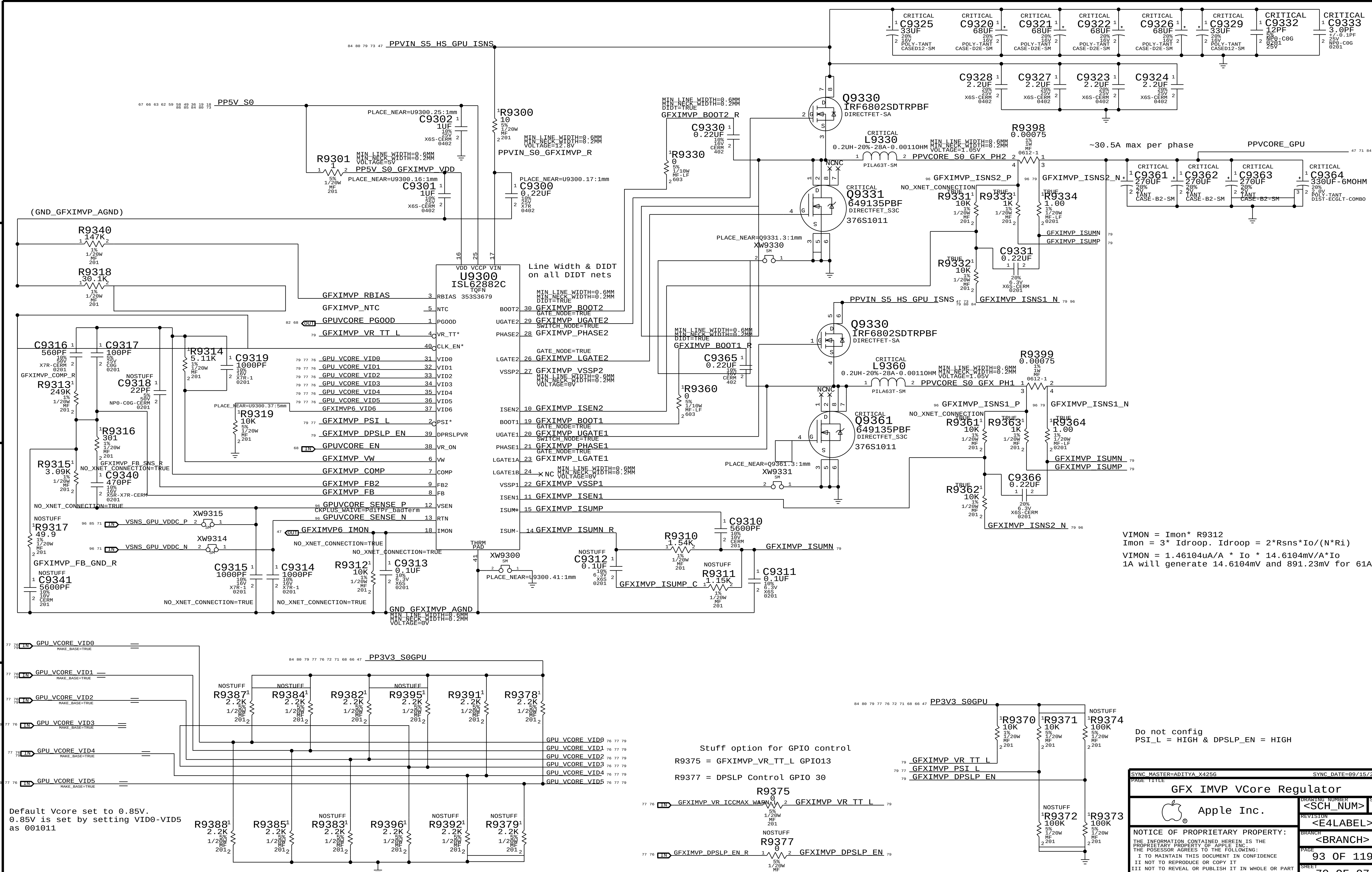

SYNC_MASTER=ADITYA_X425G SYNC_DATE=09/15/2014

PAGE TITLE

GFX IMVP VCore Regulator

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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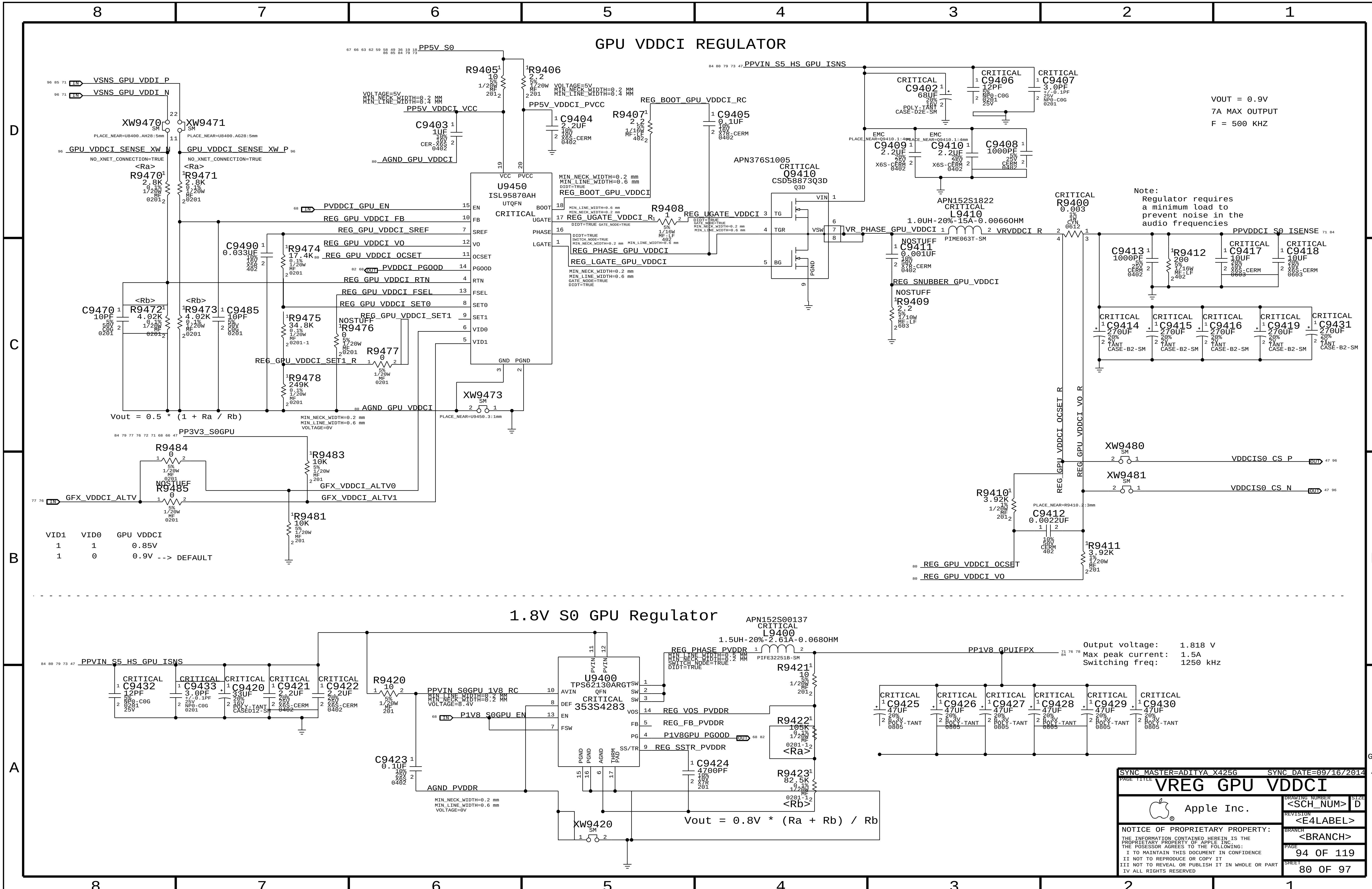
GPU VDDCI REGULATOR
VOUT = 0.9V
7A MAX OUTPUT
F = 500 KHZ
PP5V_S0
PPVIN_S5_HS_GPU_ISNS
VSNS_GPU_VDDI_P
VSNS_GPU_VDDI_N
GPU_VDDCI_SENSE_XW_N
GPU_VDDCI_SENSE_XW_P
U9450
ISL95870AH
UTQFN
CRITICAL
PVDDCI_GPU_EN
REG_GPU_VDDCI_FB
REG_GPU_VDDCI_SREF
REG_GPU_VDDCI_VO
REG_GPU_VDDCI_OCSET
PVDDCI_PGOOD
REG_GPU_VDDCI_RTN
REG_GPU_VDDCI_FSEL
REG_GPU_VDDCI_SET0
REG_GPU_VDDCI_SET1
REG_GPU_VDDCI_SET1_R
AGND_GPU_VDDCI
PP5V_VDDCI_VCC
PP5V_VDDCI_PVCC
REG_BOOT_GPU_VDDCI_RC
REG_BOOT_GPU_VDDCI
REG_UGATE_VDDCI_R
REG_UGATE_VDDCI
REG_PHASE_GPU_VDDCI
REG_LGATE_GPU_VDDCI
APN376S1005
CRITICAL
Q9410
CSD58873Q3D
APN152S1822
CRITICAL
L9410
1.0UH-20%-15A-0.0066OHM
VR_PHASE_GPU_VDDCI
VRVDDCI_R
PPVDDCI_S0_ISENSE
REG_SNUBBER_GPU_VDDCI
Note:
Regulator requires
a minimum load to
prevent noise in the
audio frequencies
Vout = 0.5 * (1 + Ra / Rb)
PP3V3_S0GPU
GFX_VDDCI_ALTV
GFX_VDDCI_ALTV0
GFX_VDDCI_ALTV1
VID1 VID0 GPU VDDCI
1 1 0.85V
1 0 0.9V --> DEFAULT
REG_GPU_VDDCI_OCSET_R
REG_GPU_VDDCI_VO_R
VDDCIS0_CS_P
VDDCIS0_CS_N
REG_GPU_VDDCI_OCSET
REG_GPU_VDDCI_VO
1.8V S0 GPU Regulator
APN152S00137
CRITICAL
L9400
1.5UH-20%-2.61A-0.068OHM
PP1V8_GPUIFPX
Output voltage: 1.818 V
Max peak current: 1.5A
Switching freq: 1250 kHz
PPVIN_S5_HS_GPU_ISNS
U9400
TPS62130ARGT
QFN
CRITICAL
353S4283
PPVIN_S0GPU_1V8_RC
P1V8_S0GPU_EN
REG_PHASE_PVDDR
REG_VOS_PVDDR
REG_FB_PVDDR
P1V8GPU_PGOOD
REG_SSTR_PVDDR
AGND_PVDDR
Vout = 0.8V * (Ra + Rb) / Rb
SYNC_MASTER=ADITYA_X425G SYNC_DATE=09/16/2014
VREG GPU VDDCI
Apple Inc.
<SCH_NUM>
D
<E4LABEL>
<BRANCH>
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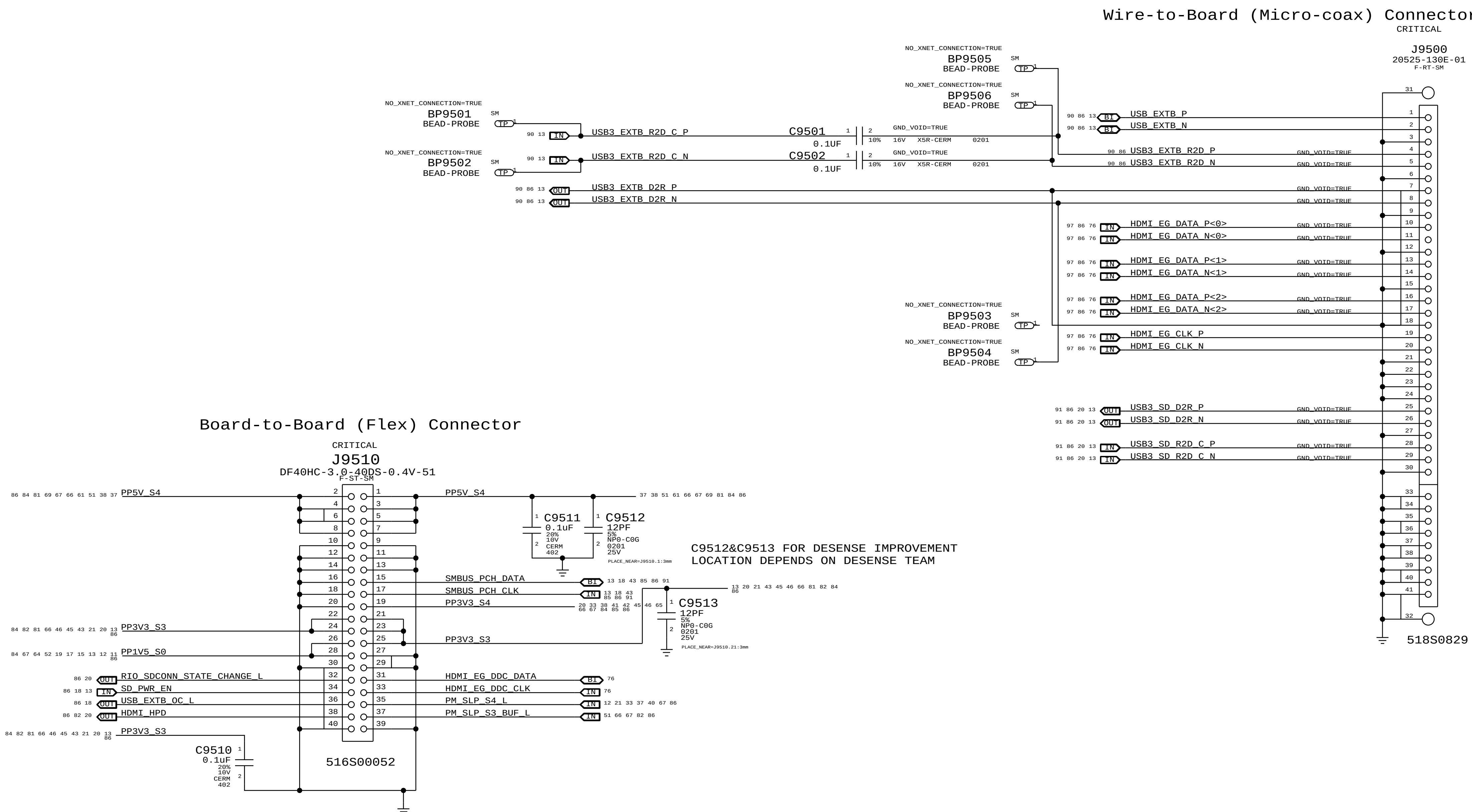

SYNC_MASTER=CLEAN_MAXWELL SYNC_DATE=07/01/2014

PAGE TITLE

RIO Connectors

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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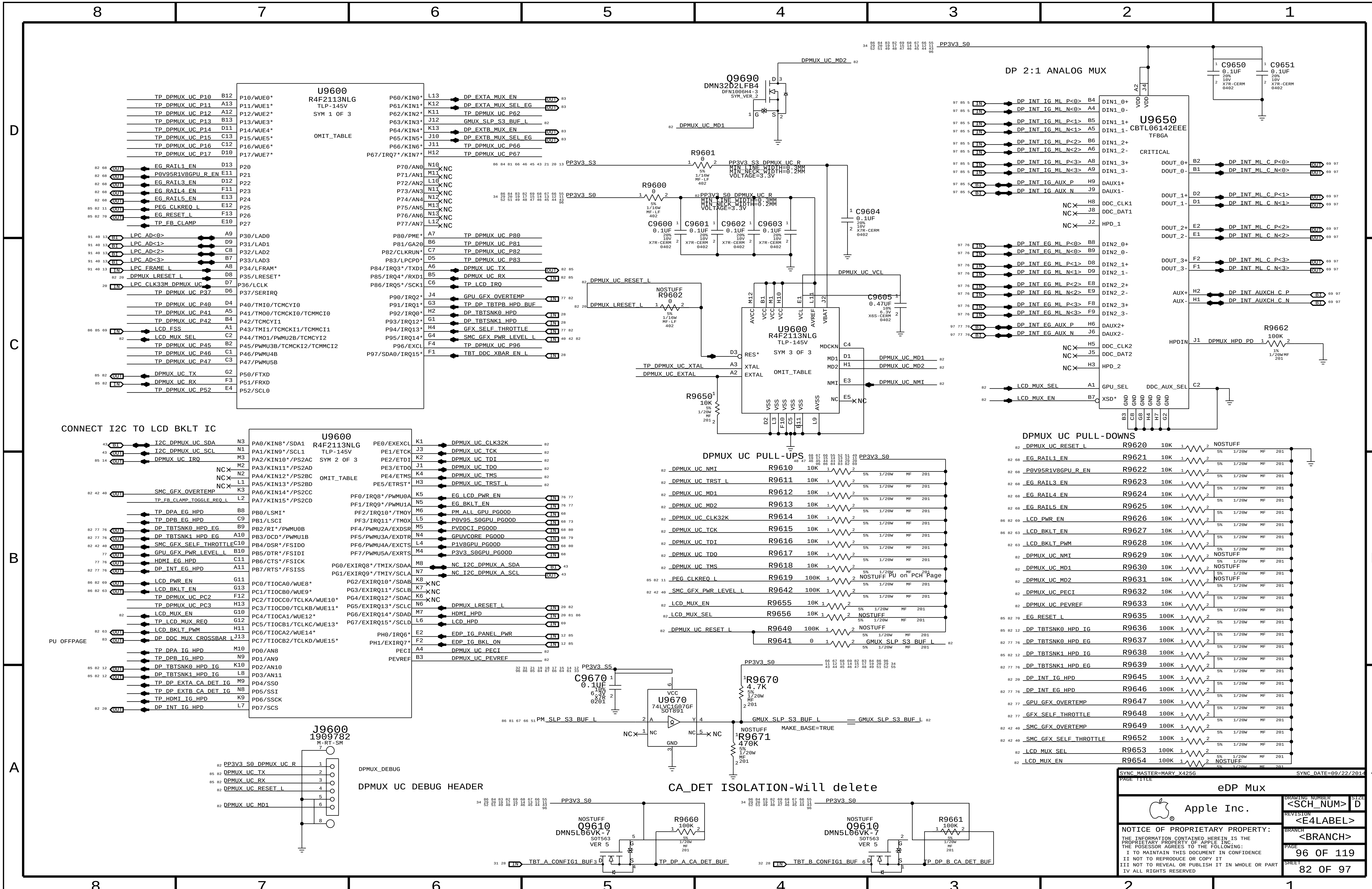

SYNC_MASTER=MARY_X425G SYNC_DATE=09/22/2014

PAGE TITLE

eDP Mux

Apple Inc.

DRAWING NUMBER <SCH_NUM>

SIZE D

REVISION <E4LABEL>

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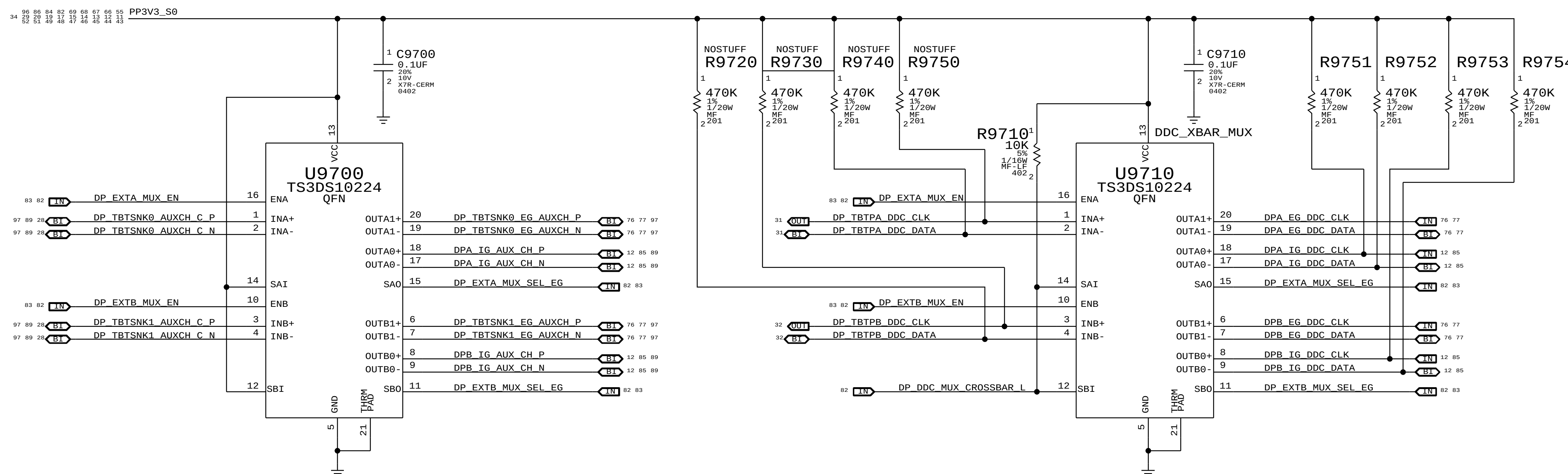
DP A & DP B AUX MUX

DP A & DP B DDC MUX



MUX TRUTH TABLE

SAI/SBI	SAO	SBO	INA	INB
0	0	0	OUTB0	OUTA0
0	0	1	OUTB1	OUTA0
0	1	0	OUTB0	OUTA1
0	1	1	OUTB1	OUTA1
1	0	0	OUTA0	OUTB0
1	0	1	OUTA0	OUTB1
1	1	0	OUTA1	OUTB0
1	1	1	OUTA1	OUTB1

SYNC_MASTER=MARY_X425G SYNC_DATE=10/15/2014

PAGE TITLE
eDP Muxed Graphics Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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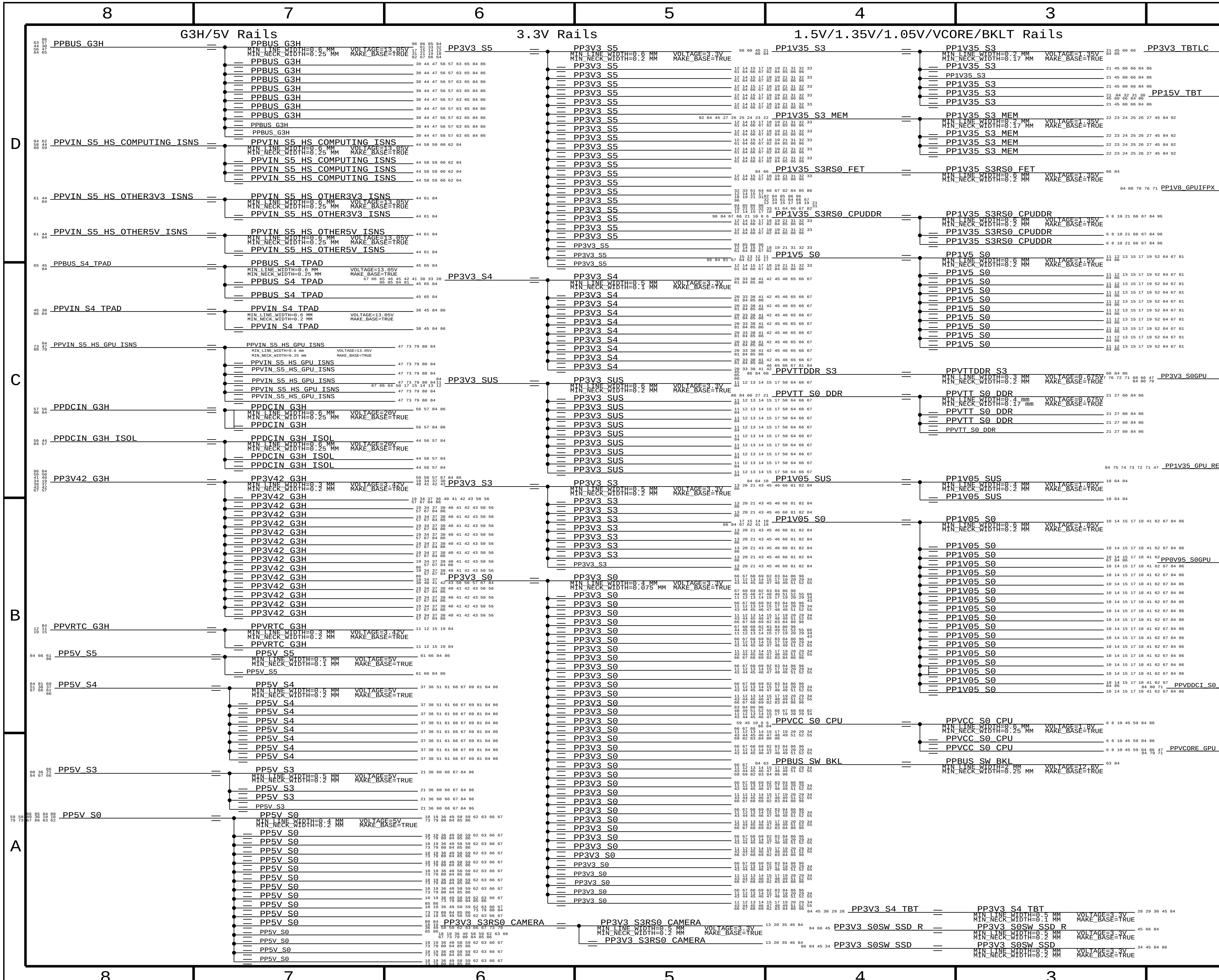

SYNC_MASTER=CLEAN_X305 SYNC_DATE=05/30/2014

PAGE TITLE

Power Aliases

DRAWING NUMBER <SCH_NUM> SIZE D

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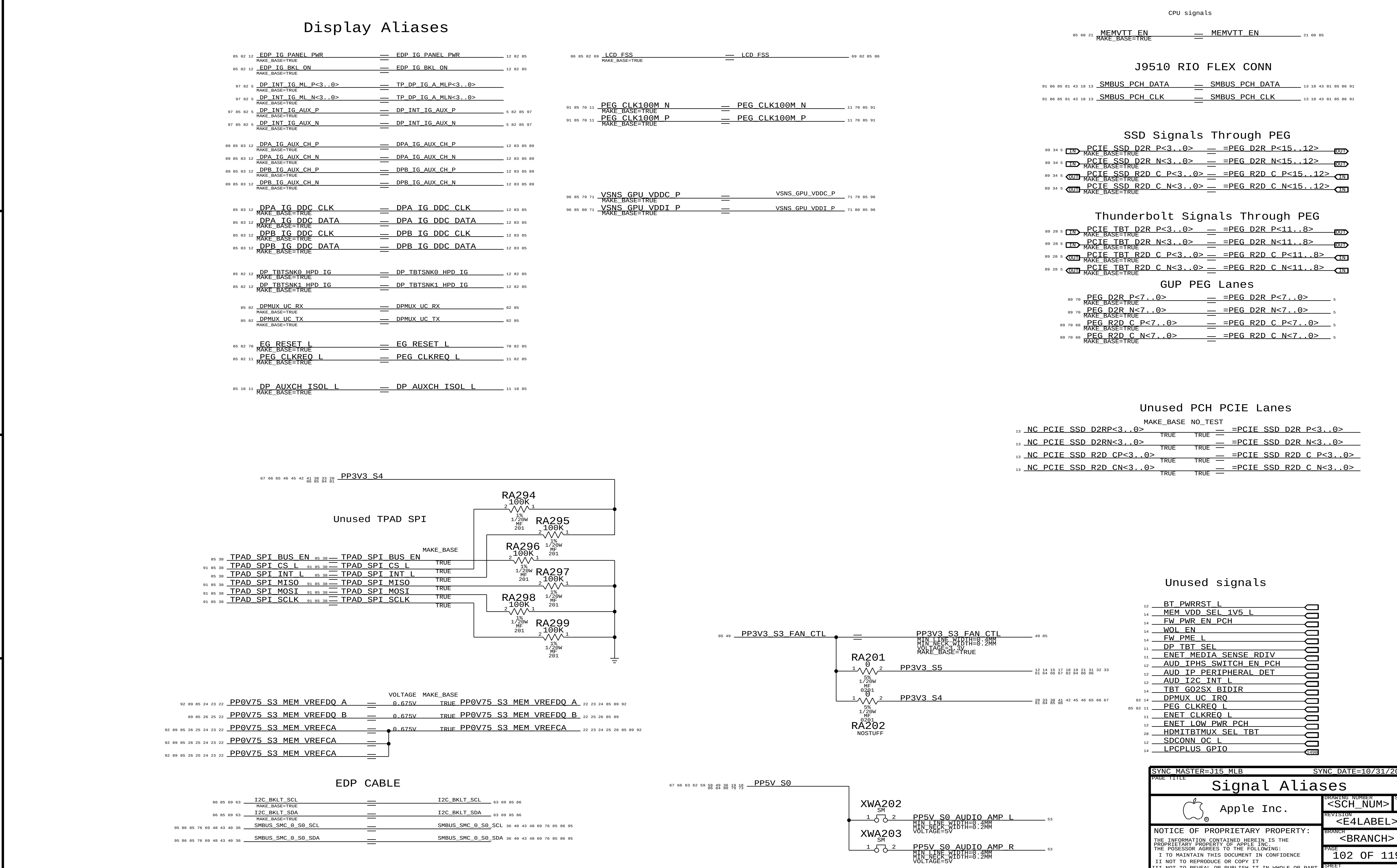

SYNC_MASTER=J15_MLB SYNC_DATE=10/31/2012

PAGE TITLE

Signal Aliases

DRAWING NUMBER <SCH_NUM> SIZE D

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Functional Test Points

FUNC_TEST J3501 - airport

AP_CLKREQ_Q_L	33
AP_RESET_CONN_L	33
PCIE_AP_D2R_PI_N	91
PCIE_AP_D2R_PI_P	91
PCIE_AP_R2D_N	33 91
PCIE_AP_R2D_P	33 91
PCIE_CLK100M_AP_CONN_N	33 91
PCIE_CLK100M_AP_CONN_P	33 91
PCIE_WAKE_L	12 33 35 91
PP3V3_S3RS4_BT_F	33
PP3V3_WLAN	33 41
USB_BT_CONN_N	33 90
USB_BT_CONN_P	33 90
WIFI_EVENT_L	33 40 41
GND	4X

J4002 - Camera

MIPI_CLK_CONN_N	36 94
MIPI_CLK_CONN_P	36 94
CAM_SENSOR_WAKE_L_CONN	36
MIPI_DATA_CONN_N	36 94
MIPI_DATA_CONN_P	36 94
SMBUS_SMC_0_S0_SDA	36 40 43 48 69 76 85 86 95
SMBUS_SMC_0_S0_SCL	36 40 43 48 69 76 85 86 95
I2C_CAM_SCK	35 36
I2C_CAM_SDA	35 36
PP5V_S3RS0_ALSCAM_F	36
GND	

J9500 - rio coax

HDMI_EG_CLK_N	76 81 97
HDMI_EG_CLK_P	76 81 97
HDMI_EG_DATA_N<0>	76 81 97
HDMI_EG_DATA_N<1>	76 81 97
HDMI_EG_DATA_N<2>	76 81 97
HDMI_EG_DATA_P<0>	76 81 97
HDMI_EG_DATA_P<1>	76 81 97
HDMI_EG_DATA_P<2>	76 81 97
USB3_SD_D2R_N	13 20 81 91
USB3_SD_D2R_P	13 20 81 91
USB3_SD_R2D_C_N	13 20 81 91
USB3_SD_R2D_C_P	13 20 81 91
USB3_EXTB_D2R_N	13 81 90
USB3_EXTB_D2R_P	13 81 90
USB3_EXTB_R2D_N	81 90
USB3_EXTB_R2D_P	81 90
USB_EXTB_N	13 81 90
USB_EXTB_P	13 81 90
GND	19X

J9510 - rio flex

SD_PWR_EN	13 18 81
HDMI_DDC_CLK	
HDMI_DDC_DATA	
HDMI_HPD	20 81 82
SMBUS_PCH_CLK	13 18 43 81 85 91
SMBUS_PCH_DATA	13 18 43 81 85 91
PM_SLP_S3_BUF_L	51 66 67 81 82
PM_SLP_S4_L	12 21 33 37 40 67 81
PP3V3_S3	3X
PP3V3_S4	
PP5V_S4	5X
RIO_SDCONN_STATE_CHANGE_L	20 81
USB_EXTB_OC_L	18 81
GND	10X

J5150 - hall effect

PP3V42_G3H	
SMC_LID_R	42
GND	

J6050 - left fan

FAN_LT_PWM	49
FAN_LT_TACH	49
PP5V_S0	3X
GND	5X

J6060 - right fan

FAN_RT_PWM	49
FAN_RT_TACH	49
PP5V_S0	3X
GND	5X

FUNC_TEST J6100 - spi

PP3V42_G3H	
SMC_RESET_L	40 41 50 57
SMC_TCK	40 41 50
SMC_TMS	40 41 50
SPIROM_USE_MLB	14 50
GND	2X

J4801 - ipd flex

USB_TPAD_N	13 38 90
USB_TPAD_P	13 38 90
IOXP2_INT_L	38
I2C_IOXP_SCL	38
I2C_IOXP_SDA	38
SMC_PME_S4_WAKE_L	33 38 40 42
TPAD_ACTUATOR_THRMTRIP_L	38 65
TPAD_VBUS_EN	38 67
SMBUS_SMC_2_S3_SCL	38 40 43 95
SMBUS_SMC_2_S3_SDA	38 40 43 95
SMC_LID	38 40 41 42
SMC_ACTUATOR_EN_L	38 40
PPVIN_S4_TPAD	4X
GND_ACTUATOR	4X 38
PP3V3_S4	
PP5V_S4	
GND	2X

J4813 - keyboard

PP3V3_S4	
PP3V42_G3H	
WS_CONTROL_KBD	38
WS_KBD1	38
WS_KBD10	38
WS_KBD11	38
WS_KBD12	38
WS_KBD13	38
WS_KBD14	38
WS_KBD15_CAP	38
WS_KBD16_NUM	38
WS_KBD17	38
WS_KBD18	38
WS_KBD19	38
WS_KBD2	38
WS_KBD20	38
WS_KBD21	38
WS_KBD22	38
WS_KBD23	38
WS_KBD3	38
WS_KBD4	38
WS_KBD5	38
WS_KBD6	38
WS_KBD7	38
WS_KBD8	38
WS_KBD9	38
WS_KBD_ONOFF_L	38
WS_LEFT_OPTION_KBD	38
WS_LEFT_SHIFT_KBD	38
GND	2X

J4915 - kbd bklt

KBDBKLT_RETURN1	2X 39 63
KBDBKLT_RETURN2	2X 39 63
PPVOUT_S0_KBDBKLT	39 63
GND	4X

J6601 - mic

DMIC_CLK3	52 55
PP3V3_S0	
DMIC_SDA2	55
DMIC_SDA3	52 55
GND	

J6602 - L speaker

SPKRCONN_L_ID	52 55
SPKRCONN_L_OUT_N	53 55 96
SPKRCONN_L_OUT_P	53 55 96
SPKRCONN_SL_OUT_N	53 55 96
SPKRCONN_SL_OUT_P	53 55 96
GND	

J6603 - R speaker

SPKRCONN_R_ID	52 55
SPKRCONN_R_OUT_N	53 55 96
SPKRCONN_R_OUT_P	53 55 96
SPKRCONN_SR_OUT_N	53 55 96
SPKRCONN_SR_OUT_P	53 55 96
GND	

J7000 - DC PWR

ADAPTER_SENSE	56
PP20V_DCIN_FUSE	2X 56
GND	2X

J7050 - battery

PPVBAT_G3H_CONN	8X 56 57
SMBUS_SMC_5_G3_SCL	40 43 56 57 95
SMBUS_SMC_5_G3_SDA	40 43 56 57 95
SYS_DETECT_L	56
GND	8X

J8300 - eDP

DP_INT_AUX_N	69 97
DP_INT_AUX_P	69 97
DP_INT_ML_N<0>	69 97
DP_INT_ML_N<1>	69 97
DP_INT_ML_N<2>	69 97
DP_INT_ML_N<3>	69 97
DP_INT_ML_P<0>	69 97
DP_INT_ML_P<1>	69 97
DP_INT_ML_P<2>	69 97
DP_INT_ML_P<3>	69 97
LCD_FSS	69 82 85
LCD_HPD_CONN	69
LCD_BKLT_PWM_R	63 69
SMBUS_SMC_0_S0_SDA	36 40 43 48 69 76 85 86 95
SMBUS_SMC_0_S0_SCL	36 40 43 48 69 76 85 86 95
I2C_BKLT_SDA	63 69 85
I2C_BKLT_SCL	63 69 85
PP5VR3V3_SW_LCD	3X 69
PPVOUT_S0_LCDBKLT	63 69
GND	16X

Power Rails

PM_SLP_S3_L	12 21 40 67
PPVTT_S0_DDR	21 27 60 84
PP3V3_S0	
PP3V3_S3	
PP3V3_S5	
PP3V3_S5_AVREF_SMC	40 41
PP3V42_G3H	
PP5V_S0	
PP5V_S3	21 36 60 66 67 84
PP5V_S5	61 66 84
PPBUS_G3H	30 44 47 56 57 63 65 84
PPDCIN_G3H	56 57 84
PPVCC_S0_CPU	6 8 10 45 59 84
PPVTTDDR_S3	60 84
PP3V3_S0SW_SSD	34 45 84
PP1V5_S0	
PP1V35_S3	21 45 60 66 84

FUNC_TEST XDP

XDP_CPU_TCK	6 18 89
XDP_PCH_TCK	11 18
XDP_CPU_TDI	6 18 89
XDP_CPU_TDO	6 18 89
XDP_CPUPCH_TRST_L	6 18 89
XDP_CPU_TMS	6 18 89
XDP_PCH_TMS	11 18
XDP_PCH_TDI	11 18
XDP_PCH_TDO	11 18
XDP_CPU_PREQ_L	6 18 89
XDP_CPU_PRDY_L	6 18 89
PM_RSMRST_L	12 67 91
PM_PCH_PWROK	12 19 91
PM_SYSRST_L	12 19 40 91
CPU_CFG<3>	6 18 89
PP1V05_S0	10 14 15 17 18 41 62 67 84
GND	2X GND

FUNC_TEST Power Sequence

SMC_ONOFF_L	38 40 41
PM_DSW_PWRGD	12 40 91
ALL_SYS_PWRGD	18 19 40 58 67
PM_PCH_SYS_PWROK	12 18 19 40 91
PLT_RESET_L	12 18 20 21
LCD_PWR_EN	69 82
LCD_BKLT_EN	63 82

FUNC_TEST GPU_VENUS JTAG

GPU_JTAG_TCK	76 77
GPU_JTAG_TDI	76 77
GPU_JTAG_TDO	76 77
GPU_JTAG_TMS	76 77
GPU_JTAG_TRST_L	76 77
GPU_PWRGOOD	

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NC NO_TESTs

PCH

Net	NO_TEST	MAKE_BASE
NC_USB3_SPARE_D2RN	TRUE	TRUE
NC_USB3_SPARE_D2RP	TRUE	TRUE
NC_USB3_SPARE_R2D_CN	TRUE	TRUE
NC_USB3_SPARE_R2D_CP	TRUE	TRUE
NC_USB3_EXTC_D2RN	TRUE	TRUE
NC_USB3_EXTC_D2RP	TRUE	TRUE
NC_USB3_EXTC_R2D_CN	TRUE	TRUE
NC_USB3_EXTC_R2D_CP	TRUE	TRUE
NC_USB3_EXTD_D2RN	TRUE	TRUE
NC_USB3_EXTD_D2RP	TRUE	TRUE
NC_USB3_EXTD_R2D_CN	TRUE	TRUE
NC_USB3_EXTD_R2D_CP	TRUE	TRUE
NC_PCIE_ENET_D2RN	TRUE	TRUE
NC_PCIE_ENET_D2RP	TRUE	TRUE
NC_PCIE_ENET_R2D_CN	TRUE	TRUE
NC_PCIE_ENET_R2D_CP	TRUE	TRUE
NC_SATA_A_D2RN	TRUE	TRUE
NC_SATA_A_D2RP	TRUE	TRUE
NC_SATA_A_R2D_CN	TRUE	TRUE
NC_SATA_A_R2D_CP	TRUE	TRUE
NC_SATA_B_D2RN	TRUE	TRUE
NC_SATA_B_D2RP	TRUE	TRUE
NC_SATA_B_R2D_CN	TRUE	TRUE
NC_SATA_B_R2D_CP	TRUE	TRUE
NC_SATA_ODD_D2RN	TRUE	TRUE
NC_SATA_ODD_D2RP	TRUE	TRUE
NC_SATA_ODD_R2D_CN	TRUE	TRUE
NC_SATA_ODD_R2D_CP	TRUE	TRUE
NC_SATA_D_D2RN	TRUE	TRUE
NC_SATA_D_D2RP	TRUE	TRUE
NC_SATA_D_R2D_CN	TRUE	TRUE
NC_SATA_D_R2D_CP	TRUE	TRUE
NC_SATA_F_D2RN	TRUE	TRUE
NC_SATA_F_D2RP	TRUE	TRUE
NC_SATA_F_R2D_CN	TRUE	TRUE
NC_SATA_F_R2D_CP	TRUE	TRUE
NC_USB_EXTCN	TRUE	TRUE
NC_USB_EXTCP	TRUE	TRUE
NC_USB_SDN	TRUE	TRUE
NC_USB_SDP	TRUE	TRUE
NC_USB_WLANN	TRUE	TRUE
NC_USB_WLANP	TRUE	TRUE
NC_USB_6N	TRUE	TRUE
NC_USB_6P	TRUE	TRUE
NC_USB_7N	TRUE	TRUE
NC_USB_7P	TRUE	TRUE
NC_USB_EXTDN	TRUE	TRUE
NC_USB_EXTDP	TRUE	TRUE
NC_USB_PSOCN	TRUE	TRUE
NC_USB_PSOCP	TRUE	TRUE
NC_USB_IRN	TRUE	TRUE
NC_USB_IRP	TRUE	TRUE
NC_ITPXDP_CLK100MN	TRUE	TRUE
NC_ITPXDP_CLK100MP	TRUE	TRUE
NC_PCI_PME_L	TRUE	TRUE
NC_PCI_CLK33M_OUT3	TRUE	TRUE
NC_HDA_SDIN1	TRUE	TRUE
NC_HDA_SDIN2	TRUE	TRUE
NC_HDA_SDIN3	TRUE	TRUE
NC_LPC_DREQ0_L	TRUE	TRUE
NC_CLINK_CLK	TRUE	TRUE
NC_CLINK_DATA	TRUE	TRUE
NC_CLINK_RESET_L	TRUE	TRUE
NC_LPC_CLK33M_LPCPLUS_R	TRUE	TRUE
NC_EDP_IG_BKL_PWM	TRUE	TRUE
NC_USB_SMCP	TRUE	TRUE
NC_USB_SMCN	TRUE	TRUE
NC_SMC_INTERFACE_2	TRUE	TRUE

Thunderbolt

Net	NO_TEST	MAKE_BASE	Net
NC_TBT_XTAL25OUT	TRUE	TRUE	NC_TBT_XTAL25OUT
TP_DP_TBTSRC_ML_CP<3..0>	TRUE	TRUE	NC_DP_TBTSRC_ML_CP<3..0>
TP_DP_TBTSRC_ML_CN<3..0>	TRUE	TRUE	NC_DP_TBTSRC_ML_CN<3..0>
NC_DP_TBTSRC_AUXCH_CP	TRUE	TRUE	NC_DP_TBTSRC_AUXCH_CP
NC_DP_TBTSRC_AUXCH_CN	TRUE	TRUE	NC_DP_TBTSRC_AUXCH_CN
NC_DP_IG_D_AUXCHN	TRUE	TRUE	NC_DP_IG_D_AUXCHN
NC_DP_IG_D_AUXCHP	TRUE	TRUE	NC_DP_IG_D_AUXCHP
NC_PCIE_CLK100M_PE5N	TRUE	TRUE	NC_PCIE_CLK100M_PE5N
NC_PCIE_CLK100M_PE5P	TRUE	TRUE	NC_PCIE_CLK100M_PE5P
NC_PCIE_CLK100M_ENETSDN	TRUE	TRUE	NC_PCIE_CLK100M_ENETSDN
NC_PCIE_CLK100M_ENETSDP	TRUE	TRUE	NC_PCIE_CLK100M_ENETSDP
NC_PCIE_CLK100M_ENETN	TRUE	TRUE	NC_PCIE_CLK100M_ENETN
NC_PCIE_CLK100M_ENETP	TRUE	TRUE	NC_PCIE_CLK100M_ENETP
NC_PCIE_CLK100M_PEGBN	TRUE	TRUE	NC_PCIE_CLK100M_PEGBN
NC_PCIE_CLK100M_PEGBP	TRUE	TRUE	NC_PCIE_CLK100M_PEGBP
NC_PCIE_CLK100M_SWN	TRUE	TRUE	NC_PCIE_CLK100M_SWN
NC_PCIE_CLK100M_SWP	TRUE	TRUE	NC_PCIE_CLK100M_SWP
NC_PCH_GPIO64_CLKOUTFLEX0	TRUE	TRUE	NC_PCH_GPIO64_CLKOUTFLEX0
NC_PCH_GPIO65_CLKOUTFLEX1	TRUE	TRUE	NC_PCH_GPIO65_CLKOUTFLEX1
NC_PCH_GPIO66_CLKOUTFLEX2	TRUE	TRUE	NC_PCH_GPIO66_CLKOUTFLEX2
NC_PCH_GPIO67_CLKOUTFLEX3	TRUE	TRUE	NC_PCH_GPIO67_CLKOUTFLEX3
NC_USB_4N	TRUE	TRUE	NC_USB_4N
NC_USB_4P	TRUE	TRUE	NC_USB_4P
TP_DVPCNTL_M<1..0>	TRUE	TRUE	NC_DVPCNTL_M<1..0>
TP_DVPCNTL<2..0>	TRUE	TRUE	NC_DVPCNTL<2..0>
NC_DVPCLK	TRUE	TRUE	NC_DVPCLK
TP_NC_DVPDATA<23..6>	TRUE	TRUE	NC_DVPDATA<23..6>
NC_FB_B0_A<9>	TRUE	TRUE	NC_FB_B0_A<9>
NC_FB_B1_A<9>	TRUE	TRUE	NC_FB_B1_A<9>
NC_FB_A0_A<9>	TRUE	TRUE	NC_FB_A0_A<9>
NC_FB_A1_A<9>	TRUE	TRUE	NC_FB_A1_A<9>

TRUE	PCIE_TBT_R2D_P<3..0>
TRUE	PCIE_TBT_R2D_N<3..0>
TRUE	PCIE_TBT_D2R_C_P<3..0>
TRUE	PCIE_TBT_D2R_C_N<3..0>
TRUE	DMI_S2N_P<3..1>
TRUE	DMI_S2N_N<3..1>
TRUE	DMI_N2S_P<3..1>
TRUE	DMI_N2S_N<3..1>

PLACEABLE BEAD-PROBES FOR TBT

TBT_A_D2R_P<1> — TP SM BEAD-PROBE BPA531 NO_XNET_CONNECTION=TRUE

TBT_A_D2R_N<1> — TP SM BEAD-PROBE BPA532 NO_XNET_CONNECTION=TRUE

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X425G BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,P65BGA	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.058 MM	?
1x_DIELECTRIC	ISL3,ISL4,ISL9,ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2,ISL5,ISL6,ISL7,ISL8,ISL11	0.101 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

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PCB Rule Definitions

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SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?
SATA_2OTHER	*	=4X_DIELECTRIC	?
SATA_RCOMP	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
SATA_RCOMP	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	=SAME	*	SATA_2SAME
SATA_R2D	SATA_D2R	*	SATA_TXRX
SATA_*	*	*	SATA_2OTHER

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?
USB_RBIAS	*	=6X_DIELECTRIC	?
BT_WAKE	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAS	TOP,BOTTOM	=10X_DIELECTRIC	?
BT_WAKE	TOP,BOTTOM	=6X_DIELECTRIC	?

USB 3.0 INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?
USB3_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	=SAME	*	USB3_2SAME
USB3_R2D	USB3_D2R	*	USB3_TXRX
USB3_*	*	*	USB3_2OTHER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.
NOTE: Latest Intel DG calls out 50ohms SE for sys clocks

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SATA_85D	SATA_R2D	NC_SATA_A_R2D_CP	11 87
	SATA_85D	SATA_R2D	NC_SATA_A_R2D_CN	11 87
	SATA_85D	SATA_D2R	NC_SATA_A_D2RP	11 87
	SATA_85D	SATA_D2R	NC_SATA_A_D2RN	11 87
	SATA_85D	SATA_R2D	NC_SATA_B_R2D_CP	11 87
	SATA_85D	SATA_R2D	NC_SATA_B_R2D_CN	11 87
	SATA_85D	SATA_D2R	NC_SATA_B_D2RP	11 87
	SATA_85D	SATA_D2R	NC_SATA_B_D2RN	11 87
PCH_SATA_RCOMP	SATA_45SE	SATA_RCOMP	PCH_SATA_RCOMP	11
USB_EXTA	USB_85D	USB	USB_EXTA_P	13 37
USB_EXTA	USB_85D	USB	USB_EXTA_N	13 37
USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_P	37
USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_N	37
USB_EXTA	USB_85D	USB	USB_LT1_P	37
USB_EXTA	USB_85D	USB	USB_LT1_N	37
USB_NC	USB_85D	USB	NC_USB_EXTCP	13 87
USB_NC	USB_85D	USB	NC_USB_EXTCN	13 87
USB_NC	USB_85D	USB	NC_USB_SDP	13 87
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	CPU_45S	CPU_ITP	SMC_DEBUGPRT_RX_L	37 40 41
	CPU_45S	CPU_ITP	SMC_DEBUGPRT_TX_L	37 40 41
USB_SMC	USB_85D	USB	NC_USB_SMCP	87
USB_SMC	USB_85D	USB	NC_USB_SMCN	87
USB_NC	USB_85D	USB	NC_USB_6P	13 87
USB_NC	USB_85D	USB	NC_USB_6N	13 87
USB_NC	USB_85D	USB	NC_USB_7P	13 87
USB_NC	USB_85D	USB	NC_USB_7N	13 87
USB_EXTB	USB_85D	USB	USB_EXTB_P	13 81 86
USB_EXTB	USB_85D	USB	USB_EXTB_N	13 81 86
USB_NC	USB_85D	USB	NC_USB_EXTDP	13 87
USB_NC	USB_85D	USB	NC_USB_EXTDN	13 87
USB_BT	USB_85D	USB	USB_BT_P	13 33
USB_BT	USB_85D	USB	USB_BT_N	13 33
	USB_85D	USB	USB_BT_CONN_P	33 86
	USB_85D	USB	USB_BT_CONN_N	33 86
USB_NC	USB_85D	USB	NC_USB_IRP	13 87
USB_NC	USB_85D	USB	NC_USB_IRN	13 87
USB_TPAD	USB_85D	USB	USB_TPAD_P	13 38 86
USB_TPAD	USB_85D	USB	USB_TPAD_N	13 38 86
	USB_85D	USB	USB_TPAD_R_P	
	USB_85D	USB	USB_TPAD_R_N	
PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS	13
USB3_EXTA_RX	USB_85D	USB3_D2R	USB3_EXTA_D2R_P	13 37
USB3_EXTA_RX	USB_85D	USB3_D2R	USB3_EXTA_D2R_N	13 37
	USB_85D	USB3_D2R	USB3_EXTA_D2R_C_P	
	USB_85D	USB3_D2R	USB3_EXTA_D2R_C_N	
USB3_EXTA_TX	USB_85D	USB3_R2D	USB3_EXTA_R2D_P	37
USB3_EXTA_TX	USB_85D	USB3_R2D	USB3_EXTA_R2D_N	37
	USB_85D	USB3_R2D	USB3_EXTA_R2D_C_P	13 37
	USB_85D	USB3_R2D	USB3_EXTA_R2D_C_N	13 37
USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_P	13 81 86
USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_N	13 81 86
	USB_85D	USB3_D2R	USB3_EXTB_D2R_C_P	
	USB_85D	USB3_D2R	USB3_EXTB_D2R_C_N	
USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_P	81 86
USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_N	81 86
	USB_85D	USB3_R2D	USB3_EXTB_R2D_C_P	13 81
	USB_85D	USB3_R2D	USB3_EXTB_R2D_C_N	13 81
NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RP	13 87
NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RN	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CP	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CN	13 87
NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RP	13 87
NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RN	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CP	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CN	13 87

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTC	11 19
SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB	11 19
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB_R	11
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	19 36
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	19 28
	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	28

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?
SPI3X	*	=3x_DIELECTRIC	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_SE	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_SE	TOP,BOTTOM	=3x_DIELECTRIC	?

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	*	=2X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?
PCIE_2OTHER	*	=4X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?
PCIECLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_R2D	PCIE_D2R	*	PCIE_TXRX
PCIE_*	*	*	PCIE_2OTHER
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_2OTHER

PCH Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	13 40 82
	LPC_FRAME_L	LPC_45S	LPC	LPC_FRAME_L	13 40 82
	SMBUS_PCH_CLK	SMB_45S	SMB	SMBUS_PCH_CLK	13 18 43 81 85 86
	SMBUS_PCH_DATA	SMB_45S	SMB	SMBUS_PCH_DATA	13 18 43 81 85 86
	SMBUS_PCH_0_CLK	SMB_45S	SMB	SML_PCH_0_CLK	13 43
	SMBUS_PCH_0_DATA	SMB_45S	SMB	SML_PCH_0_DATA	13 43
	SMBUS_PCH_1_CLK	SMB_45S	SMB	SML_PCH_1_CLK	13 43
	SMBUS_PCH_1_DATA	SMB_45S	SMB	SML_PCH_1_DATA	13 43
	HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	11 52
		HDA_45S	HDA	HDA_BIT_CLK_R	11
	HDA_SYNC	HDA_45S	HDA	HDA_SYNC	11 52
		HDA_45S	HDA	HDA_SYNC_R	11
		HDA_45S	HDA	HDA_RST_R_L	11
	HDA_RST_L	HDA_45S	HDA	HDA_RST_L	11 52
	HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	11 52
	HDA_SDIN0_R	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	52
	HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	11 52
		HDA_45S	HDA	HDA_SDOUT_R	11 19
	USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_P	13 20 81 86
	USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_N	13 20 81 86
	USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_P	13 20 81 86
	USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_N	13 20 81 86
	PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_P	33 86
	PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_N	33 86
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_P	13 33
I304		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_N	13 33
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_P	
I303		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_N	
	PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_P	13 20 33
	PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_N	13 20 33
I275		PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_P	86
I276		PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_N	86
I325	PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_P	35 36
I326	PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_N	35 36
I327		PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_P	13 36
I328		PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_N	13 36
I329	PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_P	13 20 36
I330	PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_N	13 20 36
I331		PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_P	35 36
I332		PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_N	35 36
		CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC_R	11 19
	PCH_LPC_CLK0	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC	19 40
I293		CLK_LPC_45S	CLK_LPC	NC_LPC_CLK33M_LPCPLUS_R	11 87
I298	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIIN	11 19
I258	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK14P3M_REFCLK	11
I260	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIOUT	11 19
I253	PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_P	11
I254	PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_N	11
I262	PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P	11 28
I261	PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N	11 28
I255	PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_P	11
I257	PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_N	11
I256	PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_P	11
I259	PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_N	11
	PCIE_CLK100M_ENET	PCIE_85D	CLK_PCIE	PCIE_CLK100M_SD_P	
	PCIE_CLK100M_ENET	PCIE_85D	CLK_PCIE	PCIE_CLK100M_SD_N	
	PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P	11 33
	PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N	11 33
I300		PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	33 86
I299		PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	33 86
	PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	11 36
	PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	11 36
I333		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	35 36
I334		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	35 36
I281	PCIE_CLK100M_FW	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P	11 34
I282		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N	11 34
I397	PCIE_CLK100M_GPU	CLK_PCIE_85D	CLK_PCIE	PEG_CLK100M_P	11 70 85
I396		CLK_PCIE_85D	CLK_PCIE	PEG_CLK100M_N	11 70 85

PCH Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I313	PCH_PM_NET	PCH_45S	PCH_SE	PCH_INTRUDER_L	11
I314	PCH_PM_NET	PCH_45S	PCH_SE	PCH_INTVRMEN_L	11
I315	PCH_PM_NET	PCH_45S	PCH_SE	PCH_DSWVRMEN	12
I316	PCH_PM_NET	PCH_45S	PCH_SE	PCH_SRTCRST_L	11
I308	PCH_PM_NET	PCH_45S	PCH_SE	PM_RSMRST_L	12 67 86
I307	PCH_PM_NET	PCH_45S	PCH_SE	PM_SYSRST_L	12 19 40 86
I318	PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_PWROK	12 19 86 91
I317	PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_PWROK	12 19 86 91
I320	PCH_PM_NET	PCH_45S	PCH_SE	PM_DSW_PWRGD	12 40 86
I319	PCH_PM_NET	PCH_45S	PCH_SE	PM_PCH_SYS_PWROK	12 18 19 40 86
I321	PCH_PM_NET	PCH_45S	PCH_SE	PM_PWRBTN_L	12 18 40
I322	PCH_PM_NET	PCH_45S	PCH_SE	PM_THRMTRIP_L_R	14 41 42
I323	PCH_PCIE_WAKE	PCH_45S	PCH_SE	PCIE_WAKE_L	12 33 35 86
I324	PCH_PM_NET	PCH_45S	PCH_SE	PCH_RCIN_L	14
I386	SPI_MLB	SPI_45S	SPI3X	SPI_ALT_CLK	
I369	SPI_MLB	SPI_45S	SPI	SPI_CLK	50
I368	SPI_MLB	SPI_45S	SPI	SPI_CLK_R	13 50
I383	SPI_MLB	SPI_45S	SPI3X	SPI_MLB_CLK	50
I382	SPI_MLB	SPI_45S	SPI3X	SPI_SMC_CLK	40 50
I395	SPI_MLB	SPI_45S	SPI3X	SPI_ALT_CS_L	
I364	SPI_MLB	SPI_45S	SPI	SPI_CS0_L	50
I363	SPI_MLB	SPI_45S	SPI	SPI_CS0_R_L	13 50
I392	SPI_MLB	SPI_45S	SPI3X	SPI_MLB_CS_L	50
I391	SPI_MLB	SPI_45S	SPI3X	SPI_SMC_CS_L	40 50
I390	SPI_MLB	SPI_45S	SPI3X	SPI_ALT_IO1_MISO	
I359	SPI_MLB	SPI_45S	SPI	SPI_MISO	13 50
I358	SPI_MLB	SPI_45S	SPI	SPI_MISO_R	50
I387	SPI_MLB	SPI_45S	SPI3X	SPI_MLB_IO1_MISO	50
I381	SPI_MLB	SPI_45S	SPI3X	SPI_SMC_MISO	40 50
I380	SPI_MLB	SPI_45S	SPI3X	SPI_ALT_IO0_MOSI	
I354	SPI_MLB	SPI_45S	SPI	SPI_MOSI	50
I353	SPI_MLB	SPI_45S	SPI	SPI_MOSI_R	13 50
I377	SPI_MLB	SPI_45S	SPI3X	SPI_MLB_IO0_MOSI	50
I376	SPI_MLB	SPI_45S	SPI3X	SPI_SMC_MOSI	40 50
I375	SPI_MLB_IO2	SPI_45S	SPI3X	SPI_IO<2>	13 50
I374	SPI_MLB_IO2	SPI_45S	SPI3X	SPI_MLB_IO2_WP_L	50
I371	SPI_MLB_IO2	SPI_45S	SPI3X	SPI_ALT_IO2_WP_L	
I373	SPI_MLB_IO3	SPI_45S	SPI3X	SPI_IO<3>	13 50
I372	SPI_MLB_IO3	SPI_45S	SPI3X	SPI_MLB_IO3_HOLD_L	50
I370	SPI_MLB_IO3	SPI_45S	SPI3X	SPI_ALT_IO3_HOLD_L	
I344	SPI_TPAD	SPI_45S	SPI	TPAD_SPI_SCLK	38 85
I343	SPI_TPAD_CS	SPI_45S	SPI	TPAD_SPI_CS_L	38 85
I342	SPI_TPAD	SPI_45S	SPI	TPAD_SPI_MISO	38 85
I341	SPI_TPAD	SPI_45S	SPI	TPAD_SPI_MOSI	38 85

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTRL	*	=2x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTRL2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	*	*	MEM_2OTHER
MEM_*_DQS_*	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

DDR3 (Memory Down):

DQ signals should be matched within 0.508mm of associated DQS pair
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-139.73mm] and [CLK-30.48mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-2.54mm] to [CLK+2.54mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to first DRAM device is 139.7mm max, to last DRAM device is 194.31mm max.

SOURCE: Double checked with Doc#486985 Chief River SFF Platform DG: Memory Down
SOURCE: Need to re-confirm CRW DG for memory down (Intel not yet provided)

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	MEM_A_CLK0	MEM_72D	MEM_CLK	MEM_A_CLK_P<0>	7 23 27
	MEM_A_CLK0	MEM_72D	MEM_CLK	MEM_A_CLK_N<0>	7 23 27
I130	MEM_A_CLK1	MEM_72D	MEM_CLK	MEM_A_CLK_P<1>	7 24 27
I129	MEM_A_CLK1	MEM_72D	MEM_CLK	MEM_A_CLK_N<1>	7 24 27
	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_CKE<0>	7 23 27
I124	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_CKE<1>	7 24 27
I125	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_CS_L<0>	7 23 27
I101	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_CS_L<1>	7 24 27
I126	MEM_A_CNTL0	MEM_40S	MEM_CTRL	MEM_A_ODT<0>	7 23 27
I103	MEM_A_CNTL1	MEM_40S	MEM_CTRL	MEM_A_ODT<1>	7 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_A<15..0>	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_BA<2..0>	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_RAS_L	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_CAS_L	7 23 24 27
	MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_WE_L	7 23 24 27
	MEM_A_DATA_0	MEM_45S	MEM_A_DATA_0	MEM_A_DQ<7..0>	7 23 24
	MEM_A_DATA_1	MEM_45S	MEM_A_DATA_1	MEM_A_DQ<15..8>	7 23 24
	MEM_A_DATA_2	MEM_45S	MEM_A_DATA_2	MEM_A_DQ<23..16>	7 23 24
	MEM_A_DATA_3	MEM_45S	MEM_A_DATA_3	MEM_A_DQ<31..24>	7 23 24
	MEM_A_DATA_4	MEM_45S	MEM_A_DATA_4	MEM_A_DQ<39..32>	7 23 24
	MEM_A_DATA_5	MEM_45S	MEM_A_DATA_5	MEM_A_DQ<47..40>	7 23 24
	MEM_A_DATA_6	MEM_45S	MEM_A_DATA_6	MEM_A_DQ<55..48>	7 23 24
I105	MEM_A_DATA_7	MEM_45S	MEM_A_DATA_7	MEM_A_DQ<63..56>	7 23 24
I106	MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 23 24
	MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 23 24
	MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 23 24
	MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 23 24
	MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 23 24
	MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 23 24
	MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 23 24
	MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 23 24
	MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 23 24
	MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 23 24
	MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 23 24
	MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 23 24
I117	MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 23 24
I118	MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 23 24
I116	MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 23 24
I119	MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 23 24
	MEM_B_CLK0	MEM_72D	MEM_CLK	MEM_B_CLK_P<0>	7 25 27
	MEM_B_CLK0	MEM_72D	MEM_CLK	MEM_B_CLK_N<0>	7 25 27
I127	MEM_B_CLK1	MEM_72D	MEM_CLK	MEM_B_CLK_P<1>	7 26 27
I128	MEM_B_CLK1	MEM_72D	MEM_CLK	MEM_B_CLK_N<1>	7 26 27
	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_CKE<0>	7 25 27
I120	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_CKE<1>	7 26 27
I121	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_CS_L<0>	7 25 27
I122	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_CS_L<1>	7 26 27
I123	MEM_B_CNTL0	MEM_40S	MEM_CTRL	MEM_B_ODT<0>	7 25 27
	MEM_B_CNTL1	MEM_40S	MEM_CTRL	MEM_B_ODT<1>	7 26 27
I108	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_A<15..0>	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_BA<2..0>	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_RAS_L	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_CAS_L	7 25 26 27
	MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_WE_L	7 25 26 27
	MEM_B_DATA_0	MEM_45S	MEM_B_DATA_0	MEM_B_DQ<7..0>	7 25 26
	MEM_B_DATA_1	MEM_45S	MEM_B_DATA_1	MEM_B_DQ<15..8>	7 25 26
	MEM_B_DATA_2	MEM_45S	MEM_B_DATA_2	MEM_B_DQ<23..16>	7 25 26
	MEM_B_DATA_3	MEM_45S	MEM_B_DATA_3	MEM_B_DQ<31..24>	7 25 26
	MEM_B_DATA_4	MEM_45S	MEM_B_DATA_4	MEM_B_DQ<39..32>	7 25 26
	MEM_B_DATA_5	MEM_45S	MEM_B_DATA_5	MEM_B_DQ<47..40>	7 25 26
	MEM_B_DATA_6	MEM_45S	MEM_B_DATA_6	MEM_B_DQ<55..48>	7 25 26
	MEM_B_DATA_7	MEM_45S	MEM_B_DATA_7	MEM_B_DQ<63..56>	7 25 26
	MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 25 26
	MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 25 26
	MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 25 26
	MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 25 26
	MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 25 26
	MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 25 26
	MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 25 26
	MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 25 26
	MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 25 26
	MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 25 26
	MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 25 26
	MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 25 26
I113	MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 25 26
I112	MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 25 26
I115	MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 25 26
I114	MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM_B_DQS_N<7>	7 25 26
I110			MEM_PWR	PP0V75_S3_MEM_VREFDQ_A	22 23 24 85 89
I109			MEM_PWR	PP0V75_S3_MEM_VREFCA	22 23 24 25 26 85 89
I111			MEM_PWR	PP1V35_S3_MEM	22 23 24 25 26 27 45 84

SYNC_MASTER=SIDLE_J45 SYNC_DATE=12/10/2012

PAGE TITLE
Memory Constraints

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DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SOURCE: Bill Cornelius's Thunderbolt Routing Notes

TBT_DP Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	*	=3X_DIELECTRIC	?
TBTDP_TXRX	*	=6X_DIELECTRIC	?
TBTDP_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
TBTDP_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBTDP_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_R2D	TBTDP_D2R	*	TBTDP_TXRX
TBTDP_*	*	*	TBTDP_2OTHER

Thunderbolt/DP Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_C_P<1..0>	28 31
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_C_N<1..0>	28 31
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_P<1..0>	31
	TBT_A_R2D	TBTDP_85D	TBTDP_R2D	TBT_A_R2D_N<1..0>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	28 31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	28 31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	31
	DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	31
I308	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	28 31
I310	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	28 31
I311	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	31
I309	DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_P<0>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_N<0>	31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_P<0>	28 31
	TBT_A_D2R0	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_N<0>	28 31
I312	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_P<1>	31
I313	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_C_N<1>	31
I314	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_P<1>	28 31 87
I315	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R_N<1>	28 31 87
	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R1_AUXDDC_P	31
	TBT_A_D2R1	TBTDP_85D	TBTDP_D2R	TBT_A_D2R1_AUXDDC_N	31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_P	28 31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_N	28 31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_P	31
	TBT_A_AUXCH	DP_85D		DP_TBTPA_AUXCH_N	31
I316	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_C_P<1..0>	28 32
I317	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_C_N<1..0>	28 32
I318	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_P<1..0>	32
I319	TBT_B_R2D	TBTDP_85D	TBTDP_R2D	TBT_B_R2D_N<1..0>	32
I321	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	28 32
I322	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	28 32
I320	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	32
I324	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	32
I325	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	32
I323	DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	32
I327	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	28 32
I326	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	28 32
I328	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	32
I329	DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	32
I331	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_P<0>	32
I332	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_N<0>	32
I330	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_P<0>	28 32
I334	TBT_B_D2R0	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_N<0>	28 32
I333	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_P<1>	32
I336	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_C_N<1>	32
I337	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_P<1>	28 32
I335	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R_N<1>	28 32
I339	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R1_AUXDDC_P	32
I338	TBT_B_D2R1	TBTDP_85D	TBTDP_D2R	TBT_B_D2R1_AUXDDC_N	32
I341	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_P	28 32
I340	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_N	28 32
I342	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_P	32
I343	TBT_B_AUXCH	DP_85D		DP_TBTPB_AUXCH_N	32

Only used on dual-port hosts.

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
TBT_SPI_CLK	TBT_SPI_45S	TBT_SPI	TBT_SPI_CLK	28
TBT_SPI_MOSI	TBT_SPI_45S	TBT_SPI	TBT_SPI_MOSI	28
TBT_SPI_MISO	TBT_SPI_45S	TBT_SPI	TBT_SPI_MISO	28
TBT_SPI_CS_L	TBT_SPI_45S	TBT_SPI	TBT_SPI_CS_L	28

Only used on hosts supporting Thunderbolt video-in

SYNC_MASTER=SIDLE_J45 SYNC_DATE=12/10/2012

PAGE TITLE: Thunderbolt Constraints

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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	35 36
	S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	35 36
	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	35 36
I101	S2_MEM_CNTL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	35 36
I102		S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	36
I103	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	35 36
I104	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	35 36
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	35 36
I106	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	35 36
	S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	35 36
I108	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	35 36
I107	S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	35 36
I109	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	35 36
I110	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	35 36
I147	S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	35 36
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	35 36
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	35 36
I127	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	35 36
I128	MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	35 36
I129		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	36 86
I130		MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	36 86
I134	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	35 36
I133	MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	35 36
I132		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	36 86
I131		MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	36 86
I145			S2_MEM_PWR	PP1V35_CAM	35 36
I146			S2_MEM_PWR	PP0V675_CAM_VREF	35 36
I148			S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	36
I149			S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	36

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SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
SMBUS_SMC_2_S3_SCL	SMB_45S	SMB	SMBUS_SMC_2_S3_SCL	38 40 43 86
SMBUS_SMC_2_S3_SDA	SMB_45S	SMB	SMBUS_SMC_2_S3_SDA	38 40 43 86
SMBUS_SMC_1_S0_SCL	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	40 43 48
SMBUS_SMC_1_S0_SDA	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	40 43 48
SMBUS_SMC_0_S0_SCL	SMB_45S	SMB	SMBUS_SMC_0_S0_SCL	36 40 43 48 69 76 85 86
SMBUS_SMC_0_S0_SDA	SMB_45S	SMB	SMBUS_SMC_0_S0_SDA	36 40 43 48 69 76 85 86
SMBUS_SMC_5_SCL	SMB_45S	SMB	SMBUS_SMC_5_g3_SCL	40 43 56 57 86
SMBUS_SMC_5_SDA	SMB_45S	SMB	SMBUS_SMC_5_G3_SDA	40 43 56 57 86
SMBUS_SMC_3_SCL	SMB_45S	SMB	NC_SMBUS_SMC_3_SCL	40 42
SMBUS_SMC_3_SDA	SMB_45S	SMB	NC_SMBUS_SMC_3_SDA	40 42

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
	1TO1_DIFFPAIR		CHGR_CSI_P	57
	1TO1_DIFFPAIR		CHGR_CSI_N	57
	1TO1_DIFFPAIR		CHGR_CSO_P	57
	1TO1_DIFFPAIR		CHGR_CSO_N	57

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_50S	*	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_50S	*	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
THERM_45S_CPUVRISNS1	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.2 MM	0.2 MM
THERM_1TO1_45S	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
SENSE_1TO1_45S	*	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE_*	*	GND_P2MM
GND	SATA_*	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA_*	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.1 MM	500 MIL		
CPU_27P4S	BOTTOM			0.23 MM	100 MIL		
USB3_85D	TOP			0.1 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
1TO1_DIFFPAIR	*	1:1_DIFFPAIR

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_GPUVR	*	Y	0.300 MM	0.200 MM	3.000 MM	0.400 MM	0.200 MM

AMD Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I544	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	GPUFB_CS_P	47 73
I545		THERM_1TO1_45S	THERM	GPUFB_CS_N	47 73
I546	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	GPU_TDIODE_P	48 76
I547		THERM_1TO1_45S	THERM	GPU_TDIODE_N	48 76
I549	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPU_VDDCISENSE_P	
I548	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPU_VDDCISENSE_N	
I550	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPU_VDDCI_SENSE_XW_P	80
I551	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPU_VDDCI_SENSE_XW_N	80
I552	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPUVCORE_SENSE_P	79
I553	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPUVCORE_SENSE_N	79
I554	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_PP0V95_S0GPU_R_P	47
I555	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_PP0V95_S0GPU_R_N	47
I556	SENSE_DIFFPAIR	THERM_1TO1_45S	SENSE	VSNS_GPU_0V95_XW_P	73
I557		THERM_1TO1_45S	SENSE	VSNS_GPU_0V95_XW_N	73
I558	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	VDDCIS0_CS_R_P	47
I559		SENSE_1TO1_45S	SENSE	VDDCIS0_CS_R_N	47
I560	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	GPUFB_CS_R_P	47
I561		SENSE_1TO1_45S	SENSE	GPUFB_CS_R_N	47
I562	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	VSNS_GPU_VDDC_P	71 79 85
I563	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	VSNS_GPU_VDDC_N	71 79
I564	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	VSNS_GPU_VDDI_P	71 80 85
I565	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	VSNS_GPU_VDDI_N	71 80
I568	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V8_GPU_R_P	
I569	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V8_GPU_R_N	
I566	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V8_GPU_P	
I567	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE	ISNS_1V8_GPU_N	
I570	SENSE_DIFFPAIR	THERM_1TO1_45S	SENSE	VSNS_GPU_FB_XW_P	73
I571		THERM_1TO1_45S	SENSE	VSNS_GPU_FB_XW_N	73

X425G Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	DIFFERENTIAL_PAIR	NET	PAGES
I426	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		ISNS_CPUDDR_P	46 66
I425	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		ISNS_CPUDDR_N	46 66
I421		THERM_1TO1_45S	THERM		ISNS_CPU_DDR_R_P	46
I423		THERM_1TO1_45S	THERM		ISNS_CPU_DDR_R_N	46
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		CPUTHMSNS_D2_P	48
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		CPUTHMSNS_D2_N	48
I422	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCD_PANEL_P	46 69 96
I424	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCD_PANEL_N	46 69 96
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		DDR3THMSNS_D1_P	48
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		DDR3THMSNS_D1_N	48
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		FINTHMSNS_D_P	
	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		FINTHMSNS_D_N	
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_1V35_MEM_P	45
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_1V35_MEM_N	45
I419		SENSE_1TO1_45S	SENSE		ISNS_1V35_MEM_R_P	45
I420		SENSE_1TO1_45S	SENSE		ISNS_1V35_MEM_R_N	45
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_AIRPORT_P	
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_AIRPORT_N	
		SENSE_1TO1_45S	SENSE		ISNS_AIRPORT_R_P	46
		SENSE_1TO1_45S	SENSE		ISNS_AIRPORT_R_N	46
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCDBKLT_N	63
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCDBKLT_P	63
I533	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCD_PANEL_N	46 69 96
I532	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_LCD_PANEL_P	46 69 96
I523		SENSE_1TO1_45S	SENSE		ISNS_PCH_R_P	45
I521		SENSE_1TO1_45S	SENSE		ISNS_PCH_R_N	45
I522	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_TPAD_P	45
I520	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_TPAD_N	45
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_OTHER5V_P	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_OTHER5V_N	44
I514	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_OTHER3V3_P	44
I515	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_OTHER3V3_N	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_COMPUTING_P	44
	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_COMPUTING_N	44
		SENSE_1TO1_45S	SENSE		CPUVR_ISNS_P	45
		SENSE_1TO1_45S	SENSE		CPUVR_ISNS_N	45
I510	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		P1V05_GPU_PEX_IOVDD_SNS_P	
I511	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		P1V05_GPU_PEX_IOVDD_SNS_N	
I530	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS1	CPUVR_ISNS1_P	45 59
I531	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS1	CPUVR_ISNS1_N	45 59
I528	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS2	CPUVR_ISNS2_P	45 59
I529	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS2	CPUVR_ISNS2_N	45 59
I527	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS3	CPUVR_ISNS3_P	45 59
I526	SENSE_DIFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR_ISNS3	CPUVR_ISNS3_N	45 59
I432	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		CPUVR_ISUM_R_P	45
I436	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		CPUVR_ISUM_R_N	45
I498	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GFXIMVP_ISNS1_P	79 96
I499	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GFXIMVP_ISNS1_N	79 96
I502	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GFXIMVP_ISNS1_P	79 96
I503	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GFXIMVP_ISNS1_N	79 96
I405	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO		ISNS_TBT_N	45
I406	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO		ISNS_TBT_P	45
I407		AUDIODIFF	AUDIO		ISNS_TBT_R_N	45
I408		AUDIODIFF	AUDIO		ISNS_TBT_R_P	45
I446	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		ISNS_SSD_P	45
I445		THERM_1TO1_45S	THERM		ISNS_SSD_N	45
I447	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		ISNS_SSD_R_P	45
I448		THERM_1TO1_45S	THERM		ISNS_SSD_R_N	45
I449	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		P1V05S0_CS_P	45 62
I450		THERM_1TO1_45S	THERM		P1V05S0_CS_N	45 62
I451	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM	P1V05S0_SENSE	P1V05S0_SENSE_P	62
I452		THERM_1TO1_45S	THERM	P1V05S0_SENSE	P1V05S0_SENSE_N	62
I453	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		TBT_THERMDP	28 48
I454		THERM_1TO1_45S	THERM		TBT_THERMDN	48
I455	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO		CHGR_CSI_R_P	57
I456		AUDIODIFF	AUDIO		CHGR_CSI_R_N	57
I457	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO		CHGR_CSO_R_P	57
I458		AUDIODIFF	AUDIO		CHGR_CSO_R_N	57
I534	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		GFXIMVP_ISNS2_P	79
I535	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		GFXIMVP_ISNS2_N	79
I536	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_GPU_P	47
I537	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_HS_GPU_N	47
I539	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_PP0V95_S0GPU_P	47 73
I538	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		ISNS_PP0V95_S0GPU_N	47 73
I541	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		VDDCIS0_CS_P	47 80
I540	SENSE_DIFFPAIR	SENSE_1TO1_45S	SENSE		VDDCIS0_CS_N	47 80
I543	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GPUTHMSNS_D_P	48
I542	SENSE_DIFFPAIR	THERM_1TO1_45S	THERM		GPUTHMSNS_D_N	48

X425G Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I361		AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_P	53
I362		AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_N	53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_P	53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_N	53
I459		AUDIODIFF	AUDIO	RSUBIN_P	53
I440		AUDIODIFF	AUDIO	RSUBIN_N	53
I441		AUDIODIFF	AUDIO	LSUBIN_P	53
I442		AUDIODIFF	AUDIO	LSUBIN_N	53
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	51 53
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	51 53
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	51 53
	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	51 53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_P	53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_N	53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_P	53
		AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_N	53
I461		AUDIODIFF	AUDIO	SPKRAMP_RIN_P	53
I462		AUDIODIFF	AUDIO	SPKRAMP_RIN_N	53
I460		AUDIODIFF	AUDIO	SPKRAMP_LIN_P	53
I463		AUDIODIFF	AUDIO	SPKRAMP_LIN_N	53
I358	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SL_OUT_P	53 55 86
I357	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SL_OUT_N	53 55 86
I360	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SR_OUT_P	53 55 86
I359	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_SR_OUT_N	53 55 86
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_L_OUT_P	53 55 86
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_L_OUT_N	53 55 86
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_R_OUT_P	53 55 86
	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	SPKRCONN_R_OUT_N	53 55 86
I480	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_R_P	
I481	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_R_N	
I482	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	CODEC_HS_MIC_P	51
I483	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	CODEC_HS_MIC_N	51
I484	AUDIO_DIFFPAIR	DIFFPAIR	AUDIO	AUD_MIC_IN1_L_P	
I485		DIFFPAIR	AUDIO	AUD_MIC_IN1_L_N	
I486		DIFFPAIR	AUDIO	AUD_HS_MIC_P	54 55
I487		DIFFPAIR	AUDIO	AUD_HS_MIC_N	54 55
I489		DIFFPAIR	AUDIO	HS_MIC_P	51 54
I488		DIFFPAIR	AUDIO	HS_MIC_N	51 54
I490		DIFFPAIR	AUDIO	AUD_CONN_HS_MIC_P	
I491		DIFFPAIR	AUDIO	AUD_CONN_HS_MIC_N	
I494	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_R_P	51 53
I495	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_R_N	51 53
I493	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_L_P	51 53
I492	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO3_L_N	51 53
			SB_POWER	PP3V3_S5	66 67 82 84 85 86 12 14 15 17 18 19 21 31 32 33 61 64
			SB_POWER	PP3V3_S0	66 67 68 69 82 83 84 86 11 12 13 14 15 17 19 20 29 34 43 44 45 46 47 48 49 51 52 55
			SB_POWER	PP1V35_S3RS0_CPUDDR	6 8 10 21 66 67 84
			GND	GND	

SYNC_MASTER=SIDLE_J45 SYNC_DATE=12/10/2012

PAGE TITLE
Project Specific Constraints

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GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=3x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
GDDR5_DATA	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_EDC	TOP,BOTTOM	=7x_DIELECTRIC	?

GDDR5_CMD spacing can be relaxed to 2x per AMD recommendation for x32_4.5G config.

Breakout Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR5_*	*	BGA	GDDR5_BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_BGA	*	=1.3x_DIELECTRIC	?

GDDR5 FB A Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I453	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_P	72 74
I454	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_N	72 74
I452	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_P	72 74
I451	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_N	72 74
I450	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_A<8..0>	72 74
I448	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_A<8..0>	72 74
I449	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_ABI_L	72 74
I447	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_ABI_L	72 74
I446	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_RAS_L	72 74
I445	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_RAS_L	72 74
I444	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CAS_L	72 74
I442	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CAS_L	72 74
I443	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_WE_L	72 74
I440	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_WE_L	72 74
I441	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CKE_L	72 74
I439	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CKE_L	72 74
I437	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CS_L	72 74
I438	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CS_L	72 74
I435	FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<0>	72 74
I436	FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<1>	72 74
I402	FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<2>	72 74
I400	FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<3>	72 74
I401	FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<0>	72 74
I398	FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<1>	72 74
I399	FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<2>	72 74
I397	FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<3>	72 74
I395	FB_A0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<0>	72 74
I396	FB_A0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<1>	72 74
I394	FB_A0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<2>	72 74
I392	FB_A0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<3>	72 74
I393	FB_A1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<0>	72 74
I390	FB_A1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<1>	72 74
I391	FB_A1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<2>	72 74
I389	FB_A1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<3>	72 74
I388	FB_A0_WCLK0	GDDR5_80D	GDDR5_CLK	FB_A0_WCLK_P<0>	72 74
I387	FB_A0_WCLK0	GDDR5_80D	GDDR5_CLK	FB_A0_WCLK_N<0>	72 74
I385	FB_A0_WCLK1	GDDR5_80D	GDDR5_CLK	FB_A0_WCLK_P<1>	72 74
I386	FB_A0_WCLK1	GDDR5_80D	GDDR5_CLK	FB_A0_WCLK_N<1>	72 74
I384	FB_A1_WCLK0	GDDR5_80D	GDDR5_CLK	FB_A1_WCLK_P<0>	72 74
I383	FB_A1_WCLK0	GDDR5_80D	GDDR5_CLK	FB_A1_WCLK_N<0>	72 74
I382	FB_A1_WCLK1	GDDR5_80D	GDDR5_CLK	FB_A1_WCLK_P<1>	72 74
I380	FB_A1_WCLK1	GDDR5_80D	GDDR5_CLK	FB_A1_WCLK_N<1>	72 74
I381	FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<7..0>	72 74
I379	FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<15..8>	72 74
I378	FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<23..16>	72 74
I377	FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<31..24>	72 74
I376	FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<7..0>	72 74
I375	FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<15..8>	72 74
I374	FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<23..16>	72 74
I373	FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<31..24>	72 74
I372	FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A0_RESET_L	
I371	FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A1_RESET_L	

GDDR5 FB B Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
[illegible]	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_P	72 75
[illegible]	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_N	72 75
[illegible]	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_P	72 75
[illegible]	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_N	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_A<8..0>	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_A<8..0>	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_ABI_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_ABI_L	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_RAS_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_RAS_L	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CAS_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CAS_L	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_WE_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_WE_L	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CKE_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CKE_L	72 75
[illegible]	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CS_L	72 75
[illegible]	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CS_L	72 75
[illegible]	FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<0>	72 75
[illegible]	FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<1>	72 75
[illegible]	FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<2>	72 75
[illegible]	FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<3>	72 75
[illegible]	FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<0>	72 75
[illegible]	FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<1>	72 75
[illegible]	FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<2>	72 75
[illegible]	FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<3>	72 75
[illegible]	FB_B0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<0>	72 75
[illegible]	FB_B0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<1>	72 75
[illegible]	FB_B0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<2>	72 75
[illegible]	FB_B0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<3>	72 75
[illegible]	FB_B1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<0>	72 75
[illegible]	FB_B1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<1>	72 75
[illegible]	FB_B1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<2>	72 75
[illegible]	FB_B1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<3>	72 75
[illegible]	FB_B0_WCLK0	GDDR5_80D	GDDR5_CLK	FB_B0_WCLK_P<0>	72 75
[illegible]	FB_B0_WCLK0	GDDR5_80D	GDDR5_CLK	FB_B0_WCLK_N<0>	72 75
[illegible]	FB_B0_WCLK1	GDDR5_80D	GDDR5_CLK	FB_B0_WCLK_P<1>	72 75
[illegible]	FB_B0_WCLK1	GDDR5_80D	GDDR5_CLK	FB_B0_WCLK_N<1>	72 75
[illegible]	FB_B1_WCLK0	GDDR5_80D	GDDR5_CLK	FB_B1_WCLK_P<0>	72 75
[illegible]	FB_B1_WCLK0	GDDR5_80D	GDDR5_CLK	FB_B1_WCLK_N<0>	72 75
[illegible]	FB_B1_WCLK1	GDDR5_80D	GDDR5_CLK	FB_B1_WCLK_P<1>	72 75
[illegible]	FB_B1_WCLK1	GDDR5_80D	GDDR5_CLK	FB_B1_WCLK_N<1>	72 75
[illegible]	FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<7..0>	72 75
[illegible]	FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<15..8>	72 75
[illegible]	FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<23..16>	72 75
[illegible]	FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<31..24>	72 75
[illegible]	FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<7..0>	72 75
[illegible]	FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<15..8>	72 75
[illegible]	FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<23..16>	72 75
[illegible]	FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<31..24>	72 75
[illegible]	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B0_RESET_L	
[illegible]	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B1_RESET_L	

MUXGFX & DP AUX MUX NET PROPERTIES

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I486		DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	69 82
I485		DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	69 82
I484		DP_85D	DISPLAYPORT	DP_INT_ML_F_P<3..0>	69
I483		DP_85D	DISPLAYPORT	DP_INT_ML_F_N<3..0>	69
I482	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	69 86
I481	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	69 86
I480		DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_P	69 82
I479		DP_85D	DISPLAYPORT	DP_INT_AUXCH_C_N	69 82
I477	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_P	69 86
I478	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_N	69 86
I349		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	76 77 82
I350		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	76 77 82
I345	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<3..0>	76 82
I346		DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<3..0>	76 82
I517		DP_85D	DISPLAYPORT	DP_INT_IG_AUX_P	5 82 85
I518		DP_85D	DISPLAYPORT	DP_INT_IG_AUX_N	5 82 85
I519	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_IG_ML_P<3..0>	5 82 85
I520		DP_85D	DISPLAYPORT	DP_INT_IG_ML_N<3..0>	5 82 85
I338	DP_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_EG_AUXCH_P	76 77 83
I337	DP_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_EG_AUXCH_N	76 77 83
I339	DP_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_EG_AUXCH_P	76 77 83
I340	DP_EG_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_EG_AUXCH_N	76 77 83
I502	TBTSNK_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	28 89
I501	TBTSNK_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	28 89
I500		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	28 83 89
I499		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	28 83 89
I497	TBTSNK_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	28 89
I498	TBTSNK_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	28 89
I496		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	28 83 89
I494		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	28 83 89
I495	DP_TBT_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	28 89
I492	DP_TBT_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	28 89
I493		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	28 76 89
I491		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	28 76 89
I489	DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	28 89
I490	DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	28 89
I488		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	28 76 89
I487		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	28 76 89

Kepler Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I514		1:1_DIFFPAIR		GPU_CLK_TEST_RC_P	72
I513		1:1_DIFFPAIR		GPU_CLK_TEST_RC_N	72
I515	GPU_CLK_TEST	1:1_DIFFPAIR		GPU_CLK_TEST_P	72
I516	GPU_CLK_TEST	1:1_DIFFPAIR		GPU_CLK_TEST_N	72
	HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_EG_DATA_P<2..0>	76 81 86
	HDMI_DATA	DP_85D	DISPLAYPORT	HDMI_EG_DATA_N<2..0>	76 81 86
	HDMI_CLK	DP_85D	HDMI_CLK	HDMI_EG_CLK_P	76 81 86
	HDMI_CLK	DP_85D	HDMI_CLK	HDMI_EG_CLK_N	76 81 86

SYNC_MASTER=J45G_AMD SYNC_DATE=07/01/2014

PAGE TITLE
GPU (AMD VENUS) Constraints

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BRANCH <BRANCH>

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